

REALTEK

Software Development Kit

Switch Product

Demo Board User Guide

(CONFIDENTIAL: Development Partners Only)

Rev. 2.1.4.53590

20 January 2015



Realtek Semiconductor Corp.

No. 2, Innovation Road II, Hsinchu Science Park, Hsinchu 300, Taiwan

Tel.: +886-3-578-0211 Fax: +886-3-577-6047

www.realtek.com

COPYRIGHT

© 2008-2012 Realtek Semiconductor Corp. All rights reserved. No part of this document may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language in any form or by any means without the written permission of Realtek Semiconductor Corp.

TRADEMARKS

Realtek is a trademark of Realtek Semiconductor Corporation. Other names mentioned in this document are trademarks/registered trademarks of their respective owners.

DISCLAIMER

Realtek provides this document “as is”, without warranty of any kind, neither expressed nor implied, including, but not limited to, the particular purpose. Realtek may make improvements and/or changes in this document or in the product described in this document at any time. This document could include technical inaccuracies or typographical errors.

USING THIS DOCUMENT

This document is intended for use by the system engineer when integrating with Realtek Switch Software SDK. Though every effort has been made to assure that this document is current and accurate, more information may have become available subsequent to the production of this guide. In that event, please contact your Realtek representative for additional information that may help in the development process.

REVISION HISTORY

Revision	Release Date	Summary
2.0.0	2010/6/25	Init draft
2.0.1	2010/8/14	Update menuconfig pictures
2.0.2	2010/10/19	Revision for SDK update
2.0.3	2010/10/27	Update menuconfig pictures
2.0.4	2011/02/25	Update menuconfig pictures
2.0.5	2011/4/18	Add 8328L demo board menuconfig section in 1.4. The 8328L can't support Quad SPI flash and CPU/DRAM clock is different with 8328M/8328S.
2.0.6	2011/7/4	Revision for SDK update
2.0.6.20571	2011/8/5	Add Quad IO of SPI flash option in SDK/BSP menuconfig
2.0.7.24969	2011/11/04	Add PIE input Endian option in RTL8328M/S/L, EEE compatible, NIC Rx thread, LinkMon thread mode selection, RTL8231 reset and update menuconfig pictures

Revision	Release Date	Summary
2.0.7.26226	2012/01/09	Add workaround option for Marvell 88E6063 and update menuconfig picture.
2.0.7.29481	2012/6/1	Add 8328L serial mode board model, packet buffer watchdog option for 8328x and update the menuconfig picture. Supported the CONFIG_SDK_LINKMON_MIXED_MODE option.
Pre2.1.0.30160	2012/6/29	The 2.1.0 and late SDK package is not supported the RTL8329M, RTL8389M and RTL8389LM chip. Supported the cypress and maple series MAC chips and RTL8218B PHY chip. Add the RTL8382M-FPGA and RTL8393M-FPGA demo board section and remove the RTL8389M demo board section.
Pre2.1.0.31102	2012/7/19	Add the RTL8382M demo board section. Add the Linux 2.6.32.58 configure for RTL8328M/RTL8328L demo board and RTL8393M FPGA board Remove the RTL8389M demo board and RTL8382M FPGA board.
Pre2.1.0.32476	2012/9/7	Update the section 1.4 RTL8382M Demo Board
2.1.0.33152	2012/10/3	Add the section 1.4.1.2 Linux v2.6.32.58 Menuconfig. Modify section 1.5 from 8393M FPGA to 8393M demo board.
2.1.0.33894	2012/11/02	Update 1.5.2 section, RTL8393M/52M demo board U-Boot 2011.12 menuconfig Remove 1.4.2, RTL8382M/32M demo board U-Boot 1.3.0 menuconfig Add 1.4.2 section, RTL8382M/52M demo board U-Boot 2011.12 menuconfig Update 1.4 menuconfig figure and description. Update 1.4.2 menuconfig figures. Update 1.5 menuconfig figure and description.
2.1.0.36032	2013/01/11	Update all menu with U-Boot-2011.12 in section 1.2, 1.3, 1.4, 1.4.2, 1.5 and 1.5.2 Modify all the loader item description for U-Boot-2011.12 in section 1.2, 1.3, 1.4 and 1.5. Add U-Boot-2011.12 misc. menu in section 1.4.2 and 1.5.2.
2.1.0.33693	2013/01/30	Update 1.4.2 pictures.
2.1.1.38074	2013/03/29	Updated all menus with Linux kernel 2.6.19 and 2.6.32.58. Updated all menus with U-Boot-2011.12. Modify the section 1.5 description.
2.1.1.39239	2013/05/16	Updated RTL8380/90 figures of U-Boot-2011.12 sections.

Revision	Release Date	Summary
2.1.2.41872	2013/08/13	Modify the SDK version number. Updated RTL839x/8x U-Boot-2011.12 Board Probe Configuration figures. Updated RTL839x/8x U-Boot-2011.12 Miscellaneous figures. Updated RTL838x.9x SDK BSP Option figures.
2.1.3.pre458 42	2014/01/13	Modify the SDK version number. Updated RTL839x/8x U-Boot-2011.12 Board Probe Configuration figures. Updated RTL839x/8x U-Boot-2011.12 Miscellaneous figures. Added RTL839x/8x U-Boot-2011.12 System figures.
2.1.3.46351	2014/02/06	Change SDK version from 2.1.3.pre45842 to 2.1.3.46351.
2.1.4.53590	2015/01/20	Change SDK version from 2.1.3.46351 to 2.1.4.53590. Updated RTL839x/8x U-Boot-2011.12 Board Probe Configuration figures.

Table of Contents

Chapter 1	Menuconfig Setting	6
1.1.	Introduction	6
1.2.	RTL8328M Demo Board	6
1.2.1	Kernel Menuconfig	11
1.2.1.1	Linux v2.6.19 Menuconfig	11
1.2.1.2	Linux v2.6.32.58 Menuconfig	12
1.2.2	U-Boot v1.3.0 Menuconfig	14
1.2.3	SDK Menuconfig	18
1.3.	RTL8328L Demo Board	24
1.3.1	Kernel Menuconfig	28
1.3.1.1	Linux v2.6.19 Menuconfig	28
1.3.1.2	Linux v2.6.32.58 Menuconfig	29
1.3.2	U-Boot v1.3.0 Menuconfig	31
1.3.3	SDK Menuconfig	35
1.4.	RTL8382M/RTL8332M Demo Board	41
1.4.1	Kernel Menuconfig	46
1.4.1.1	Linux v2.6.19 Menuconfig	46
1.4.1.2	Linux v2.6.32.58 Menuconfig	47
1.4.2	U-Boot 2011.12 Menuconfig	49
1.4.3	SDK Menuconfig	54
1.5.	RTL8393M/RTL8353M Demo Board	60
1.5.1	Kernel Menuconfig	65
1.5.1.1	Linux v2.6.19 Menuconfig	65
1.5.1.2	Linux v2.6.32.58 Menuconfig	66
1.5.2	U-Boot 2011.12 Menuconfig	68
1.5.3	SDK Menuconfig	73

Chapter 1 Menuconfig Setting

1.1. Introduction

- In SDK 2.1.4.53590 package, the default menuconfig is for RTL8393M/RTL8353M demo board with SDK driver and NIC driver in kernel mode. You need to change the configure for your specified board if the default menuconfig is not meet your expected.

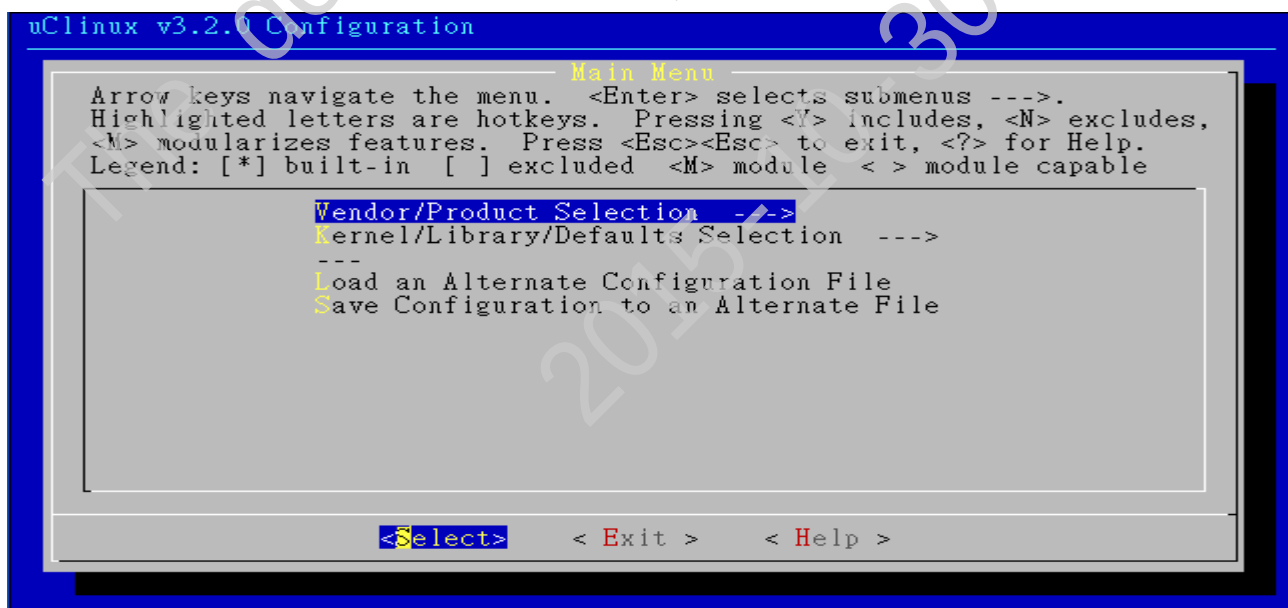
1.2. RTL8328M Demo Board

The default diag-v2 command tree of the SDK package is for 8390 series chip. Please remove following 2 files and re-generate the new diag-v2 command for the chip.

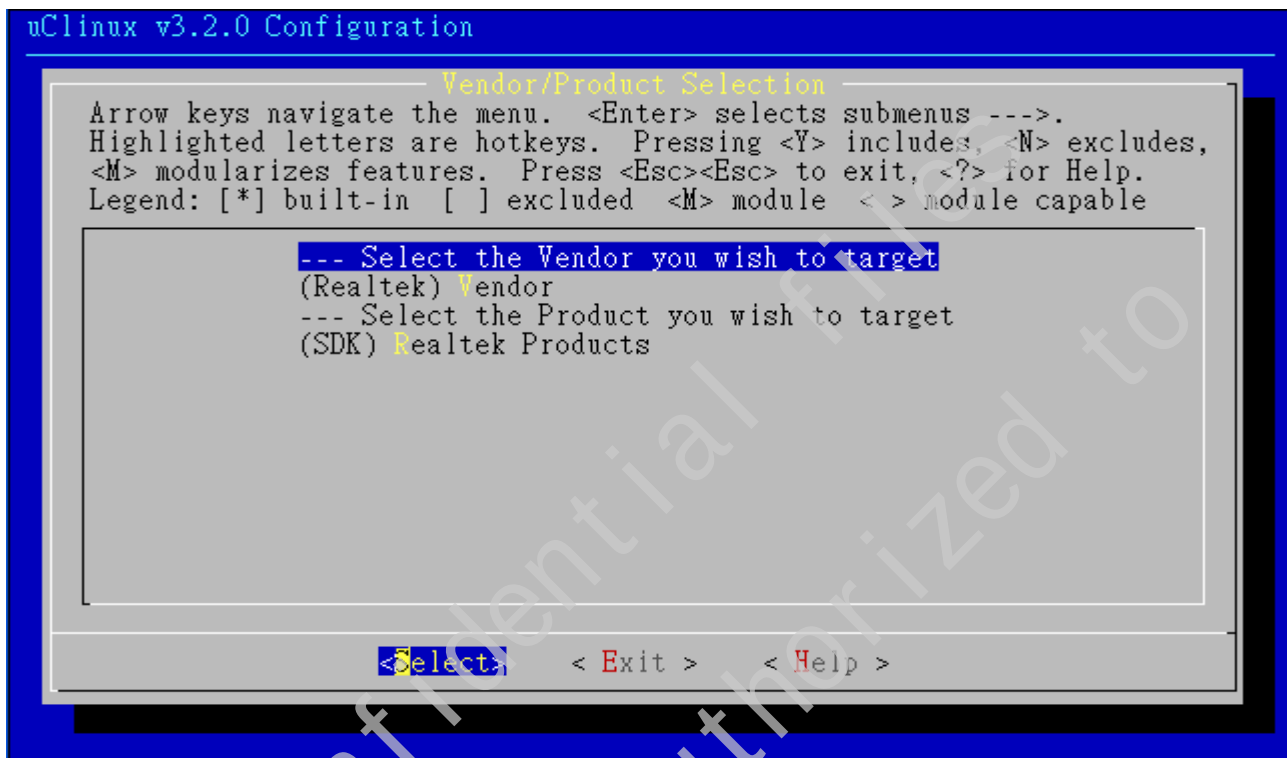
- sdk\src\app\diag_v2\include\parser\cparser_tree.h
- sdk\src\app\diag_v2\src\parser\cparser_tree.c

Following example is SDK driver and NIC driver run at Linux kernel mode

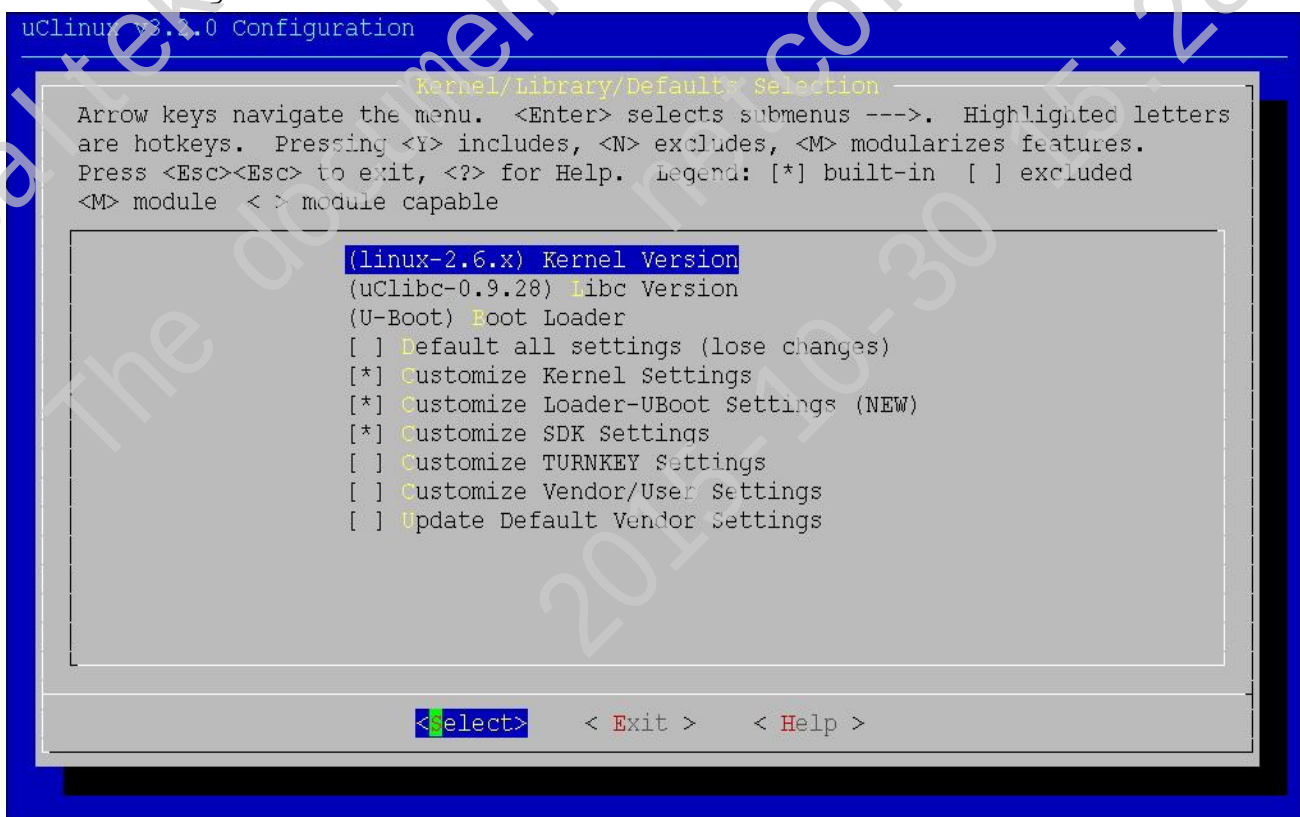
- When type 'make menuconfig' command, you can see the 'Main Menu' as following:



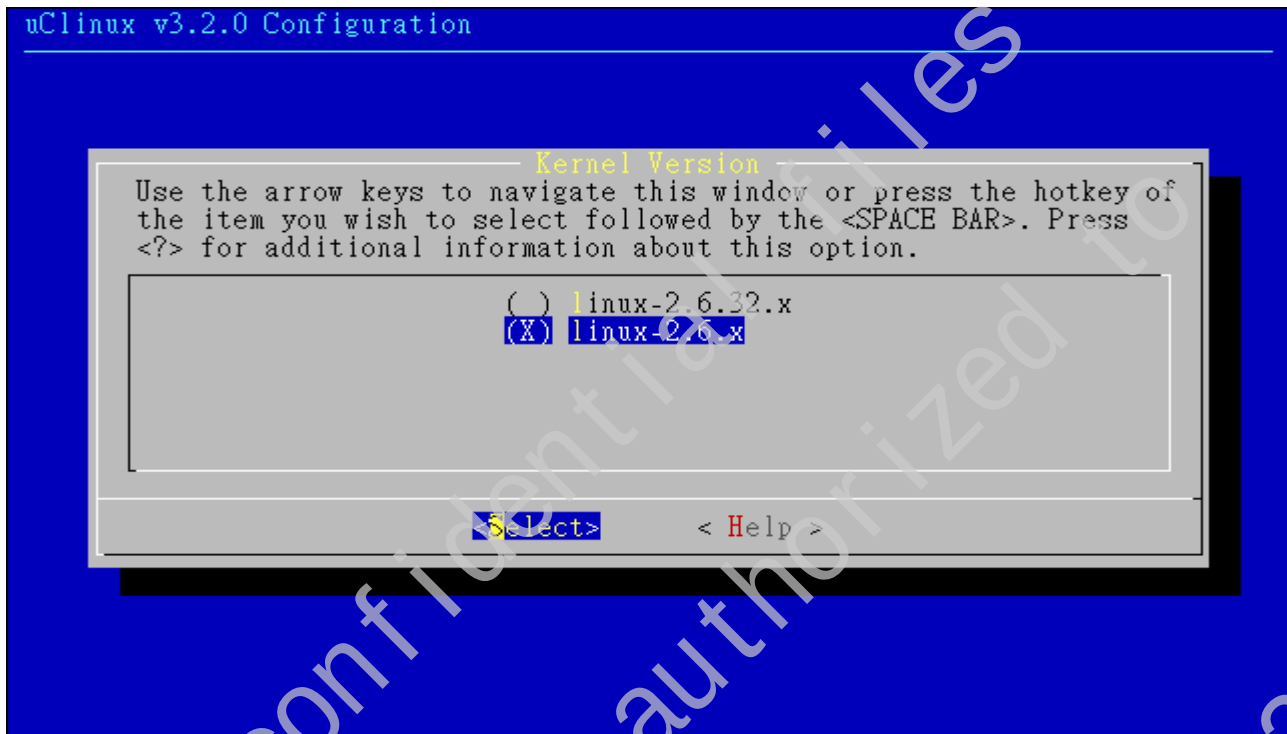
- “Vendor/Product Selection” setting as following:



- Select some items in “Kernel/Library/Defaults Selection” and “Exit” for configure Kernel, Loader-UBoot and SDK setting:



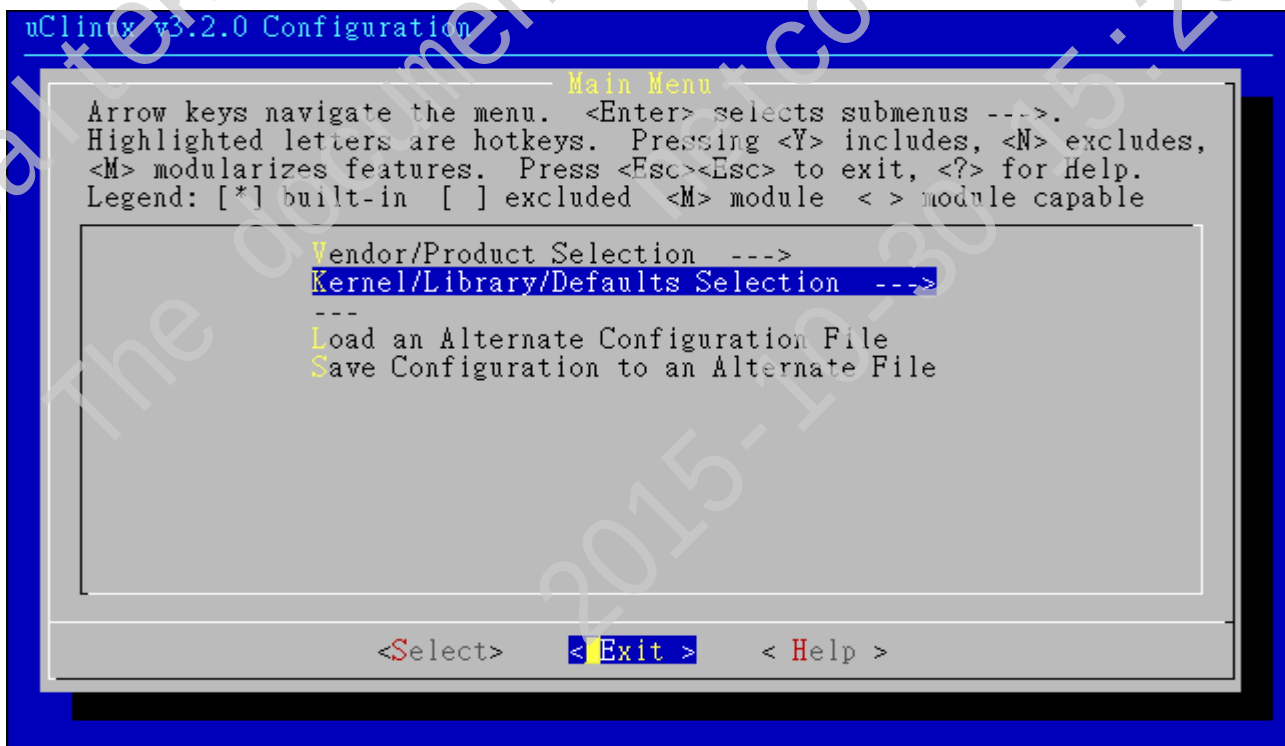
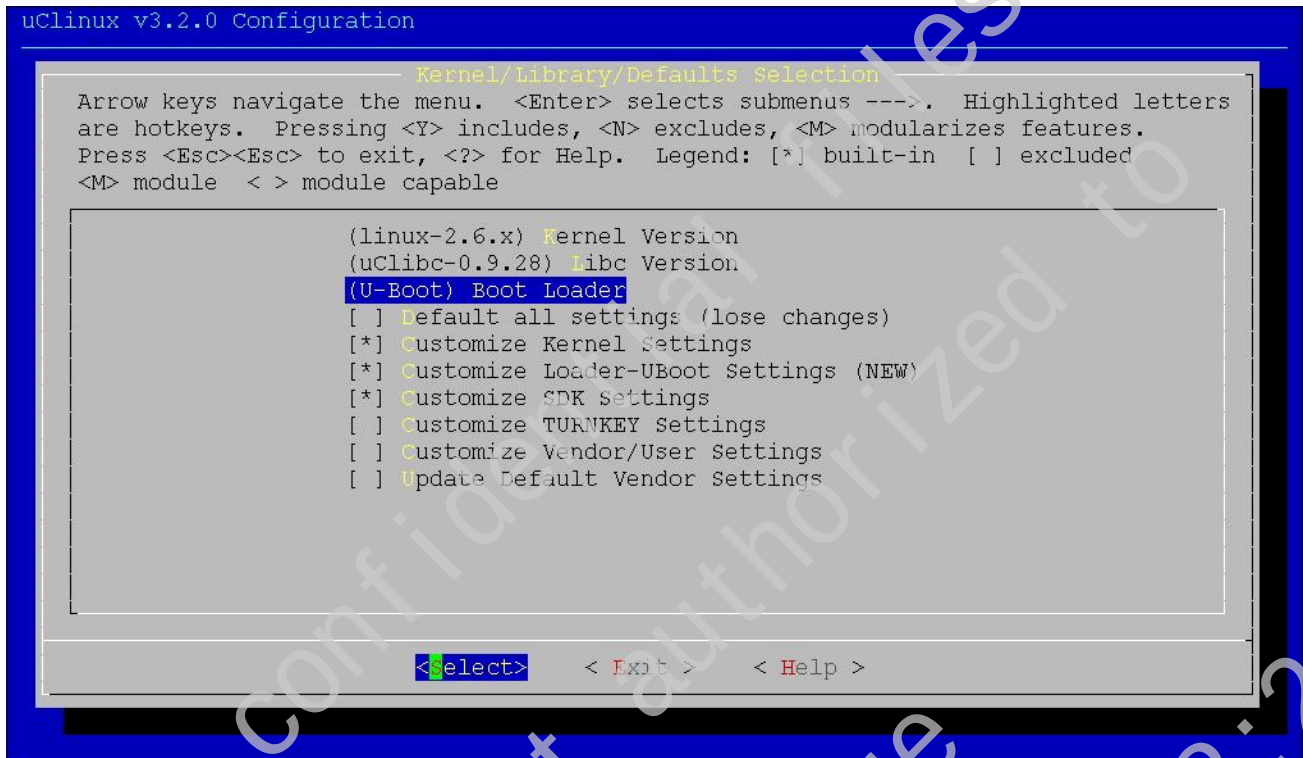
- In "Kernel/Library/Defaults Selection" menu, there are two Linux kernel version could be selected. The linux-2.6.x is 2.6.19 and the linux-2.6.32.x is 2.6.32.58:



- In "Kernel/Library/Defaults Selection" menu, there are two uClibc version could be selected. The linux-2.6.x should use 0.9.28 and the linux-2.6.32.x should used 0.9.30:



- In "Kernel/Library/Defaults Selection" menu, there are two boot code version could be selected. The U-Boot is old style and the U-Boot-2011.12 is new one. For the RTL8328M, only U-Boot is supported, and please selects the U-Boot.



uClinux v3.2.0 Configuration

Do you wish to save your new kernel configuration?

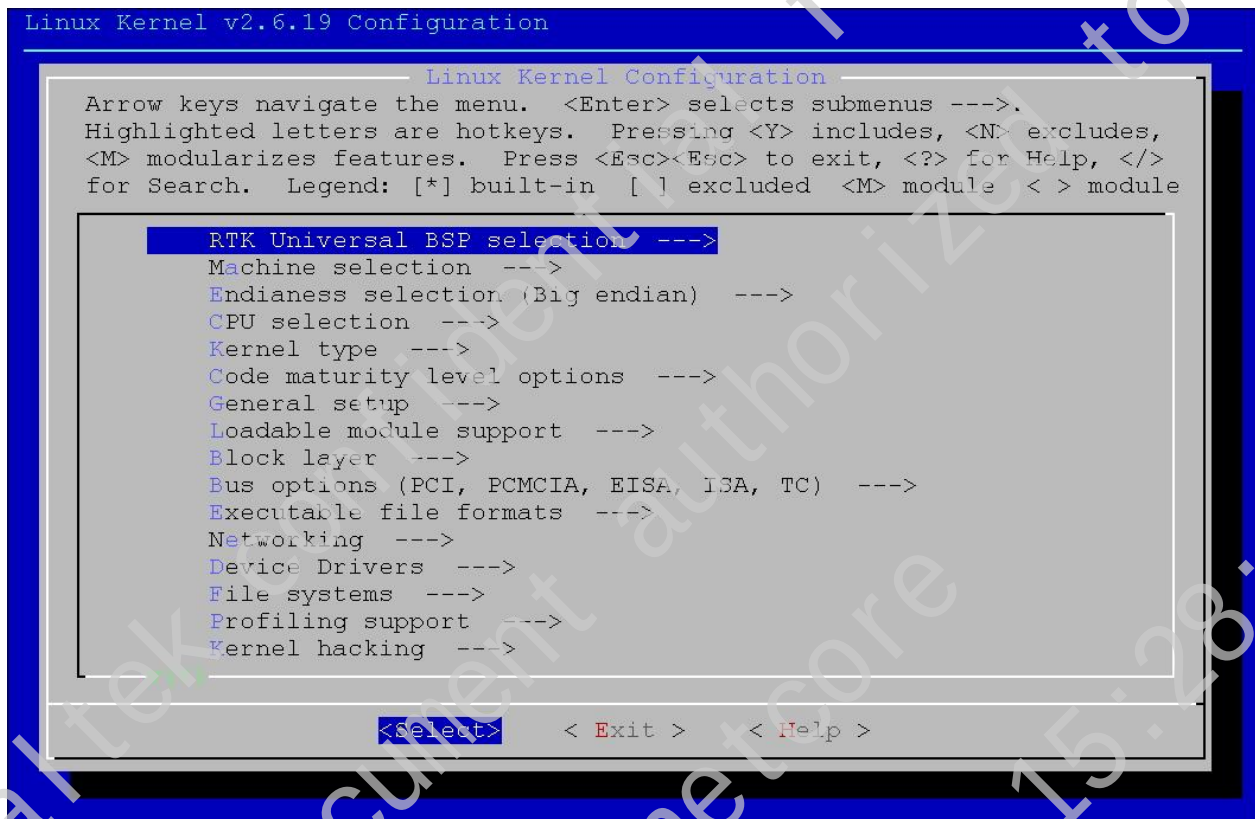
< Yes >

< No >

1.2.1 Kernel Menuconfig

1.2.1.1 Linux v2.6.19 Menuconfig

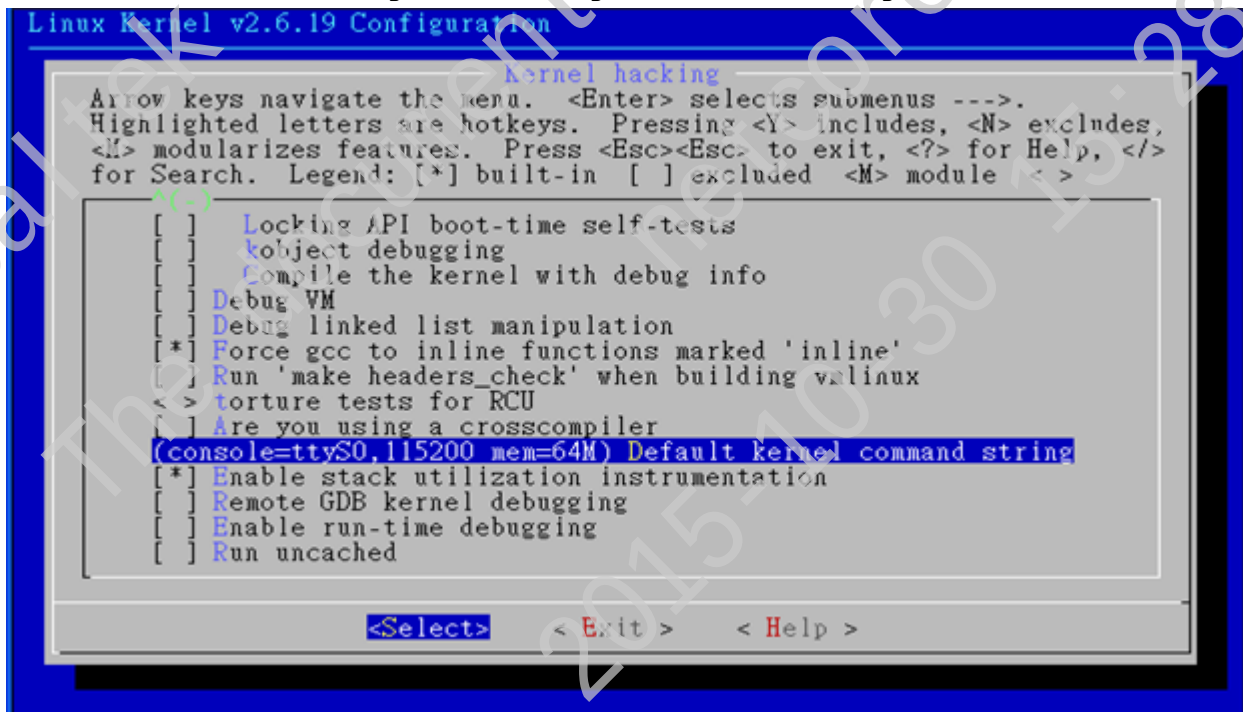
- Enter the "Linux Kernel v2.6.19 Configuration" as following:



- "RTK Universal BSP selection" setting as following: (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.



- "Kernel hacking" setting as following: (64MB DDR RAM)



1.2.1.2 Linux v2.6.32.58 Menuconfig

- Enter the "Linux Kernel v2.6.32.58 Configuration" as

following:

```
.config - Linux Kernel v2.6.32.58 Configuration

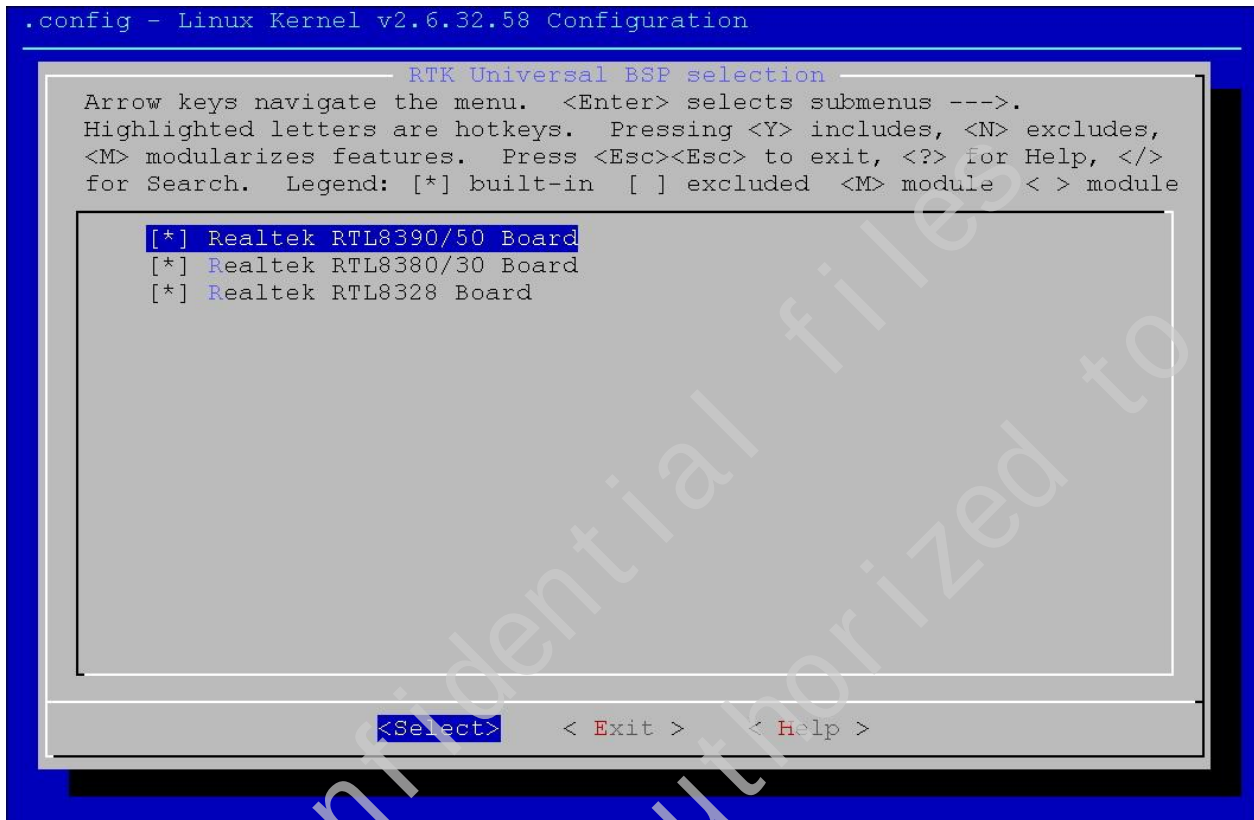
Linux Kernel Configuration

Arrow keys navigate the menu. <Enter> selects submenus --->.
Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes,
<M> modularizes features. Press <Esc><Esc> to exit, <?> for Help, </>
for Search. Legend: [*] built-in [ ] excluded <M> module < > module

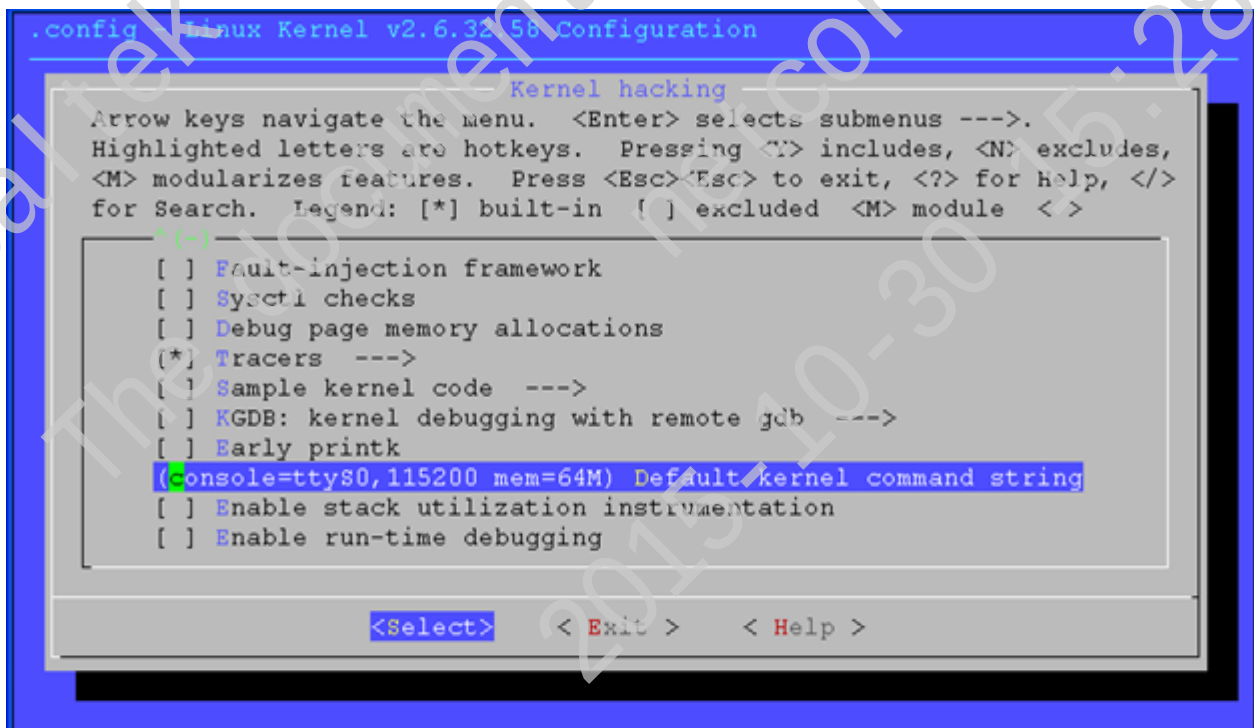
BSP configuration --->
Endianess selection (Big endian) --->
RTK Universal BSP selection ---->
CPU selection --->
Kernel type --->
General setup --->
[*] Enable loadable module support --->
[*] Enable the block layer --->
[ ] Probe initrd header created by addinitrd
Bus options (PCI, PCMCIA, EISA, ISA, TC) --->
Executable file formats --->
Power management options --->
[*] Networking support --->
Device Drivers --->
File systems --->
Kernel hacking --->

<Select> < Exit > < Help >
```

- “RTK Universal BSP selection” setting as following. (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.

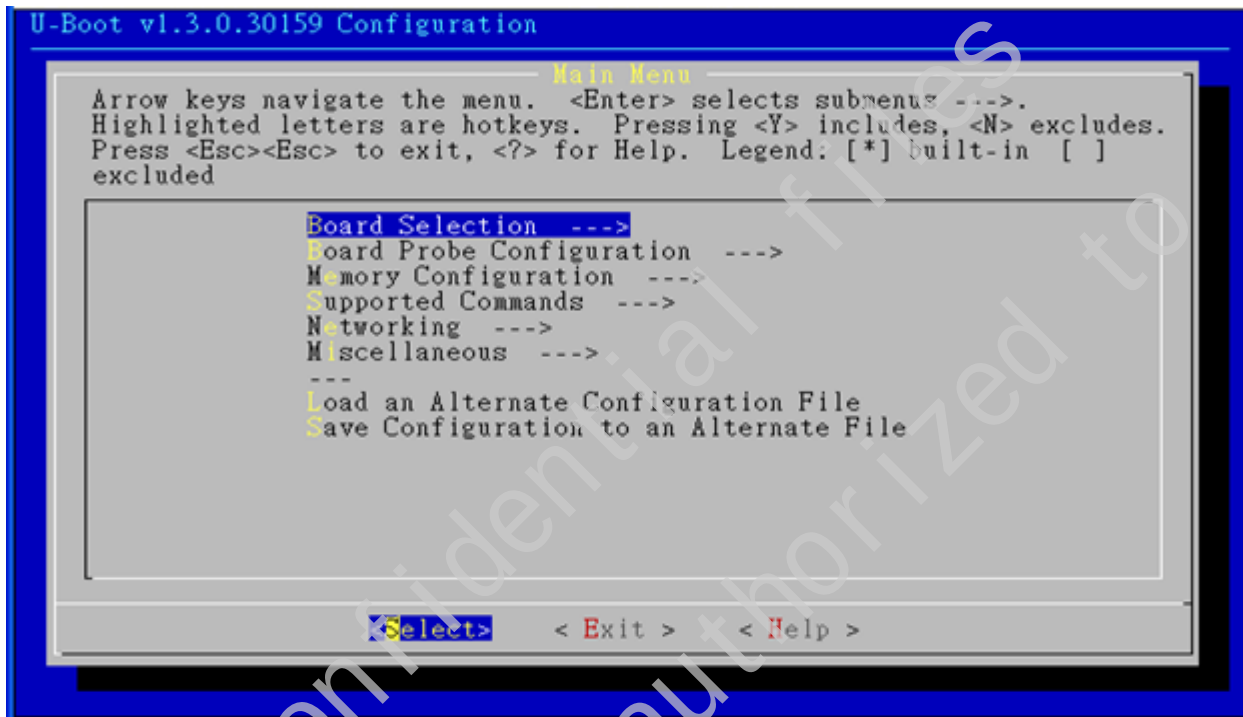


- “Kernel hacking” setting as following: (64MB DDR RAM)



1.2.2 U-Boot v1.3.0 Menuconfig

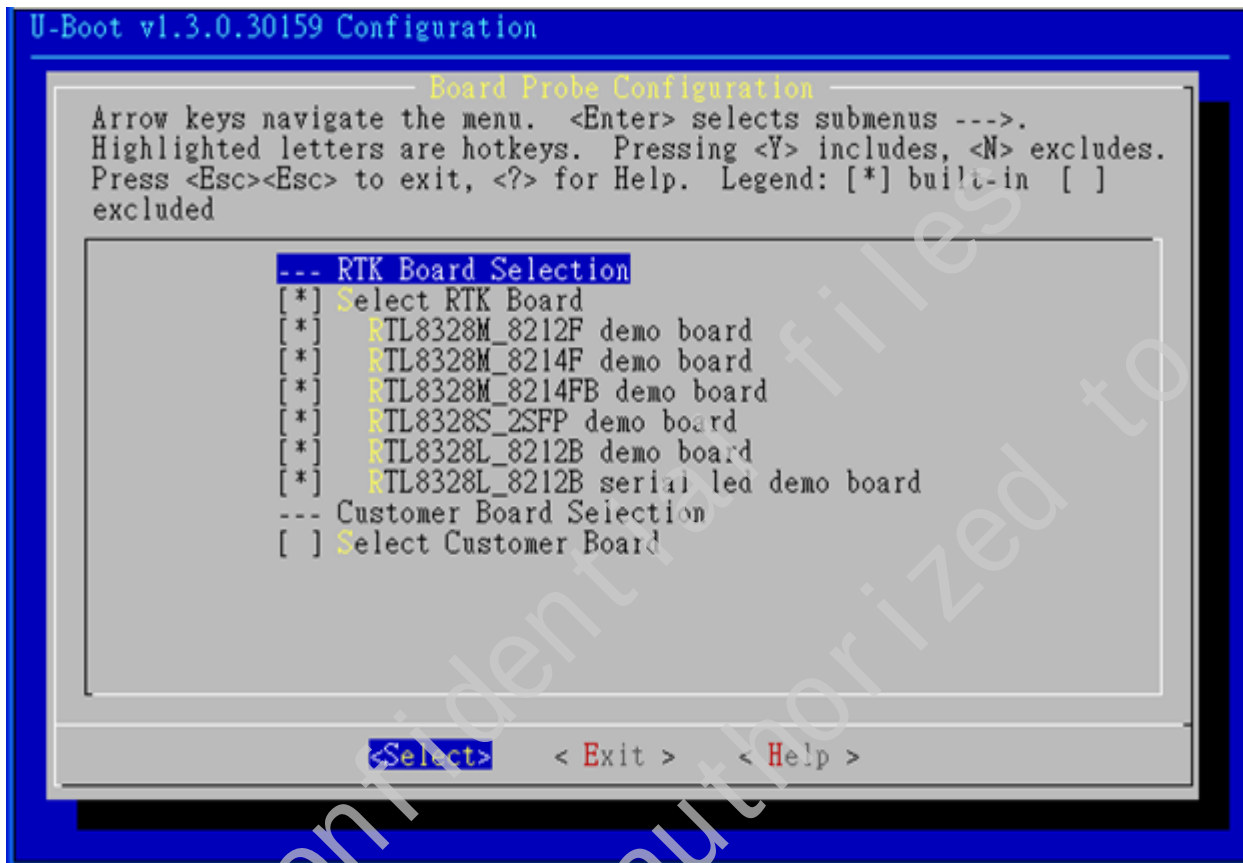
- After exit and save "Linux Kernel Configuration" setting, you can enter "U-Boot v1.3.0.xxxxx Configuration" Main Menu as following:



- "Board Selection" setting as following:



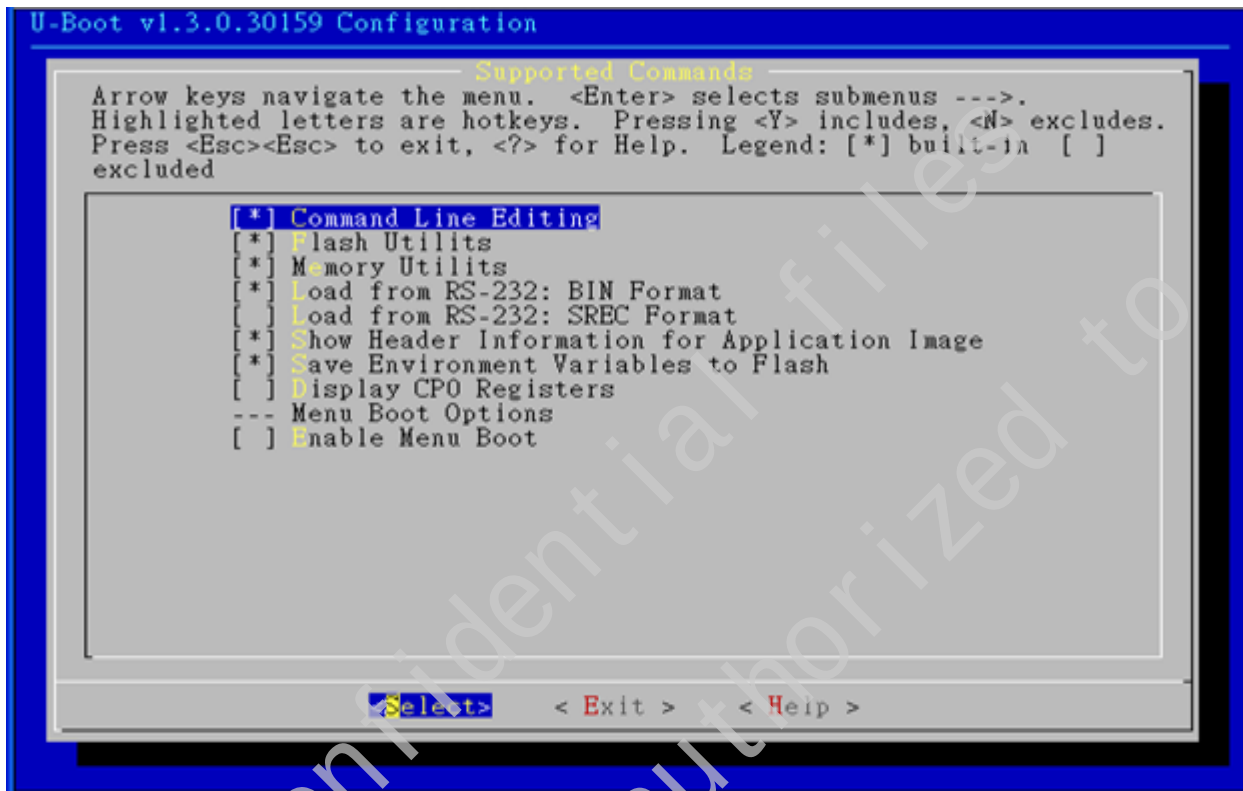
- "Board Probe Configuration" setting as following:



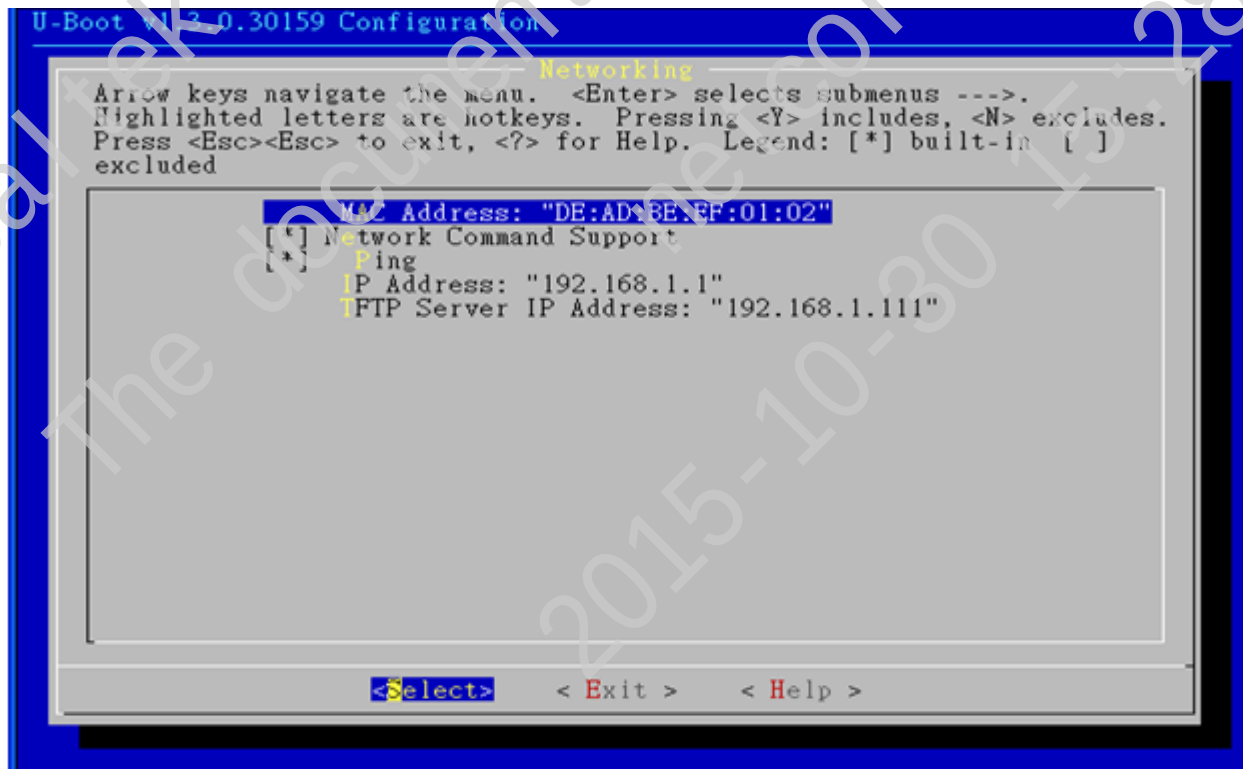
➤ "Memory Configuration" setting as following:



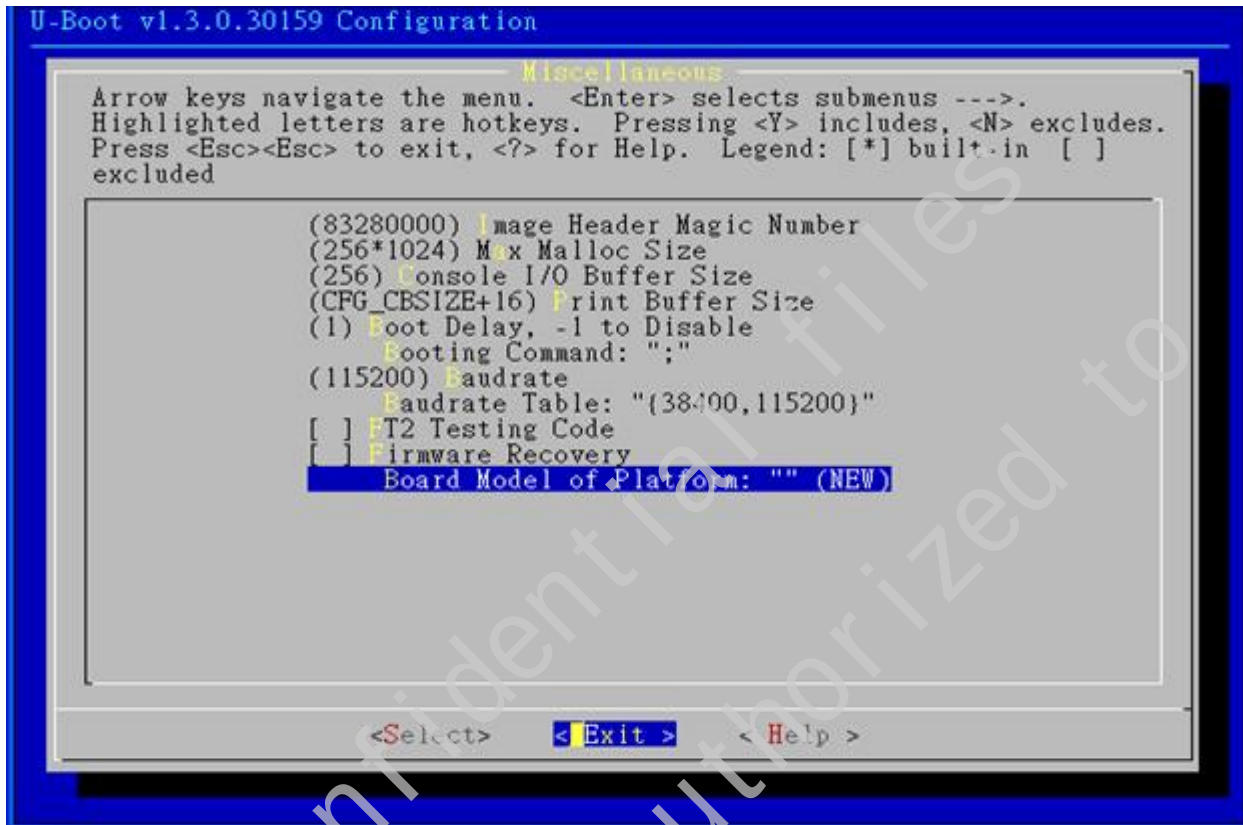
- “Supported Commands” setting as following:



- “Networking” setting as following: (Default)

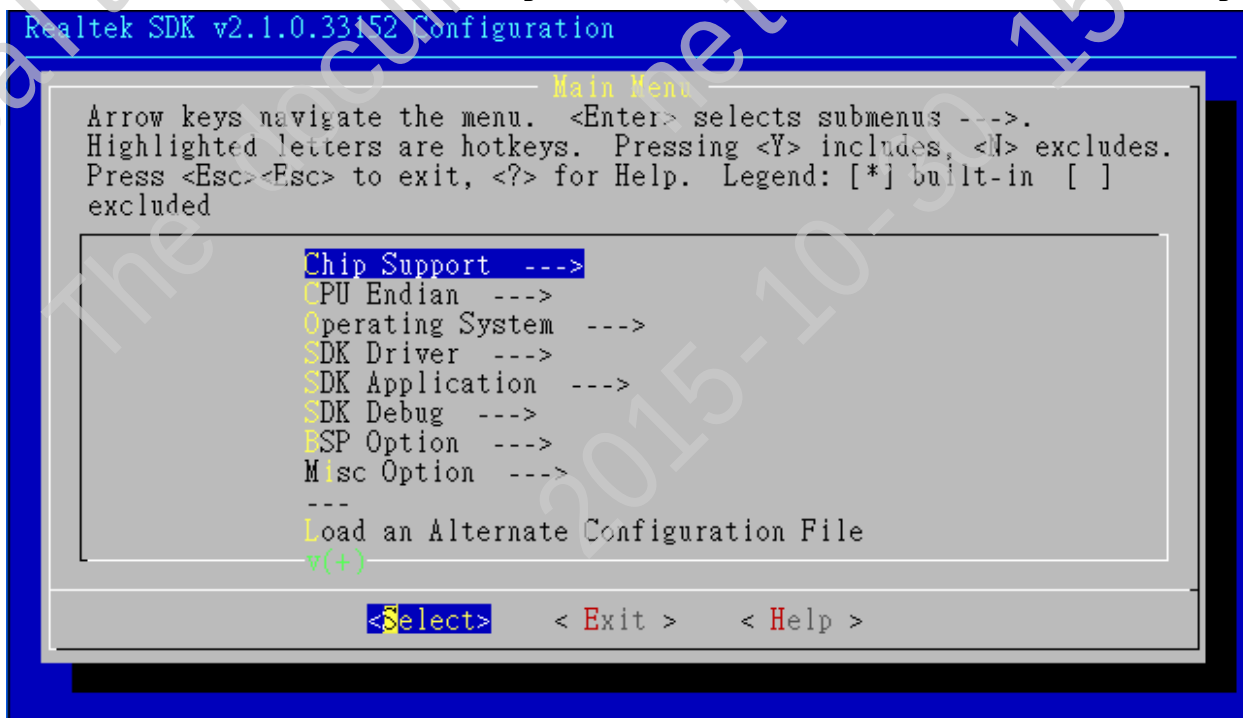


- “Miscellaneous” setting as following:



1.2.3 SDK Menuconfig

- After exit and save "U-Boot Configuration" setting, you can enter "SDK Configuration" Main Menu as following:



- “Chip Support” setting as following:

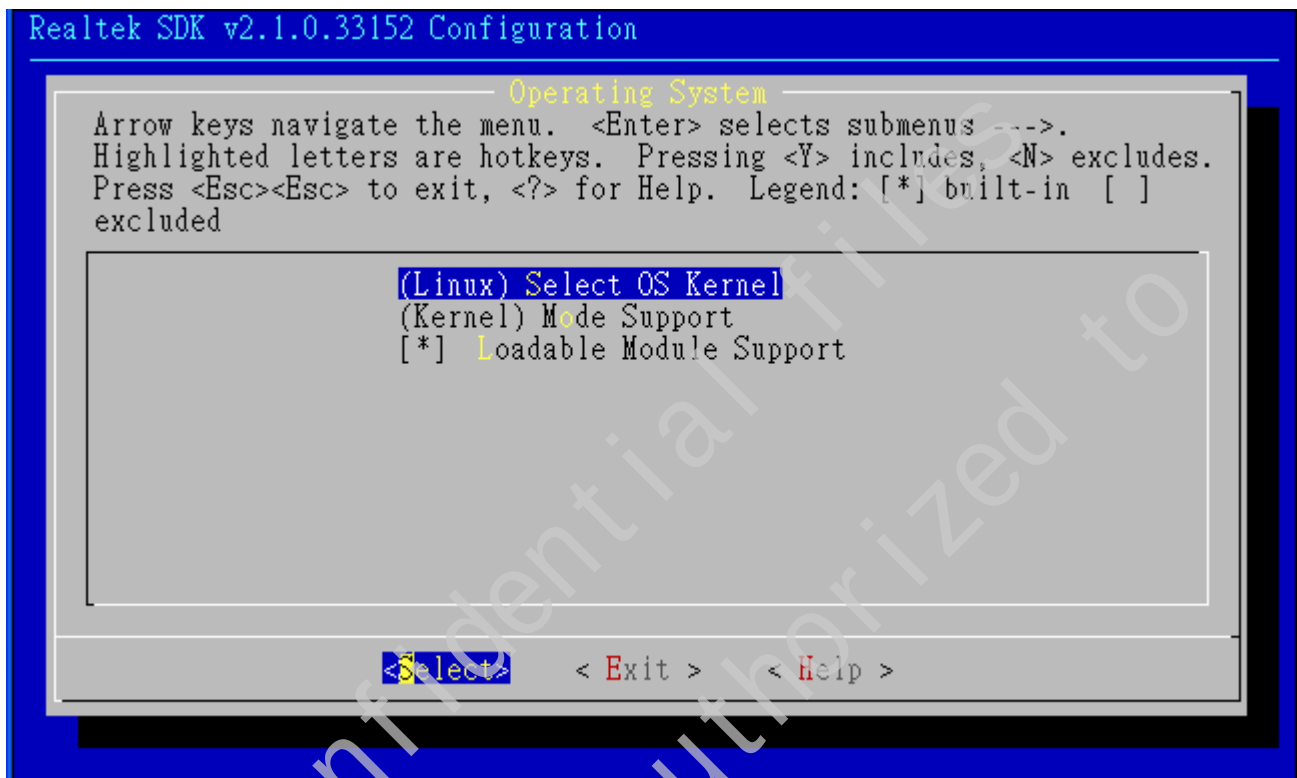


- “CPU Endian” setting as following:

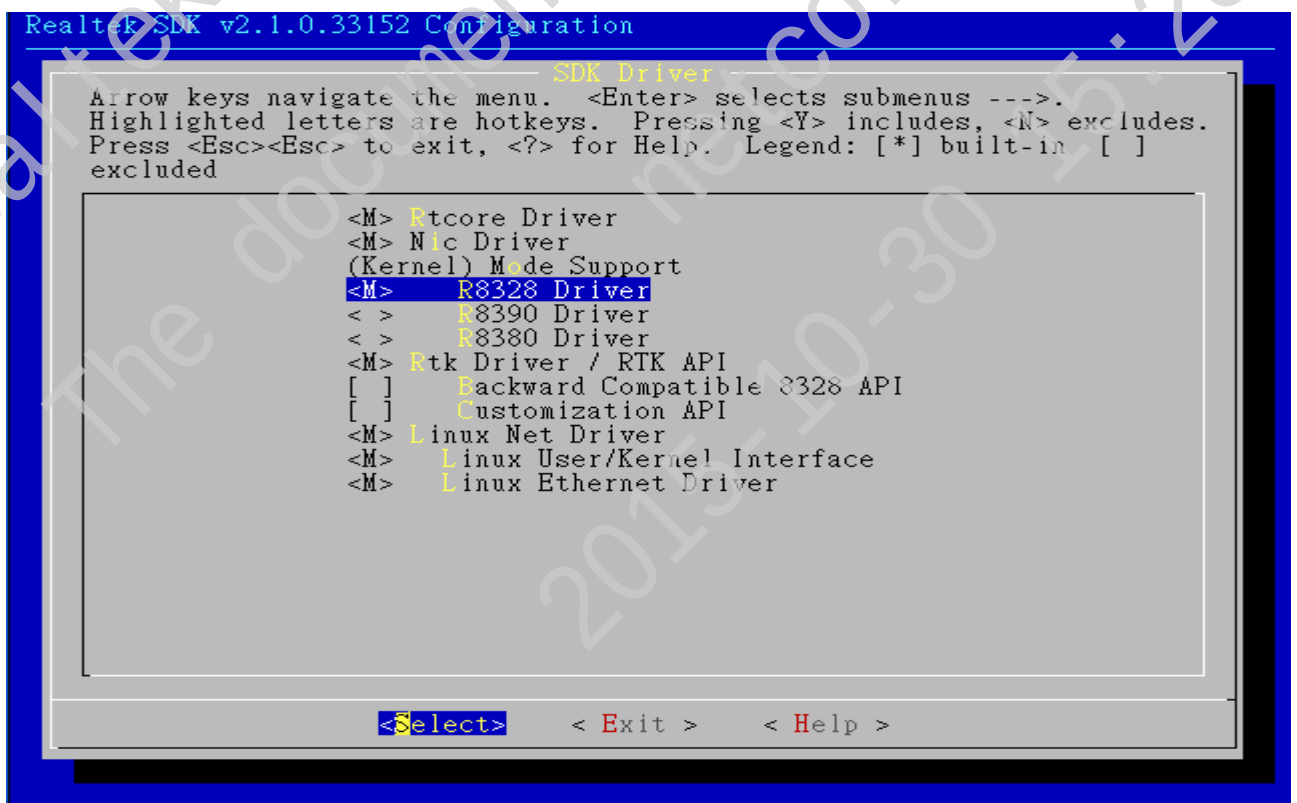


- “Operation System” setting as following for SDK driver run

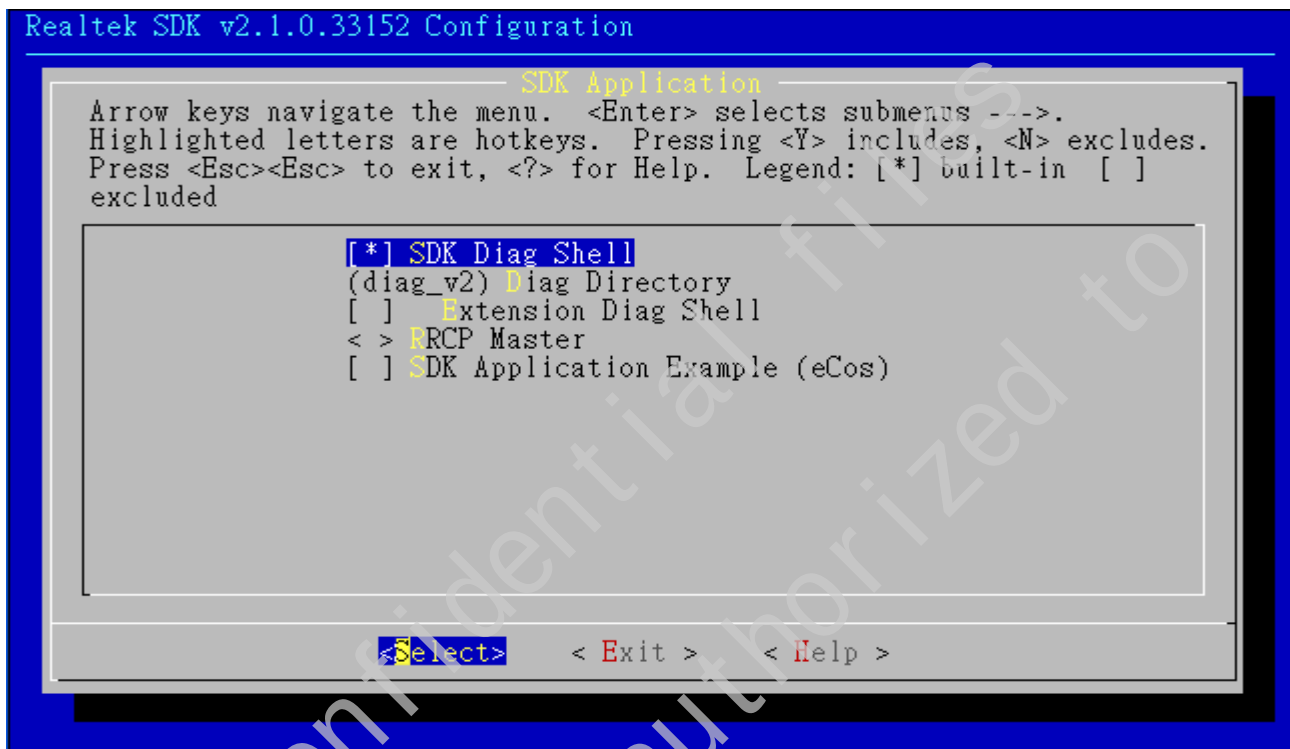
at Linux kernel mode:



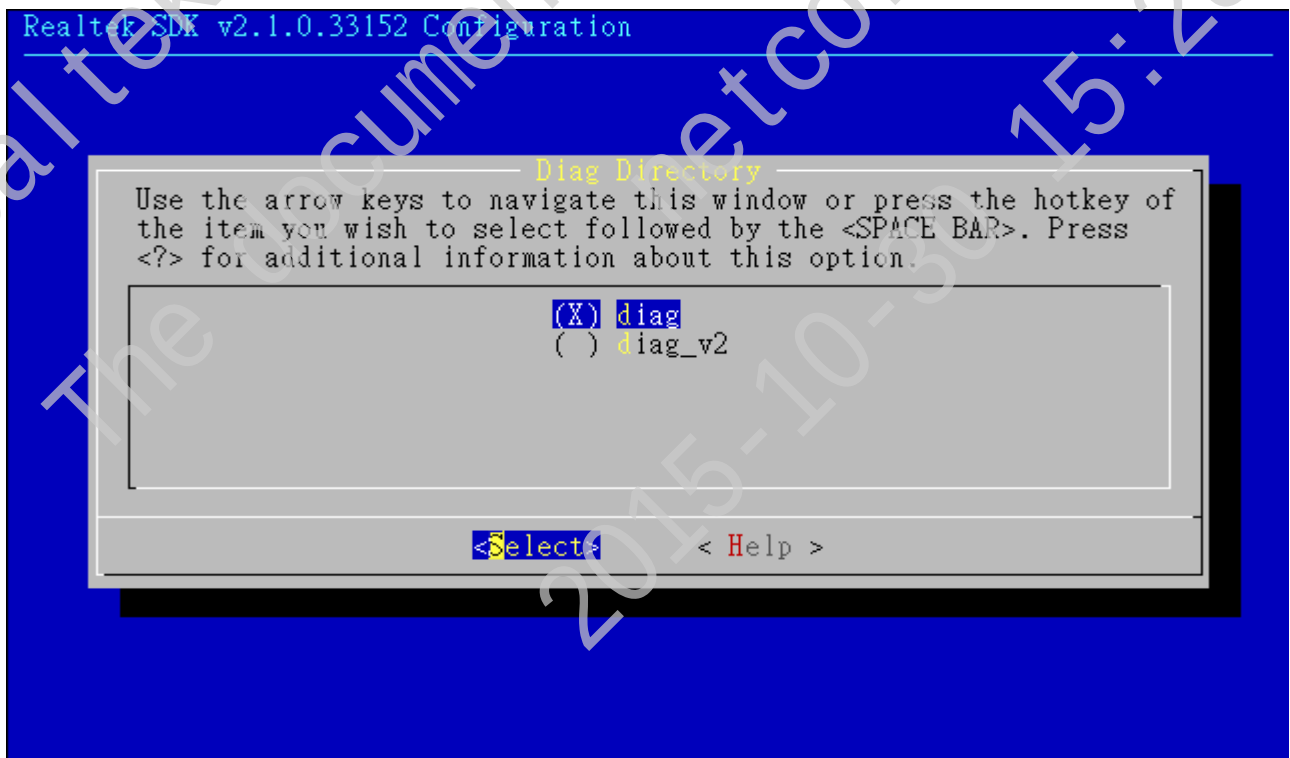
- "SDK Driver" setting as following for NIC driver run at Linux kernel mode:



- “SDK Application” setting as following for turn on diagnostic shell:



- “Diag Directory” setting can select diag or diag_v2 as following:



- “SDK Debug” setting as following:

Realtek SDK v2.1.0.33152 Configuration

Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes. Press <Esc><Esc> to exit, <?> for Help. Legend: [*] built-in [] excluded

[*] **Enable SDK Debug Message**
(Level) **D**ebug Log Type
(3) **L**og Level
(7FFFFFFF) **D**ebug Log Module Mask

< **S**elect > < **E**xit > < **H**elp >

➤ "BSP Option" setting as following:

Realtek SDK v2.1.0.33152 Configuration

Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes. Press <Esc><Esc> to exit, <?> for Help. Legend: [*] built-in [] excluded

[*] **S**MI/GPIO Driver
[*] **F**lash Common Setting
(10000) **F**lash Block Size
(16MB) **F**lash Size
[*] **D**ual Image
(6D0000) **P**artition Size (NEW)
(Type1) **F**lash Layout
<*> **L**inux Mtd Driver
[*] **S**PI Flash Driver
[*] **Q**UAD IO (NEW)
[] **N**OR Flash Driver

< **S**elect > < **E**xit > < **H**elp >

➤ "Misc Option" setting as following:



- After exit and save "SDK Configuration" setting, please type "make" for build out u-boot and runtime images

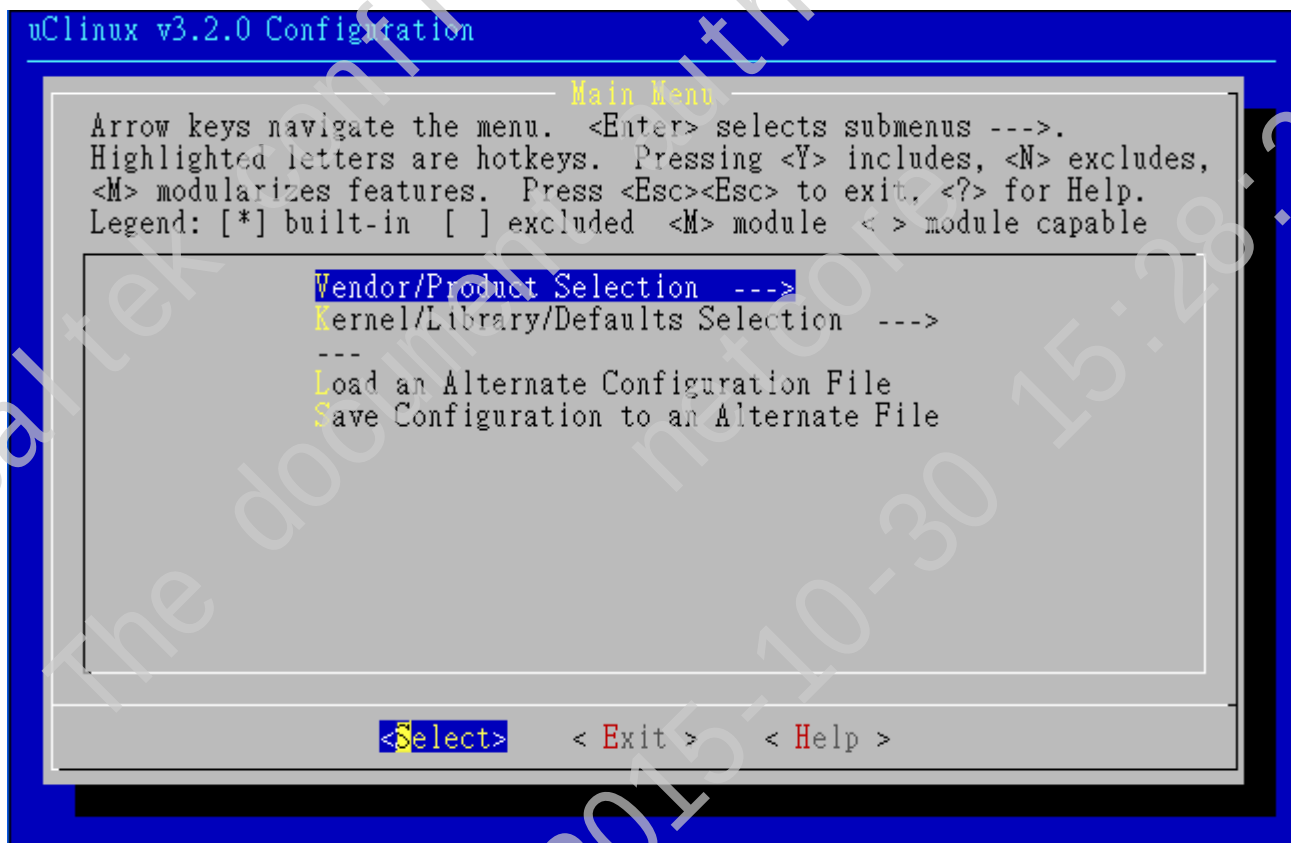
1.3. RTL8328L Demo Board

The default diag-v2 command tree of the SDK package is for 8390 series chip. Please remove following 2 files and re-generate the new diag-v2 command for the chip.

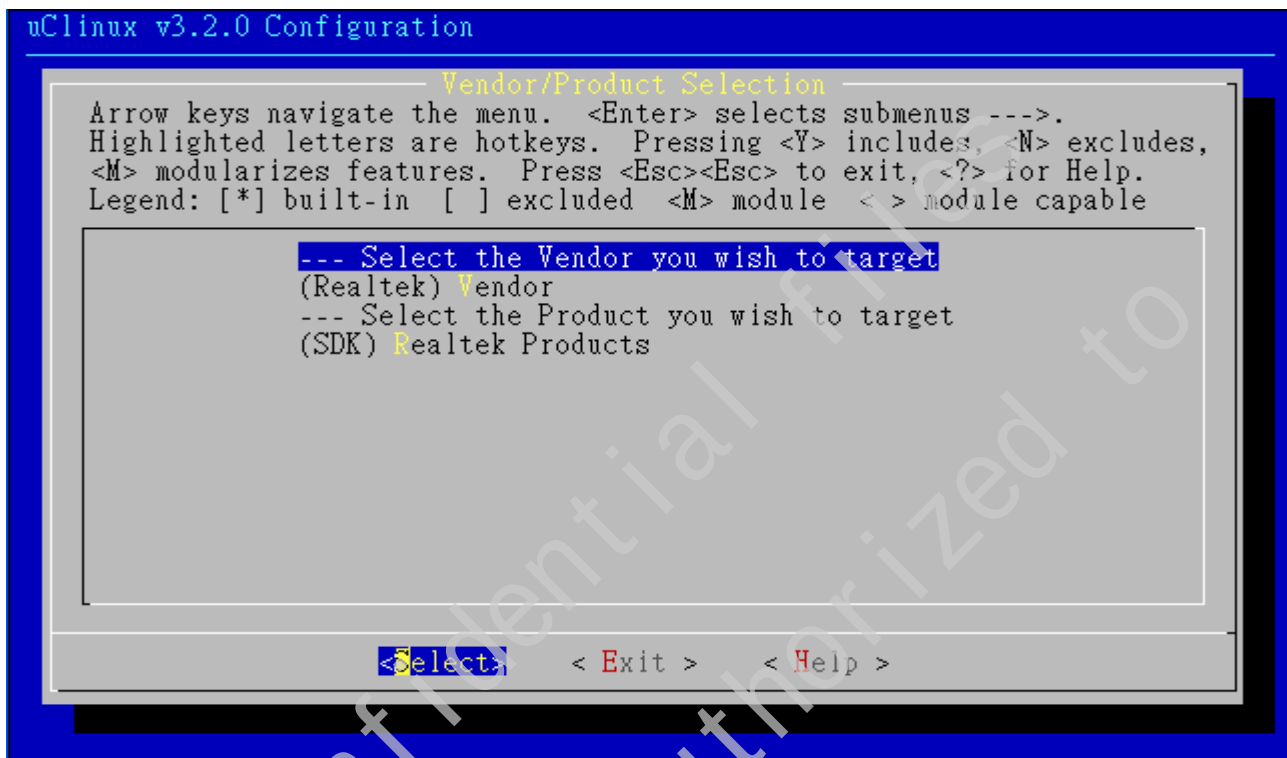
- `sdk\src\app\diag_v2\include\parser\cparser_tree.h`
- `sdk\src\app\diag_v2\src\parser\cparser_tree.c`

Following example is SDK driver and NIC driver run at Linux kernel mode. 8328L can't support the Quad SPI flash; only support the dual and single flash.

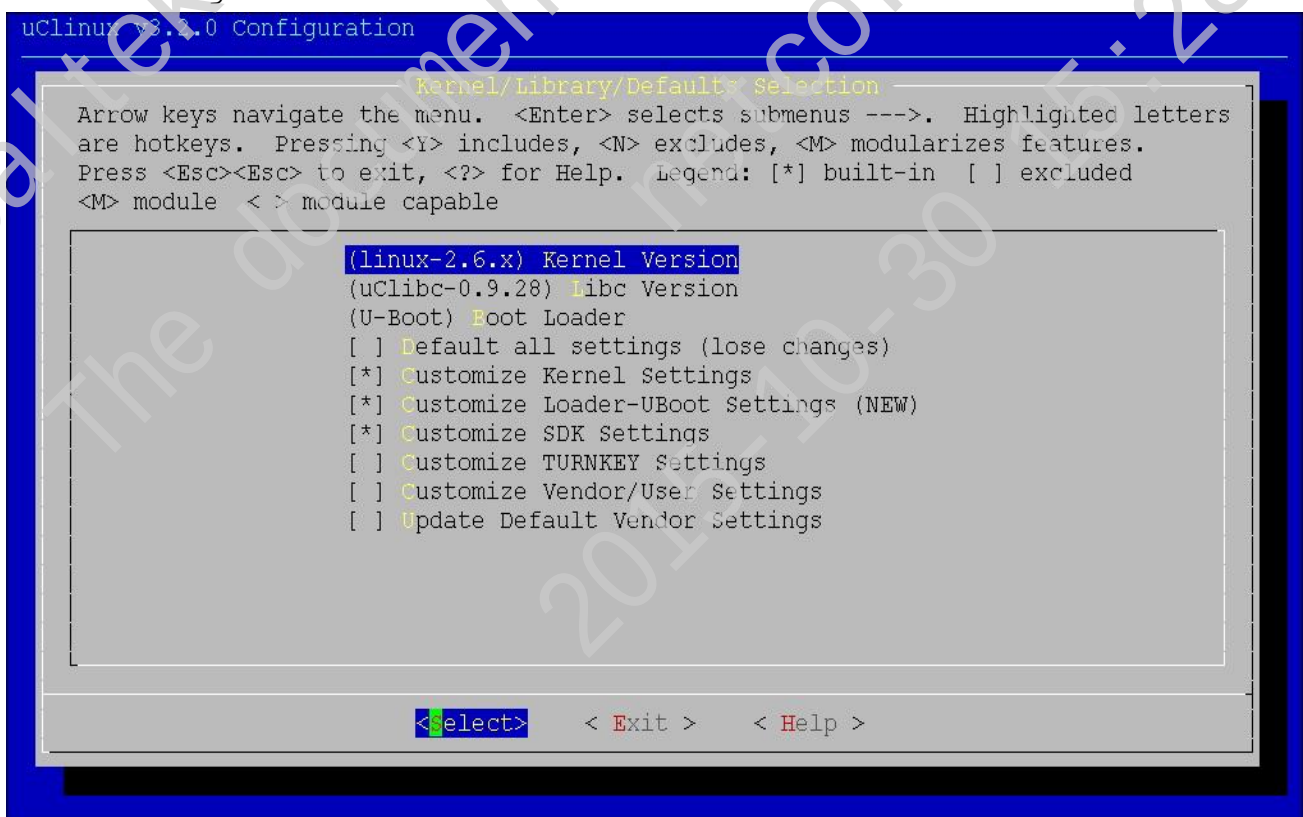
- When type 'make menuconfig' command, you can see the 'Main Menu' as following:



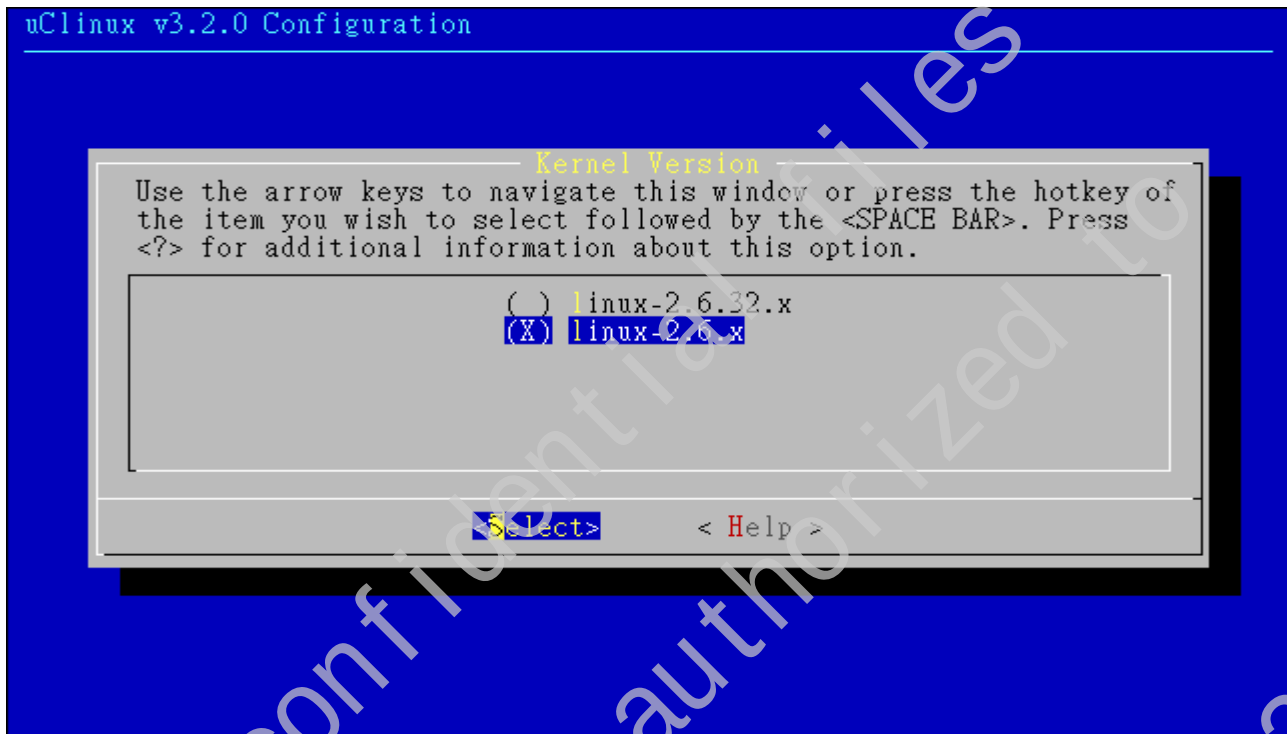
- “Vendor/Product Selection” setting as following:



- Select some items in “Kernel/Library/Defaults Selection” and “Exit” for configure Kernel, Loader-UBoot and SDK setting:



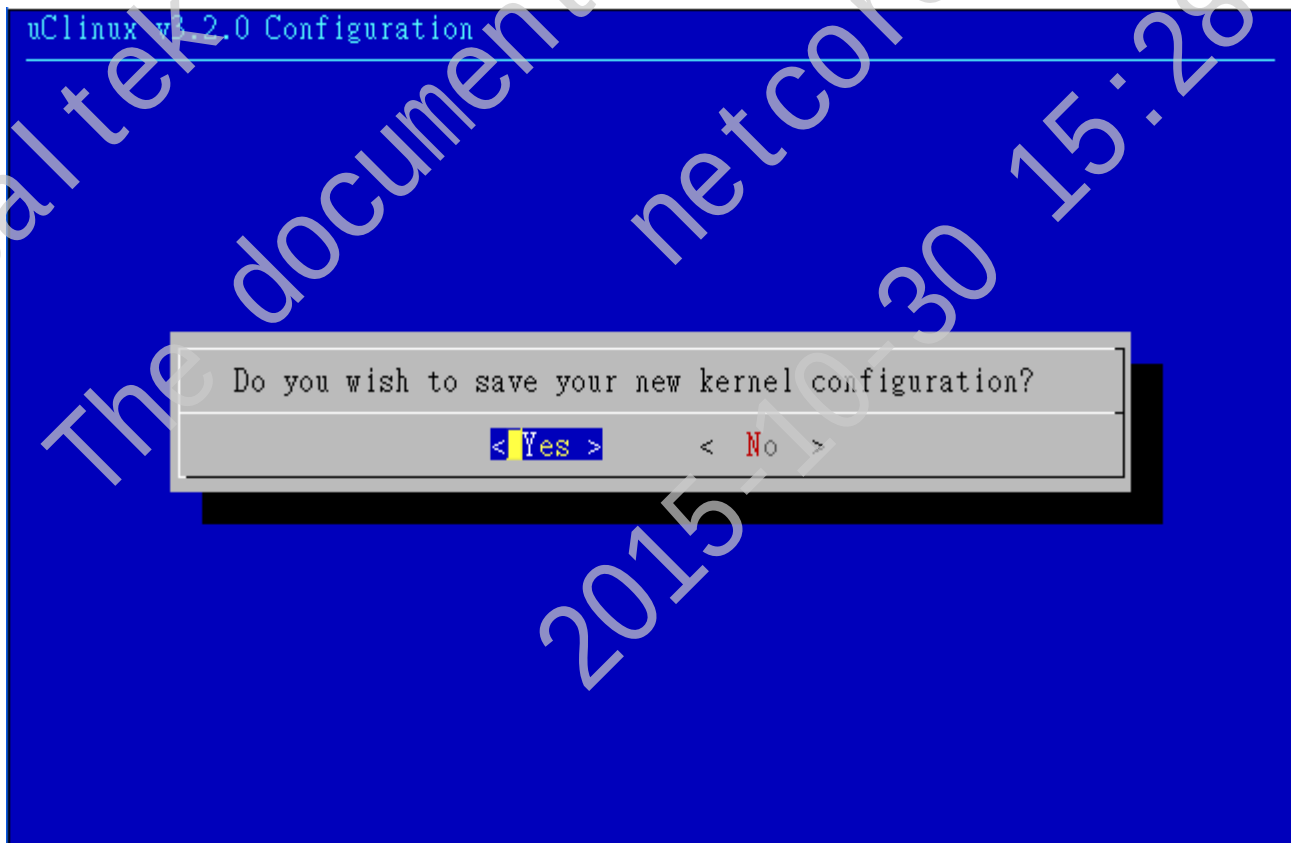
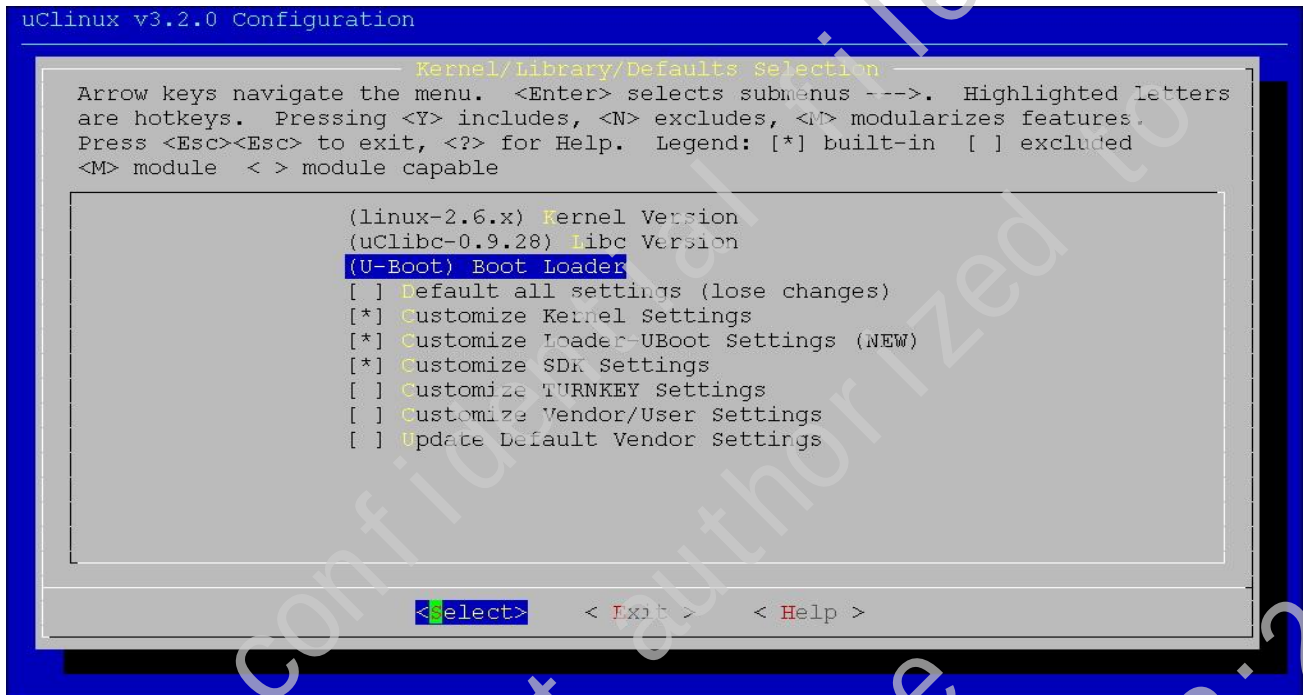
- In "Kernel/Library/Defaults Selection" menu, there are two Linux kernel version could be selected. The linux-2.6.x is 2.6.19 and the linux-2.6.32.x is 2.6.32.58:



- In "Kernel/Library/Defaults Selection" menu, there are two uClibc version could be selected. The linux-2.6.x should use 0.9.28 and the linux-2.6.32.x should used 0.9.30:



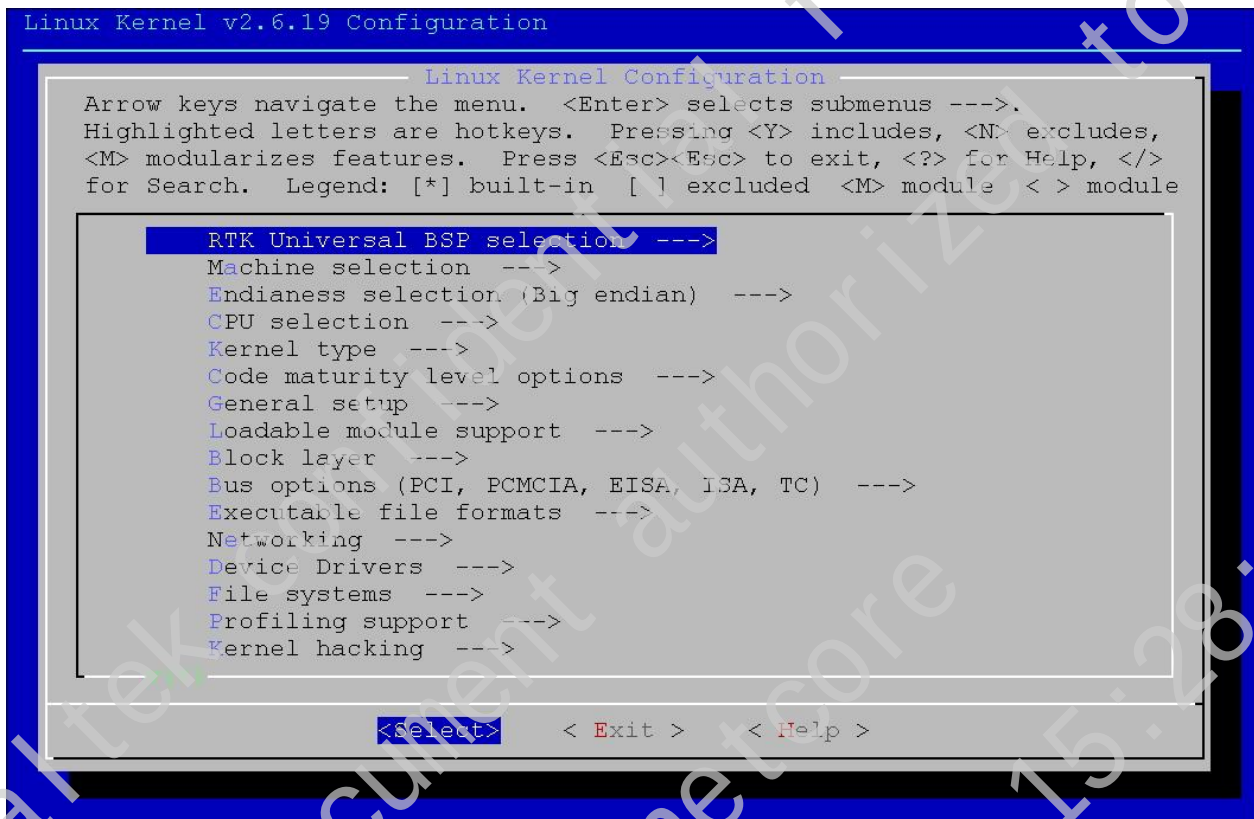
- In "Kernel/Library/Defaults Selection" menu, there are two boot code version could be selected. The U-Boot is old one and the U-Boot-2011.12 is new one. For the RTL8328L, only U-Boot is supported, and please selects the U-Boot.



1.3.1 Kernel Menuconfig

1.3.1.1 Linux v2.6.19 Menuconfig

- Enter the "Linux Kernel v2.6.19 Configuration" as following:



- "RTK Universal BSP selection" setting as following: (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.

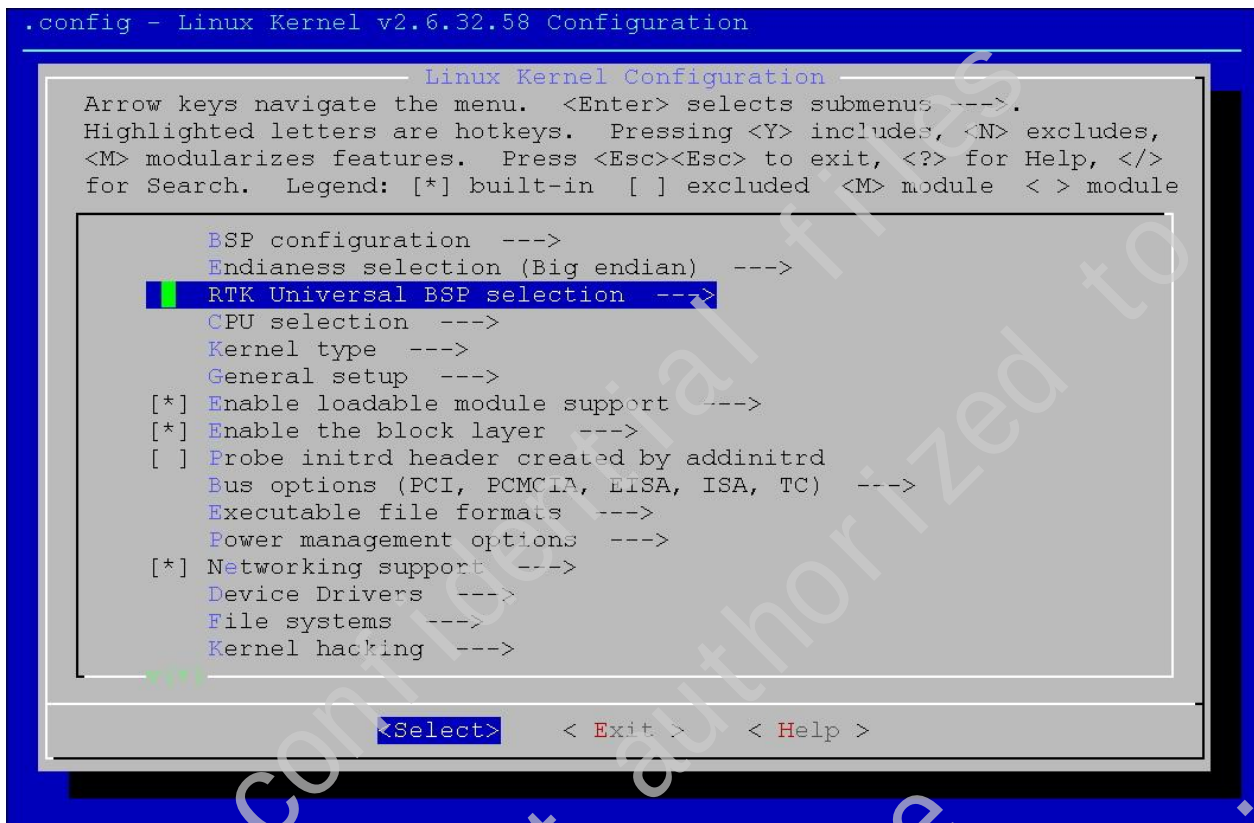


- "Kernel hacking" setting as following: (64MB DDR RAM)

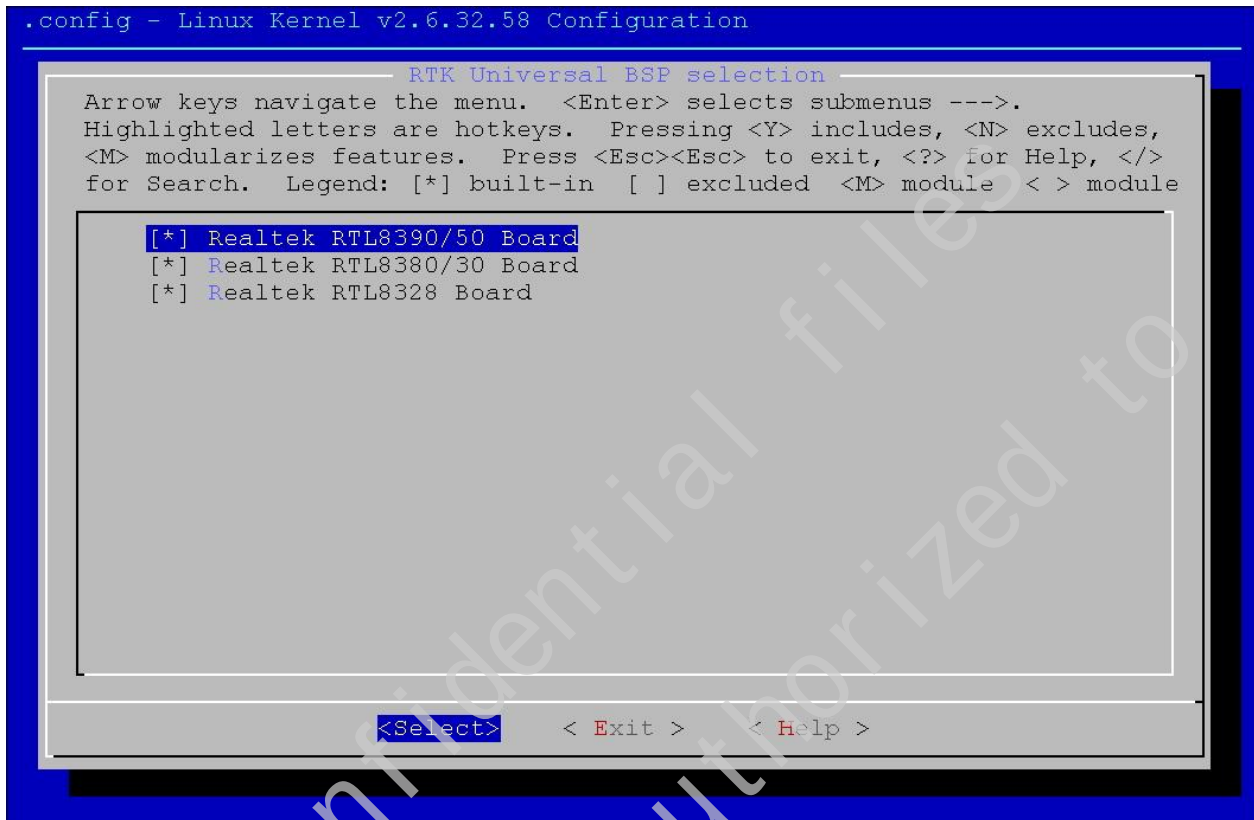


1.3.1.2 Linux v2.6.32.58 Menuconfig

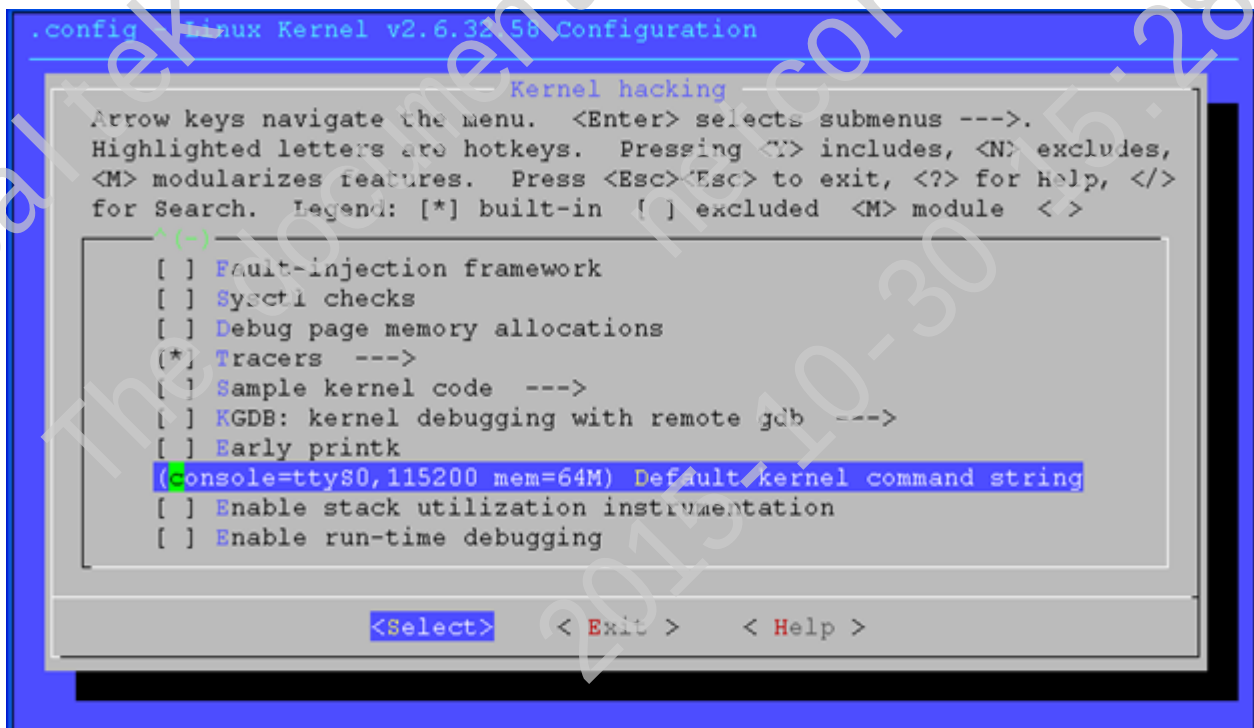
- Enter the “Linux Kernel v2.6.32.58 Configuration” as following:



- “RTK Universal BSP selection” setting as following: (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.

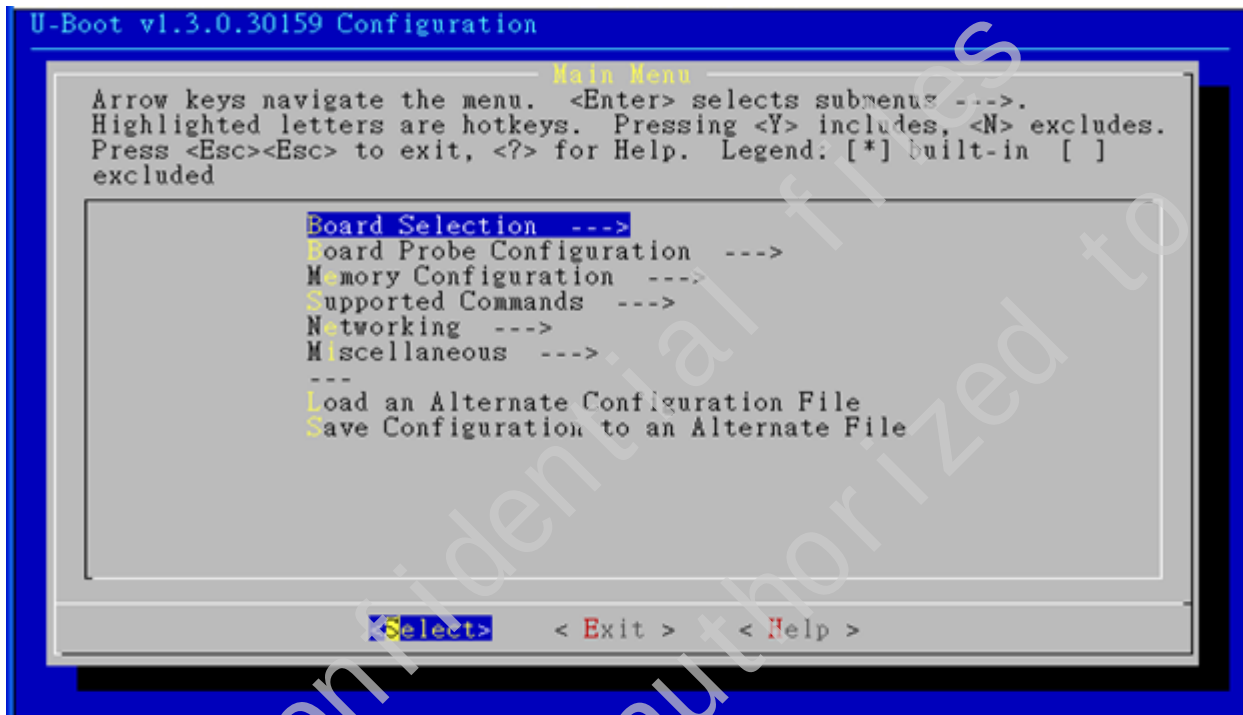


- “Kernel hacking” setting as following: (64MB DDR RAM)



1.3.2 U-Boot v1.3.0 Menuconfig

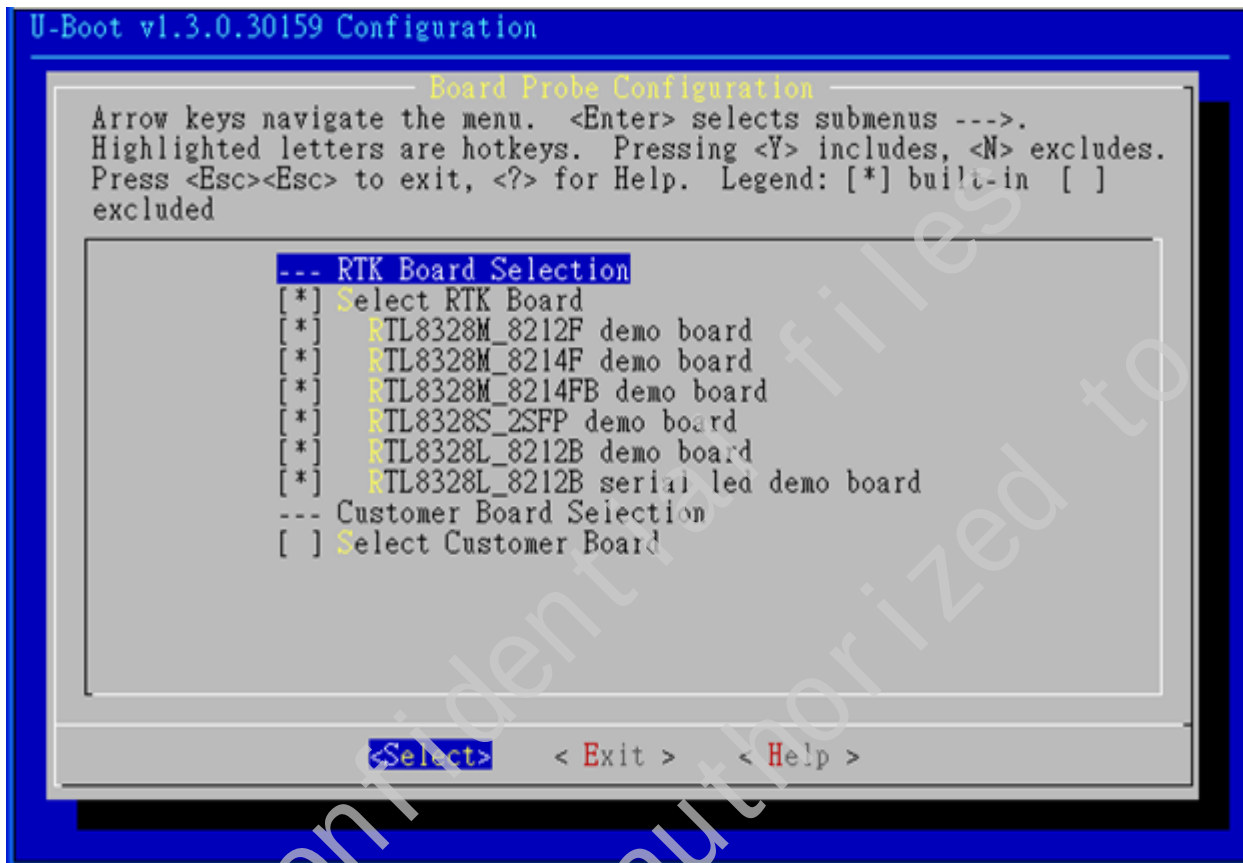
- After exit and save "Linux Kernel Configuration" setting, you can enter "U-Boot v1.3.0.xxxxx Configuration" Main Menu as following:



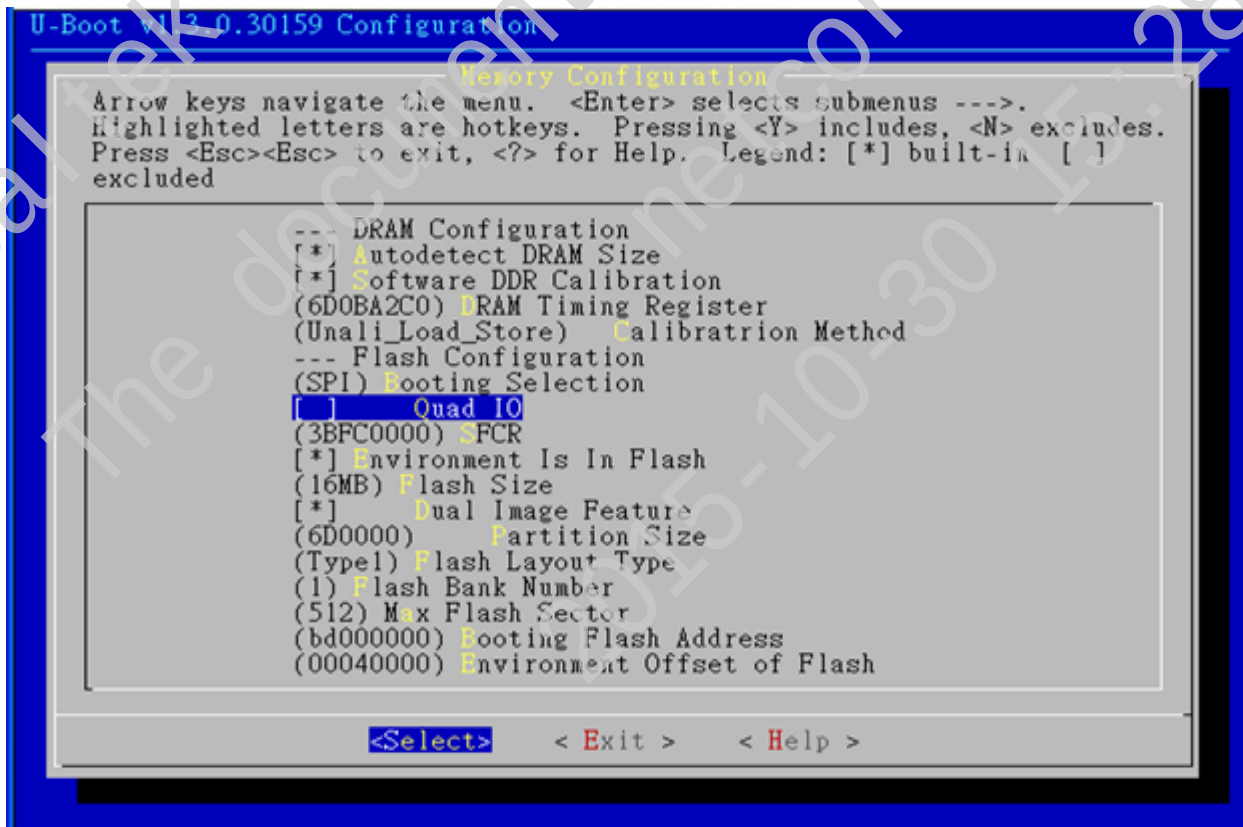
- "Board Selection" setting as following:



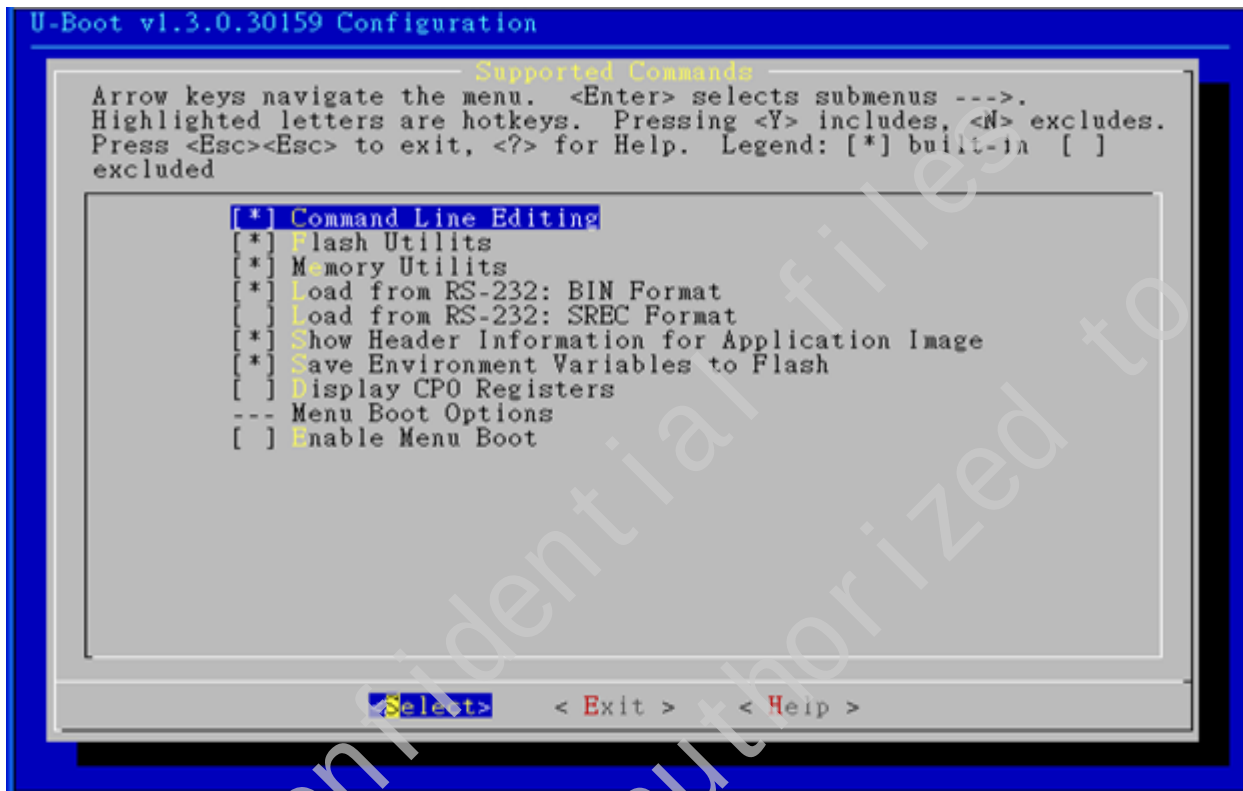
- "Board Probe Configuration" setting as following:



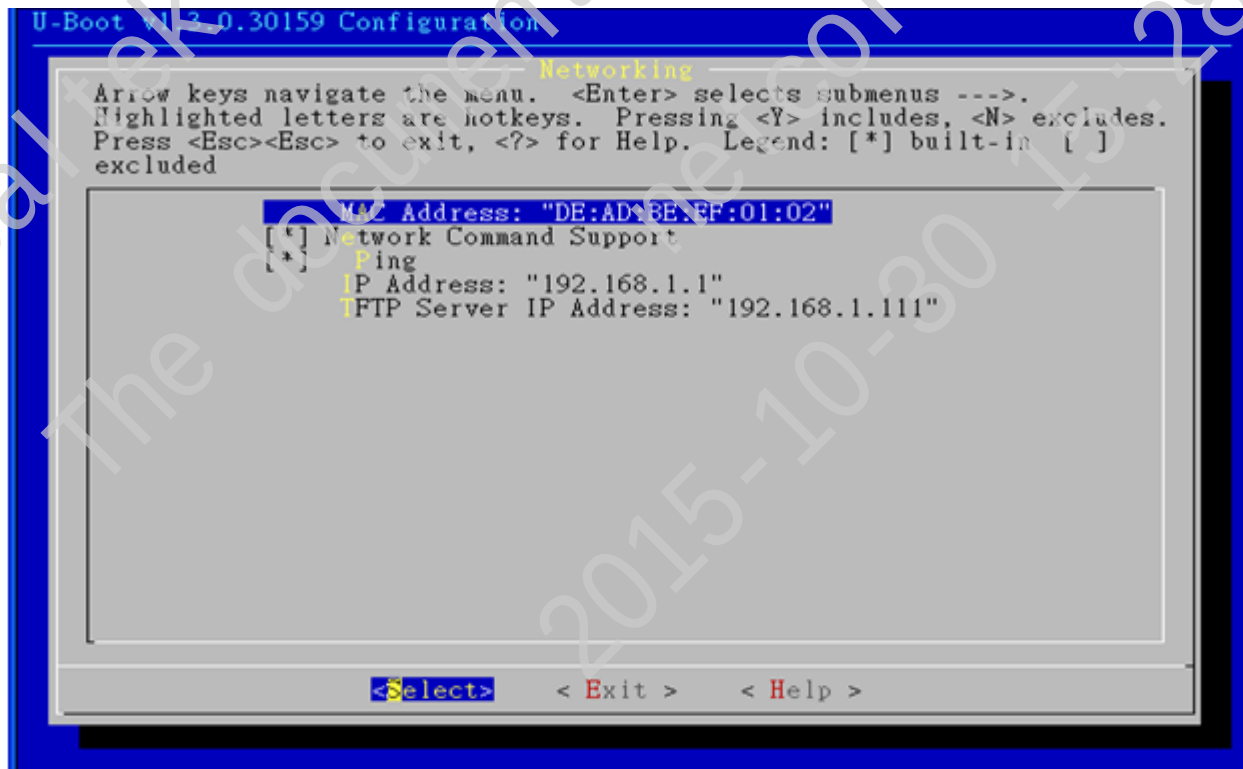
- "Memory Configuration" setting as following:



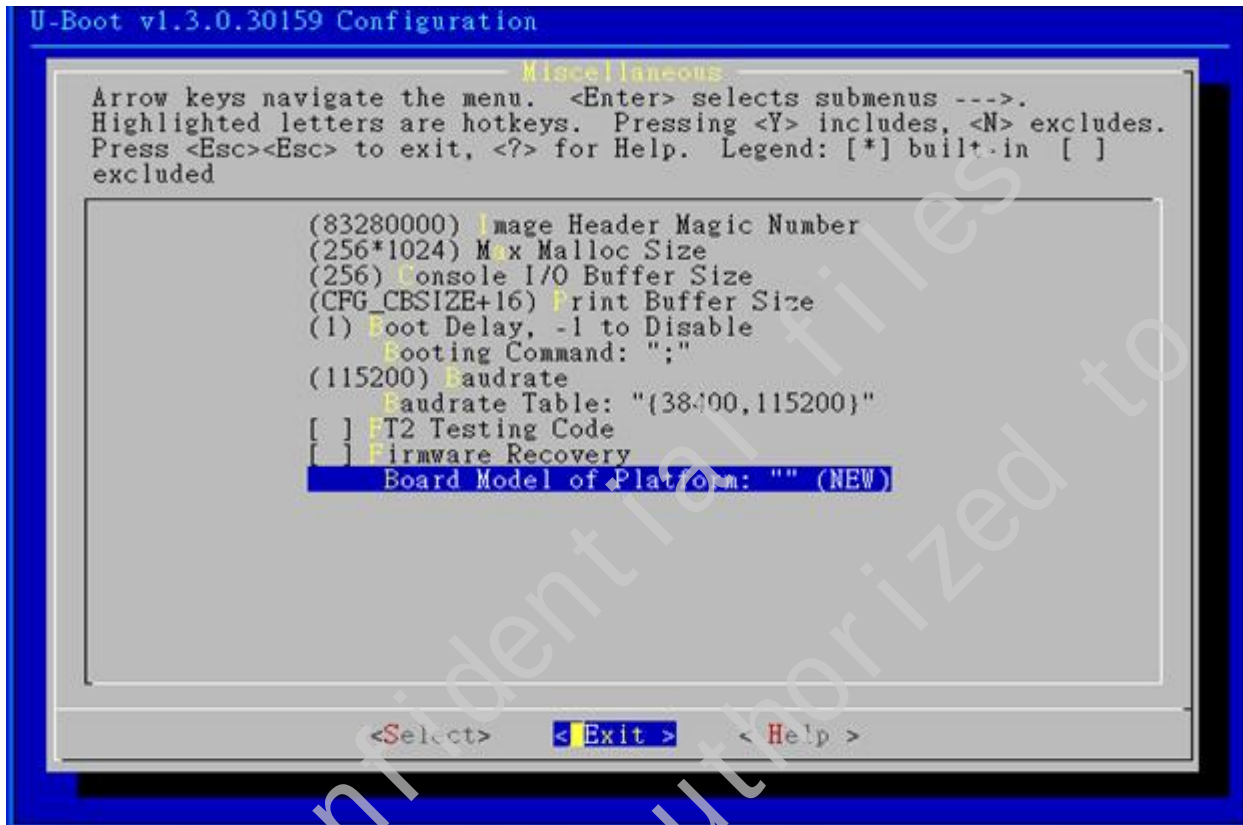
- “Supported Commands” setting as following:



- “Networking” setting as following: (Default)

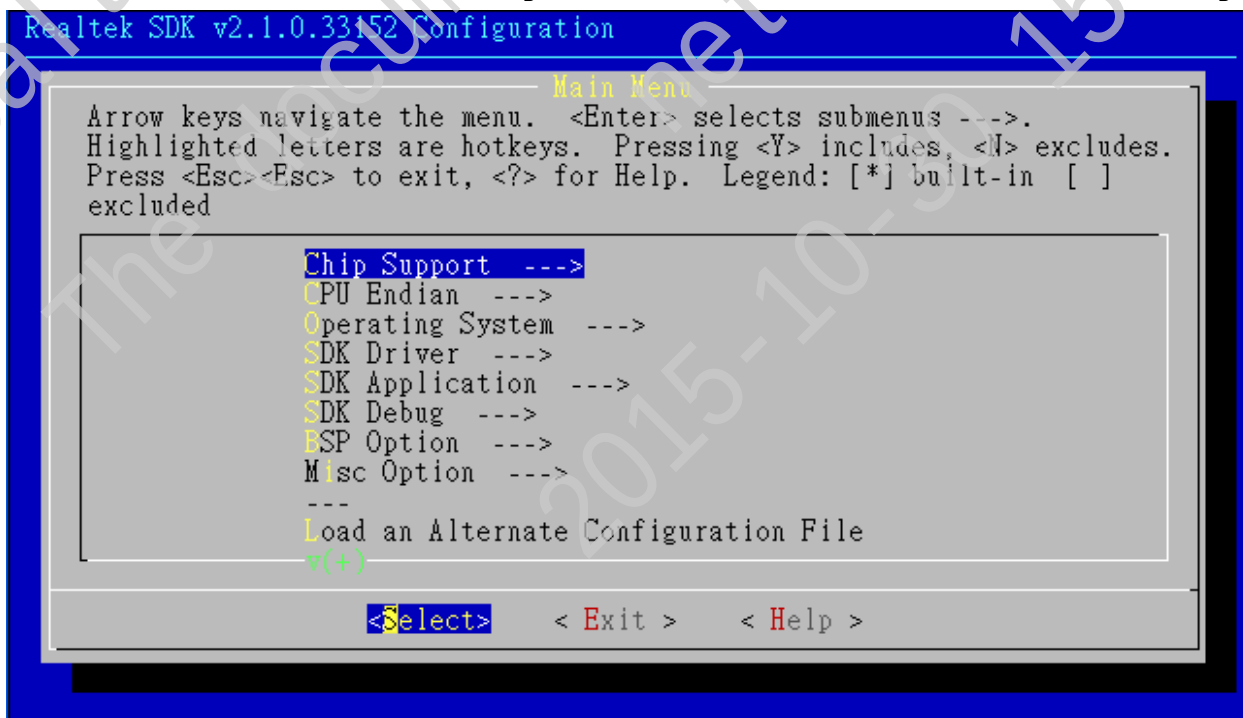


- “Miscellaneous” setting as following:

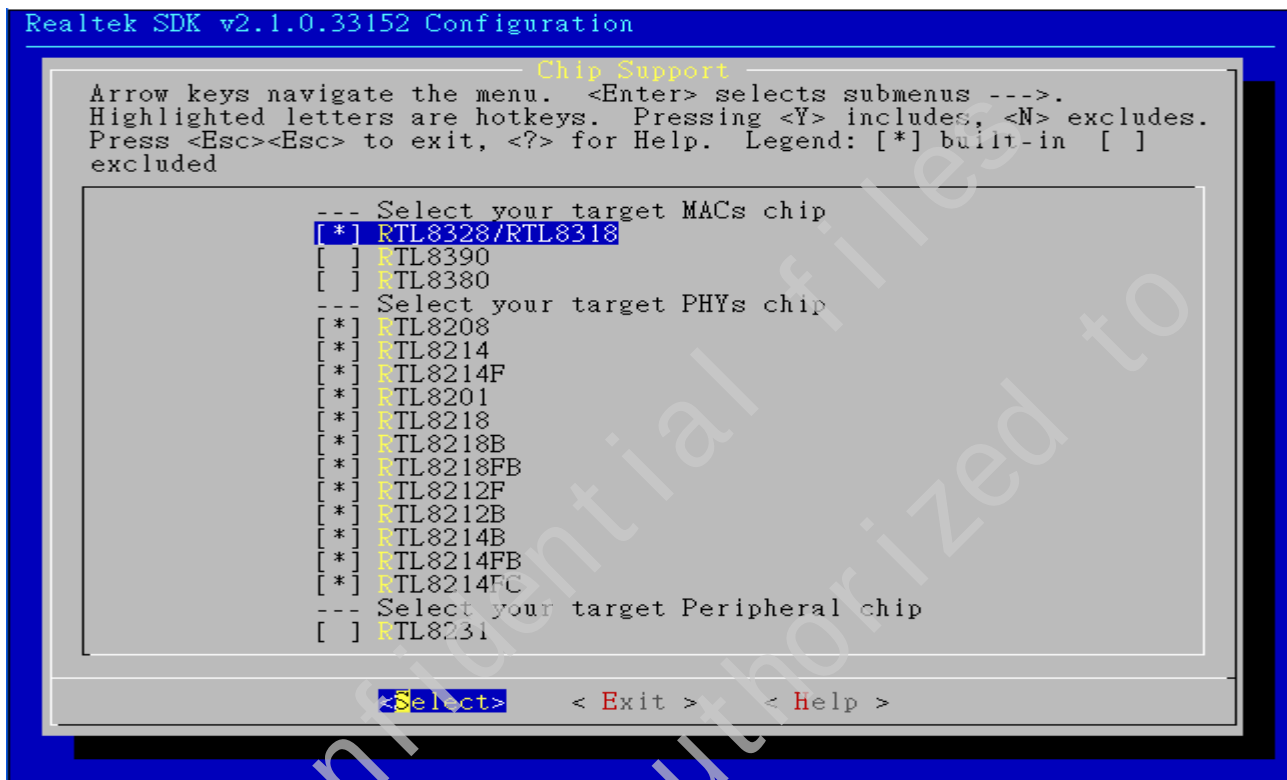


1.3.3 SDK Menuconfig

- After exit and save "U-Boot Configuration" setting, you can enter "SDK Configuration" Main Menu as following:



- “Chip Support” setting as following:

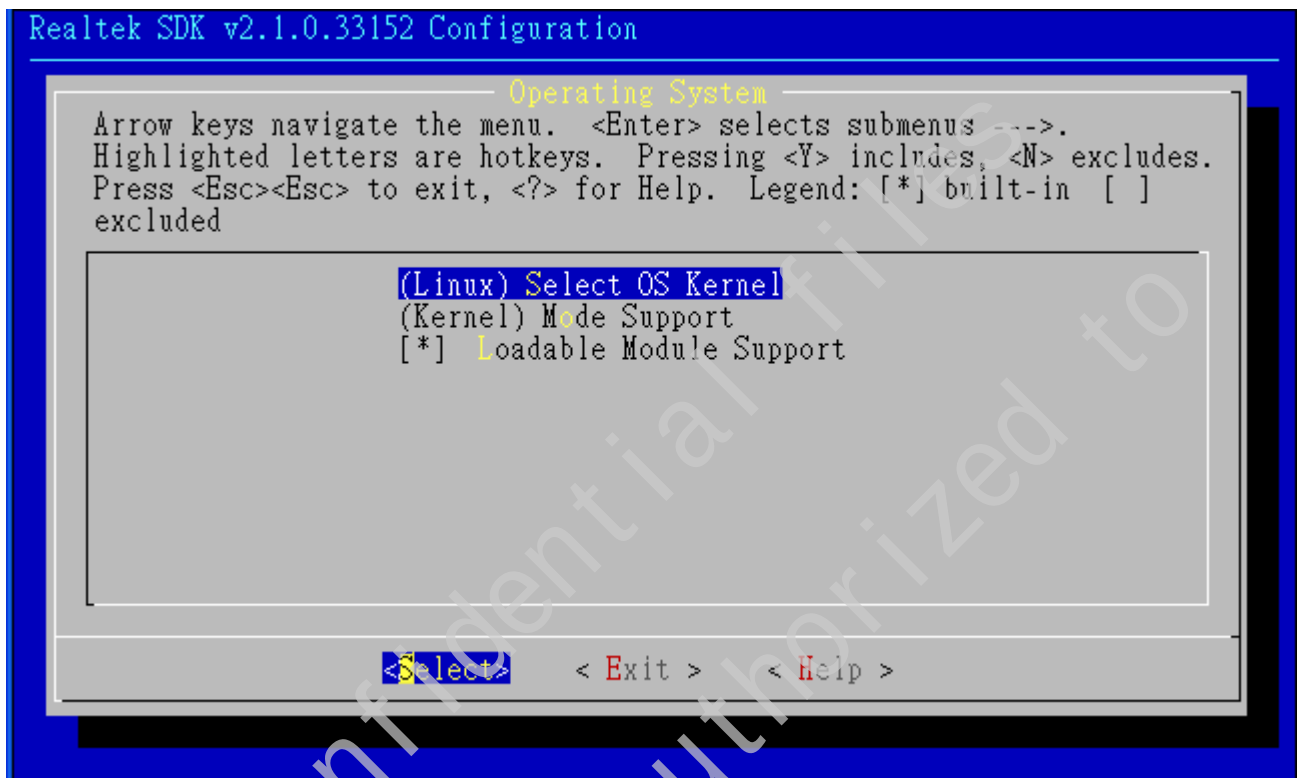


- “CPU Endian” setting as following:

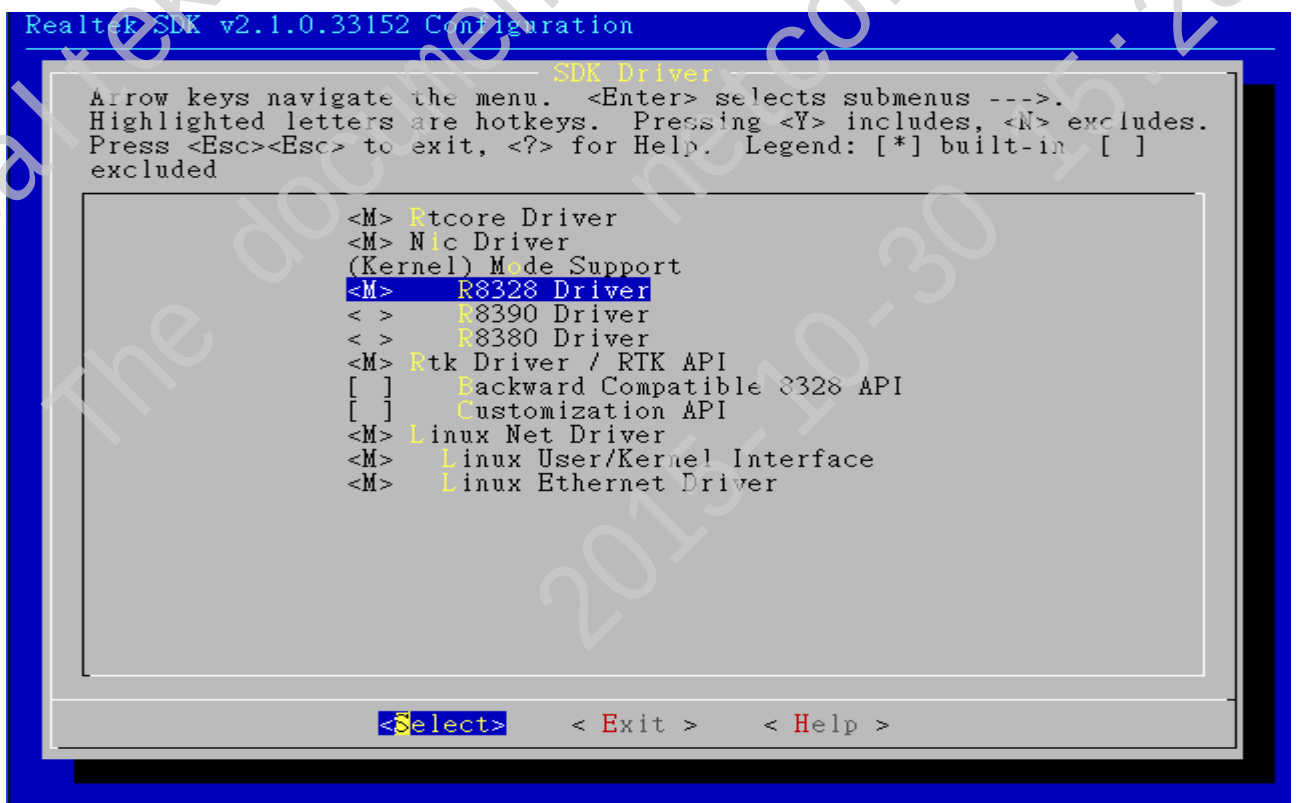


- “Operation System” setting as following for SDK driver run

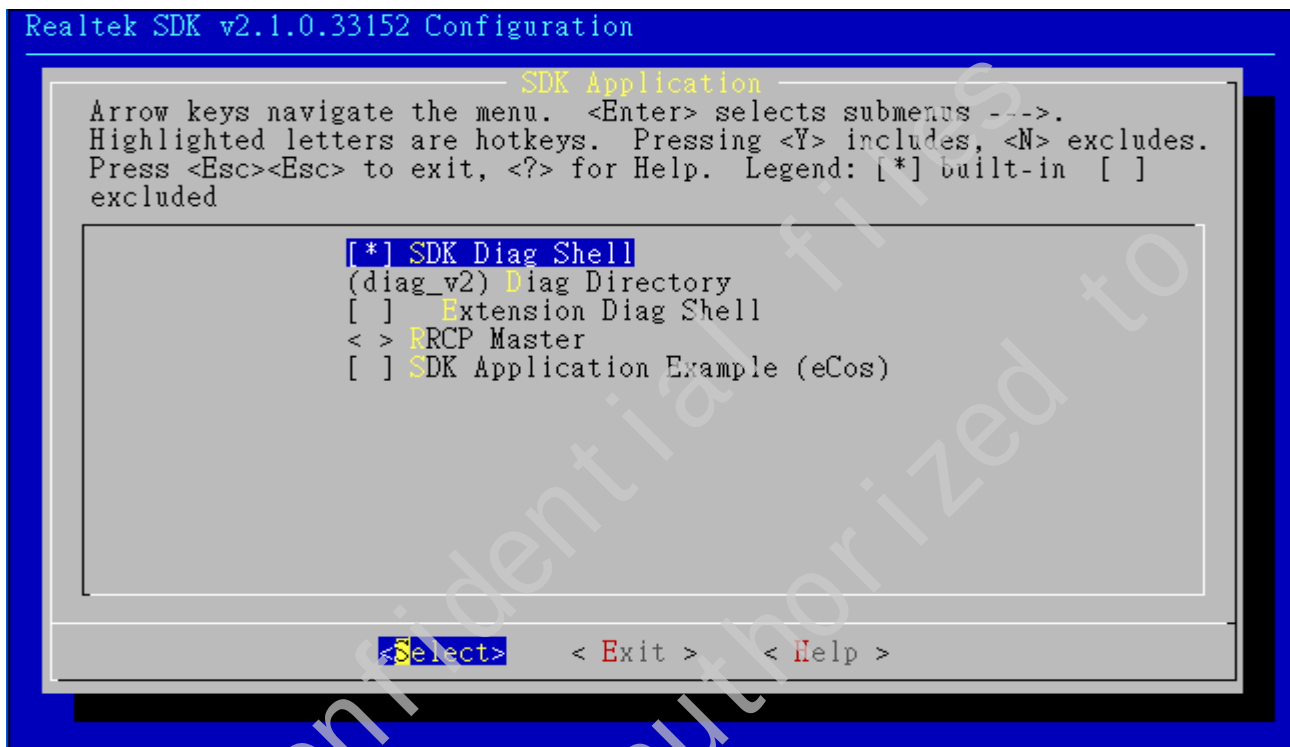
at Linux kernel mode:



- "SDK Driver" setting as following for NIC driver run at Linux kernel mode:



- “SDK Application” setting as following for turn on diagnostic shell:



- “Diag Directory” setting can select diag or diag_v2 as following:



- “SDK Debug” setting as following:

Realtek SDK v2.1.0.33152 Configuration

SDK Debug

Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes. Press <Esc><Esc> to exit, <?> for Help. Legend: [*] built-in [] excluded

```
[*] Enable SDK Debug Message
(Level) Debug Log Type
(3) Log Level
(7FFFFFFF) Debug Log Module Mask
```

<Select> <Exit> <Help>

➤ "BSP Option" setting as following:

Realtek SDK v2.1.0.33152 Configuration

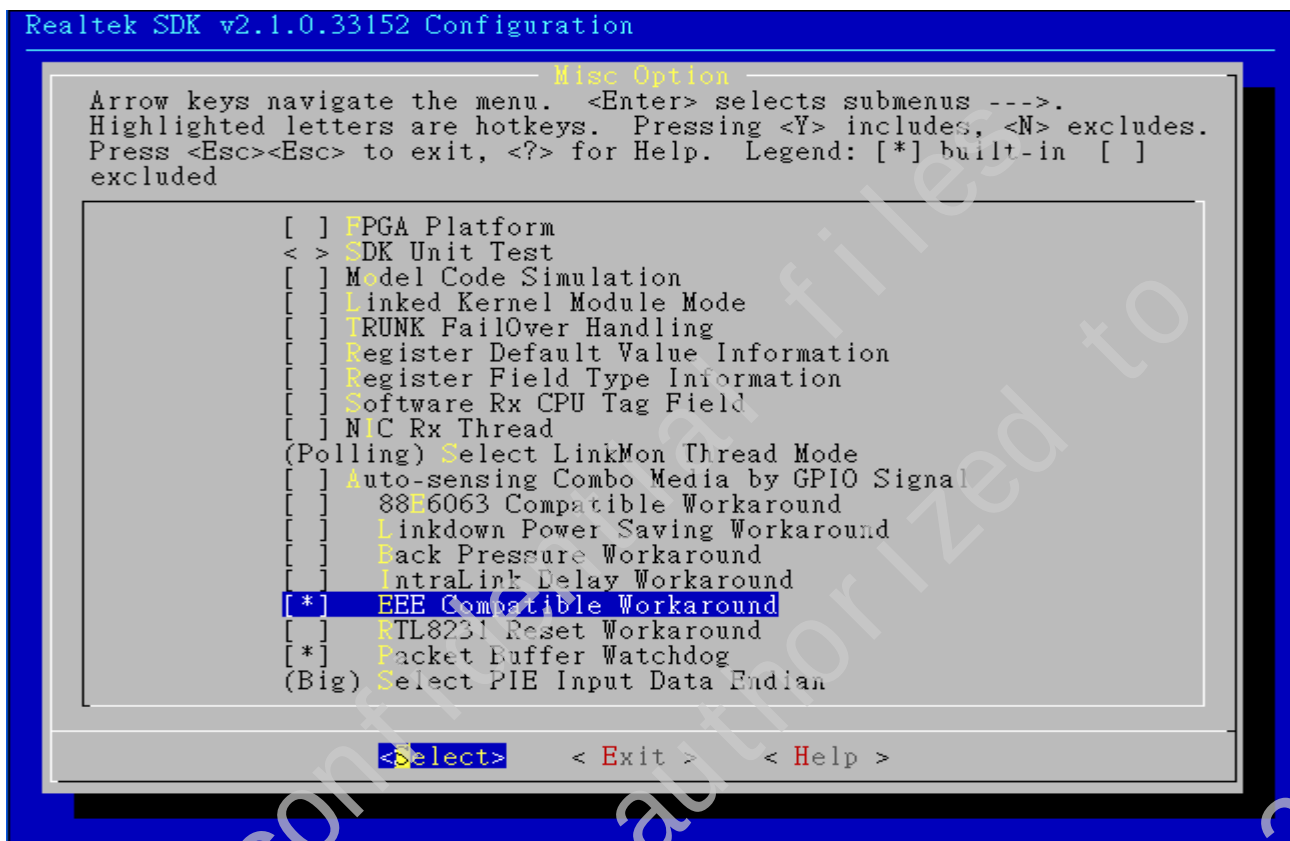
BSP Option

Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes. Press <Esc><Esc> to exit, <?> for Help. Legend: [*] built-in [] excluded

```
[*] SMI/GPIO Driver
[*] Flash Common Setting
(10000) Flash Block Size
(16MB) Flash Size
[*] Dual Image
(6D0000) Partition Size
(Type1) Flash Layout
<*> Linux Mtd Driver
[*] SPI Flash Driver
[ ] QUAD IO
[ ] NOR Flash Driver
```

<Select> <Exit> <Help>

- “Misc Option” setting as following:



- After exit and save “SDK Configuration” setting, please type “make” for build out u-boot and runtime images

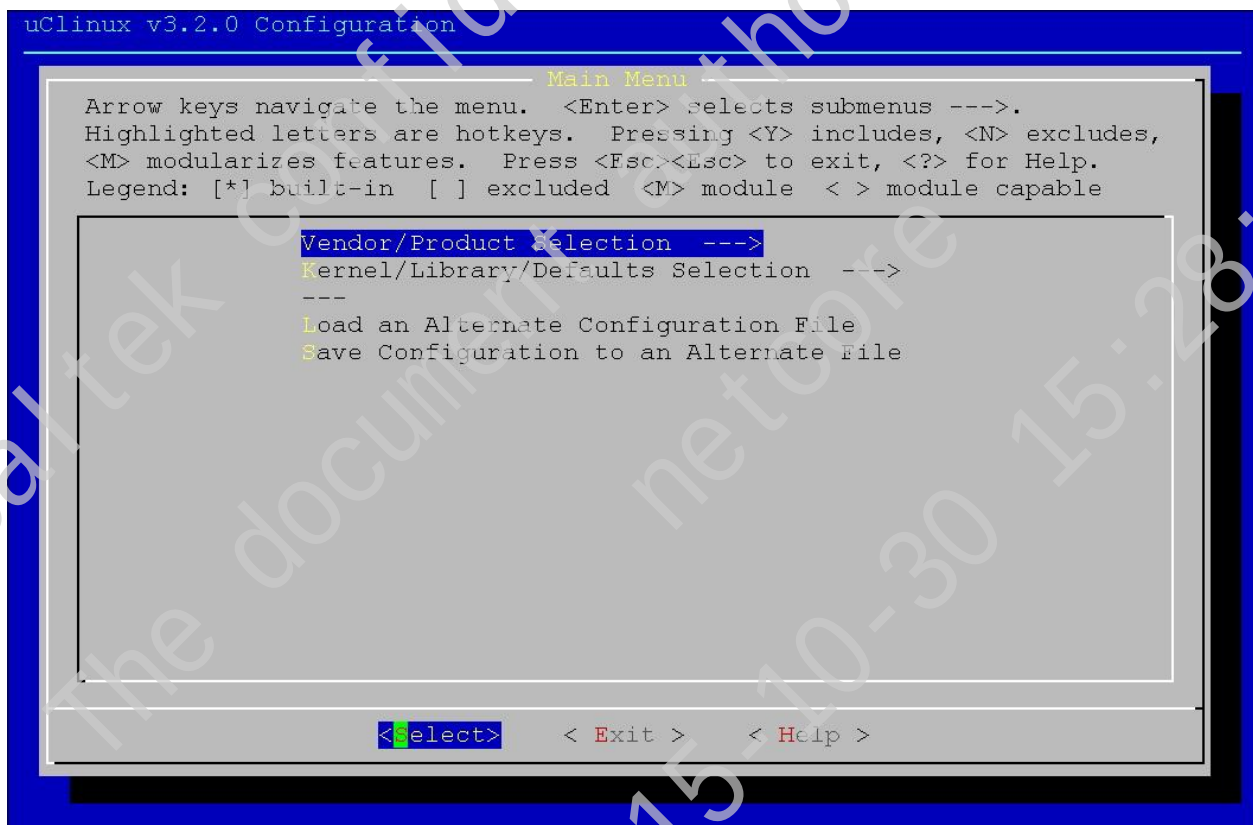
1.4. RTL8382M/RTL8332M Demo Board

The default diag-v2 command tree of the SDK package is for 8390 series chip. Please remove following 2 files and re-generate the new diag-v2 command for the chip.

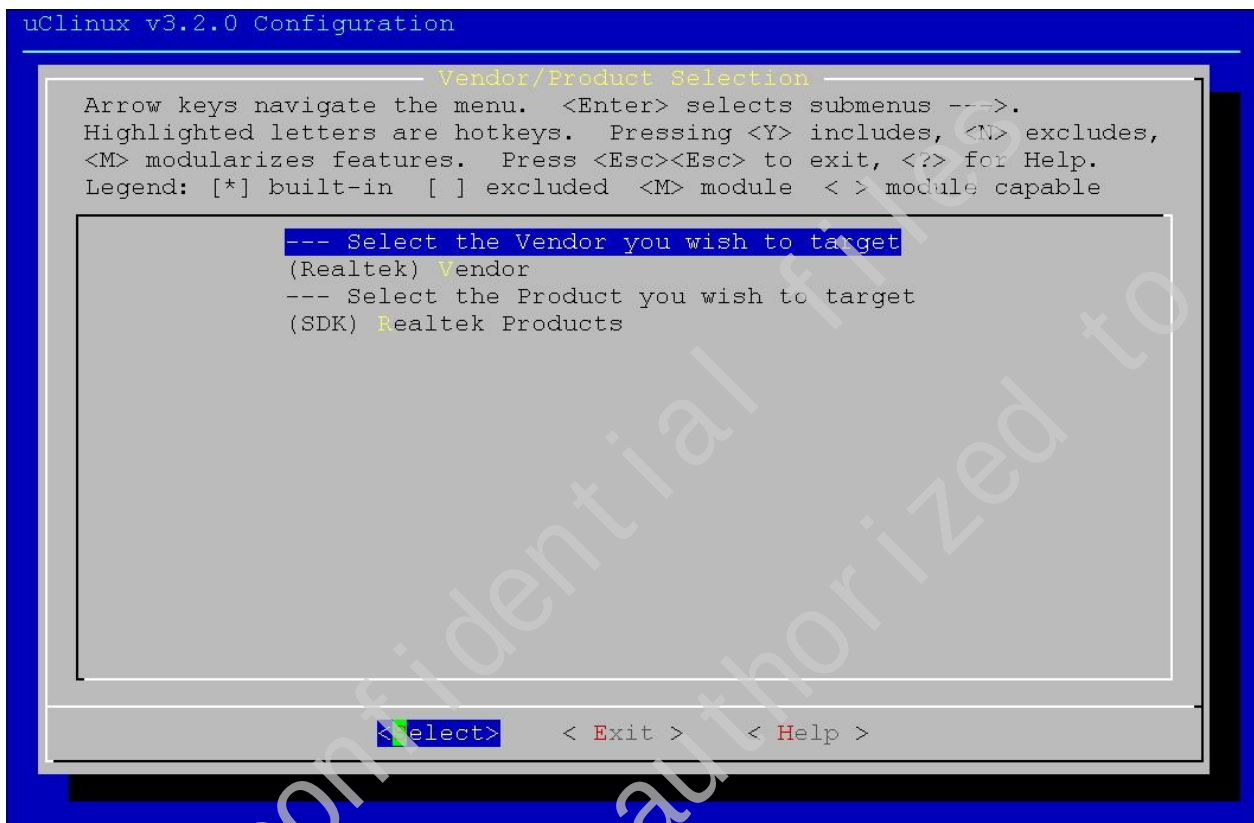
- sdk\src\app\diag_v2\include\parser\cparser_tree.h
- sdk\src\app\diag_v2\src\parser\cparser_tree.c

Following example is SDK driver and NIC driver run at Linux kernel mode

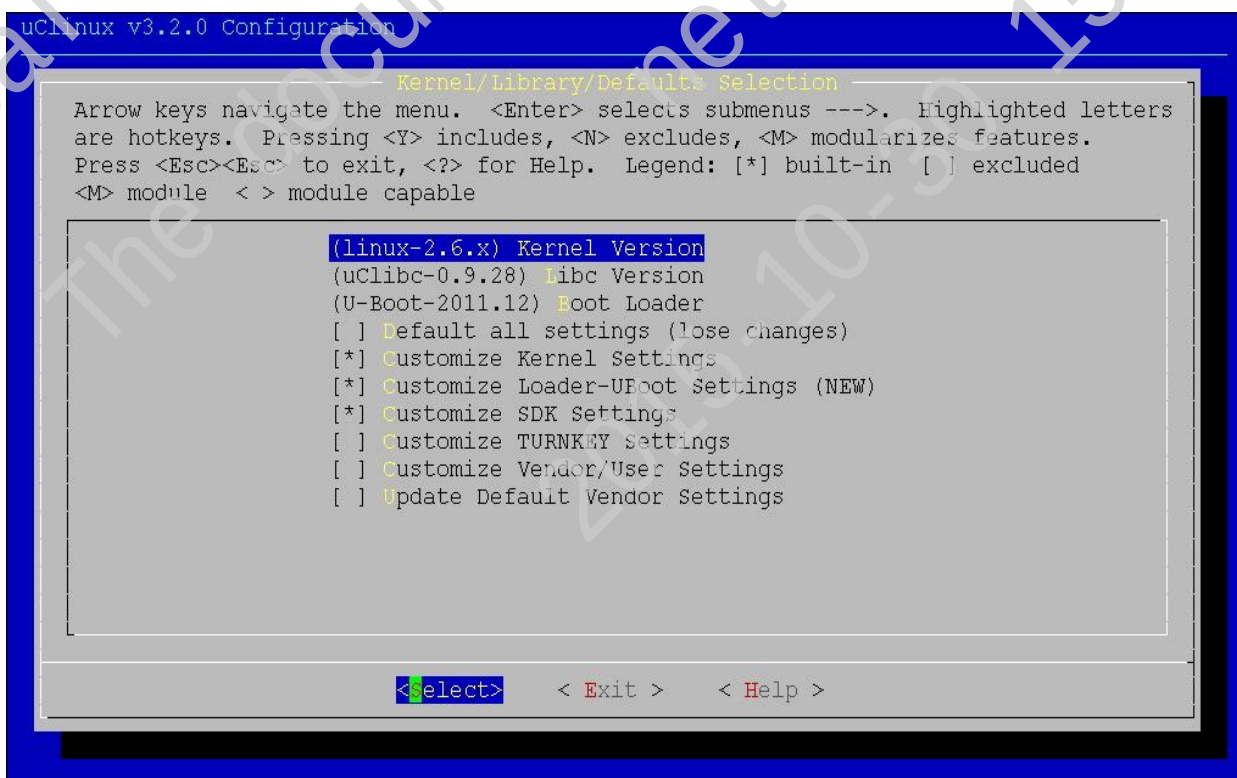
- When type 'make menuconfig' command, you can see the 'Main Menu' as following:



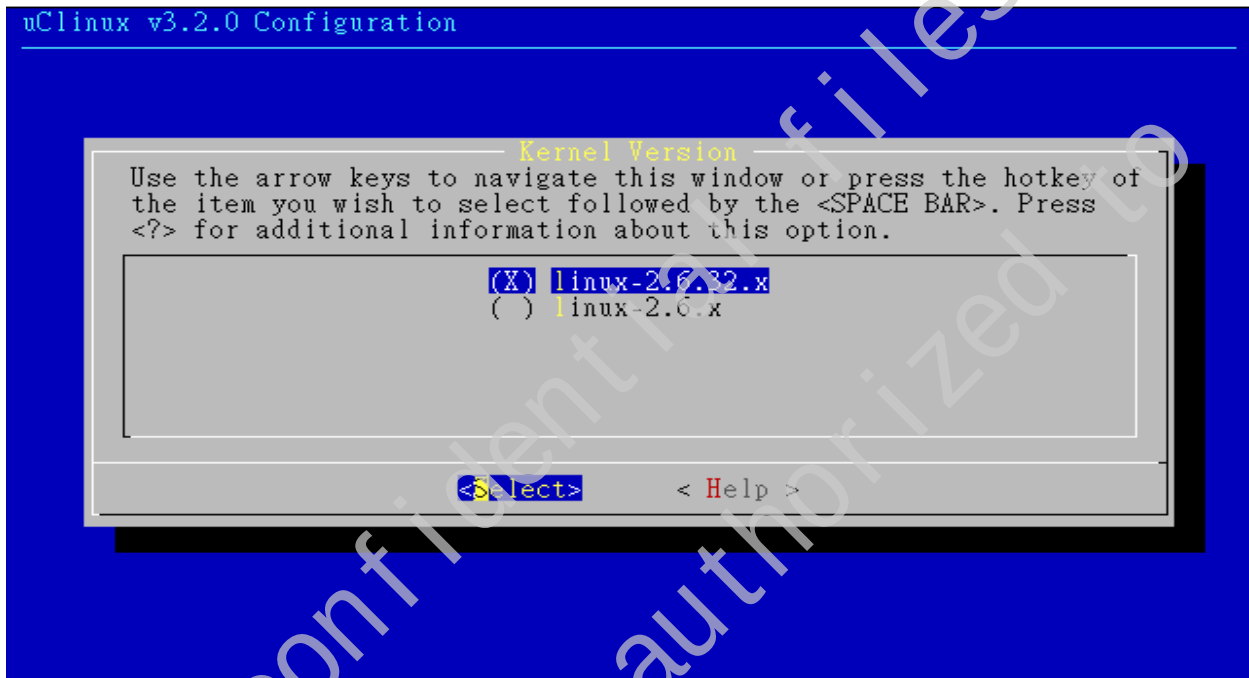
- “Vendor/Product Selection” setting as following:



- Select some items in “Kernel/Library/Defaults Selection” and “Exit” for configure Kernel, Loader-UBoot and SDK setting:



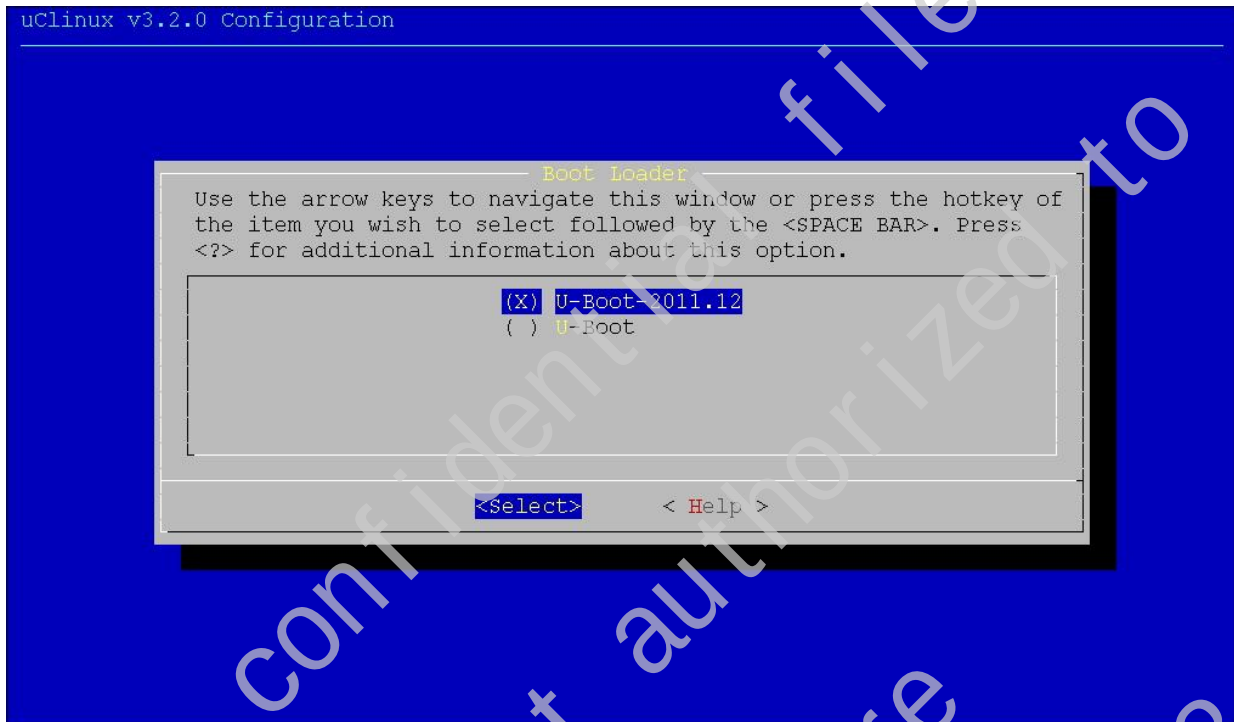
- In "Kernel/Library/Defaults Selection" menu, there are two Linux kernel version could be selected. The linux-2.6.x is 2.6.19 and the linux-2.6.32.x is 2.6.32.58:

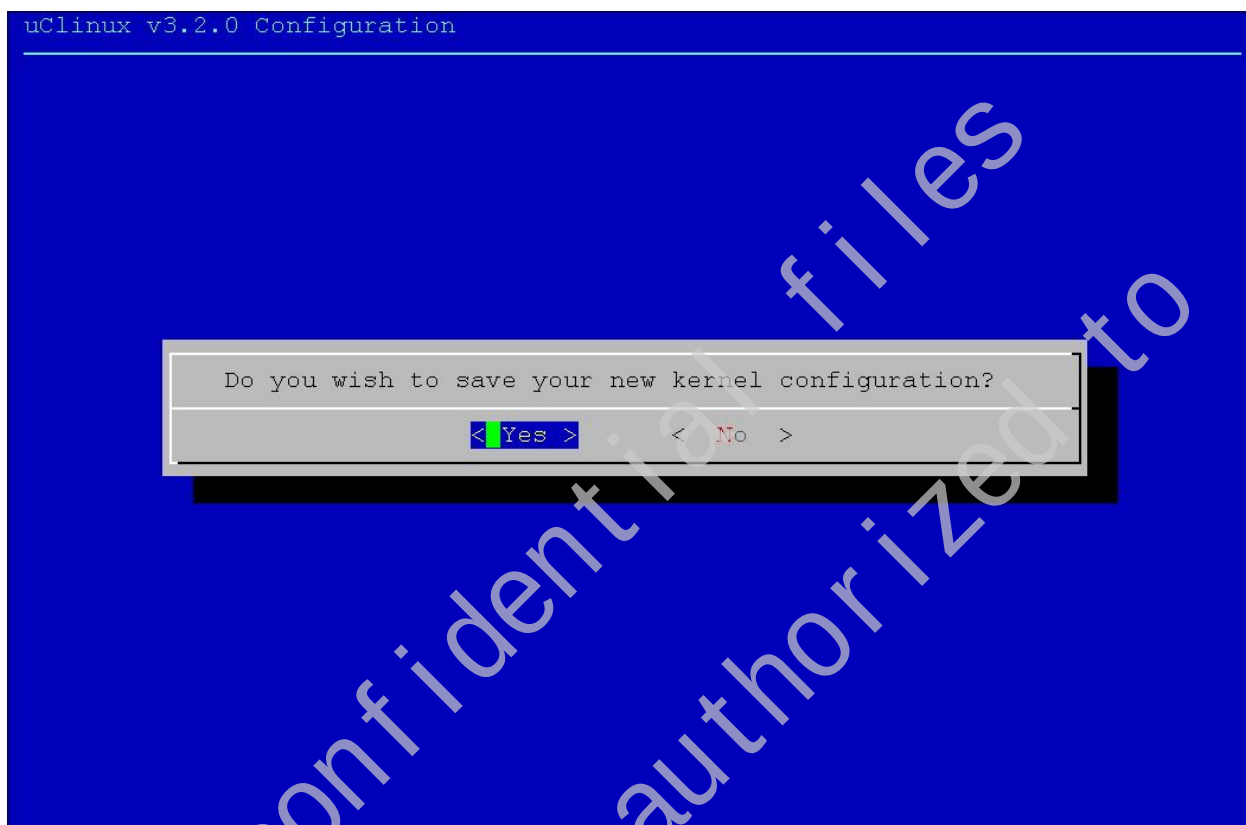


- In "Kernel/Library/Defaults Selection" menu, there are two uClibc version could be selected. The linux-2.6.x should use 0.9.28 and the linux-2.6.32.x should use 0.9.30:



- In "Kernel/Library/Defaults Selection" menu, there are two boot code version could be selected. The "U-Boot-2011.12" is new one. For the RTL8382M/RTL8332M, please selects the "U-Boot-2011.12".

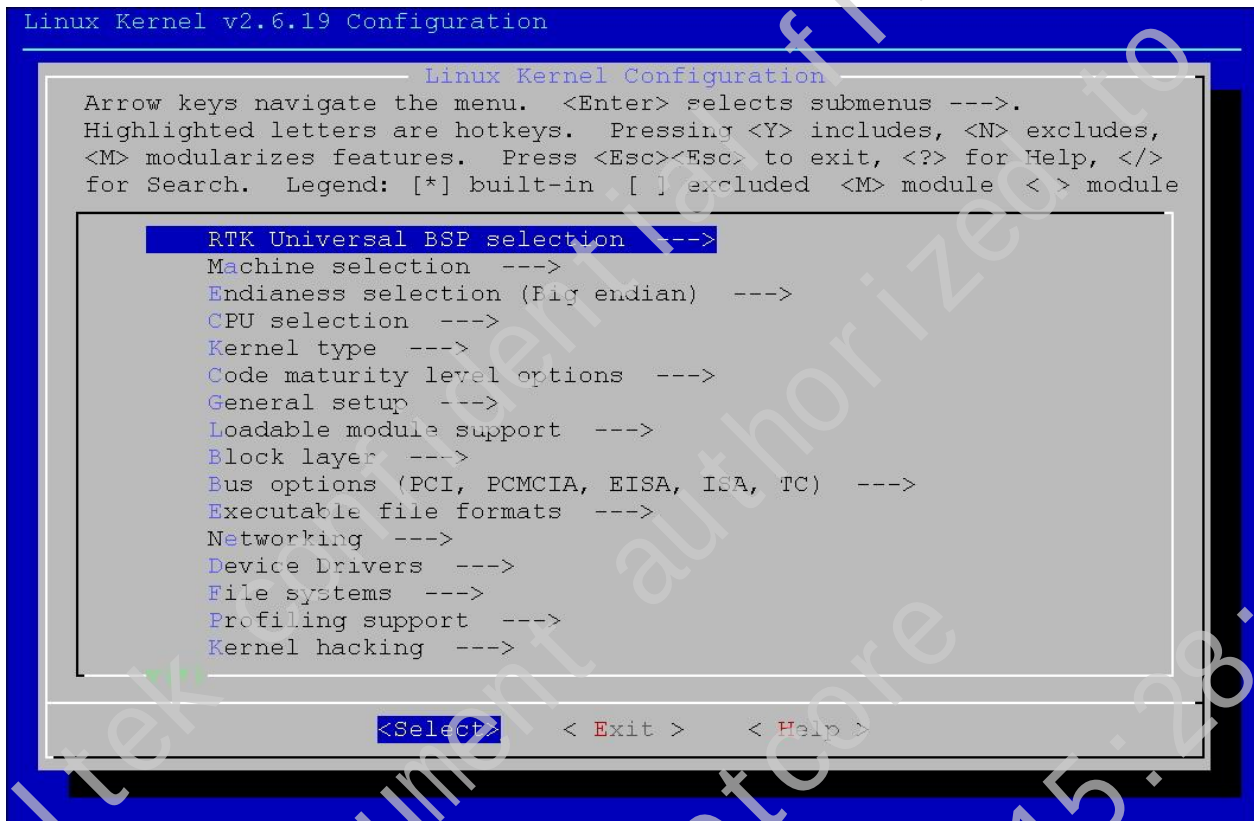




1.4.1 Kernel Menuconfig

1.4.1.1 Linux v2.6.19 Menuconfig

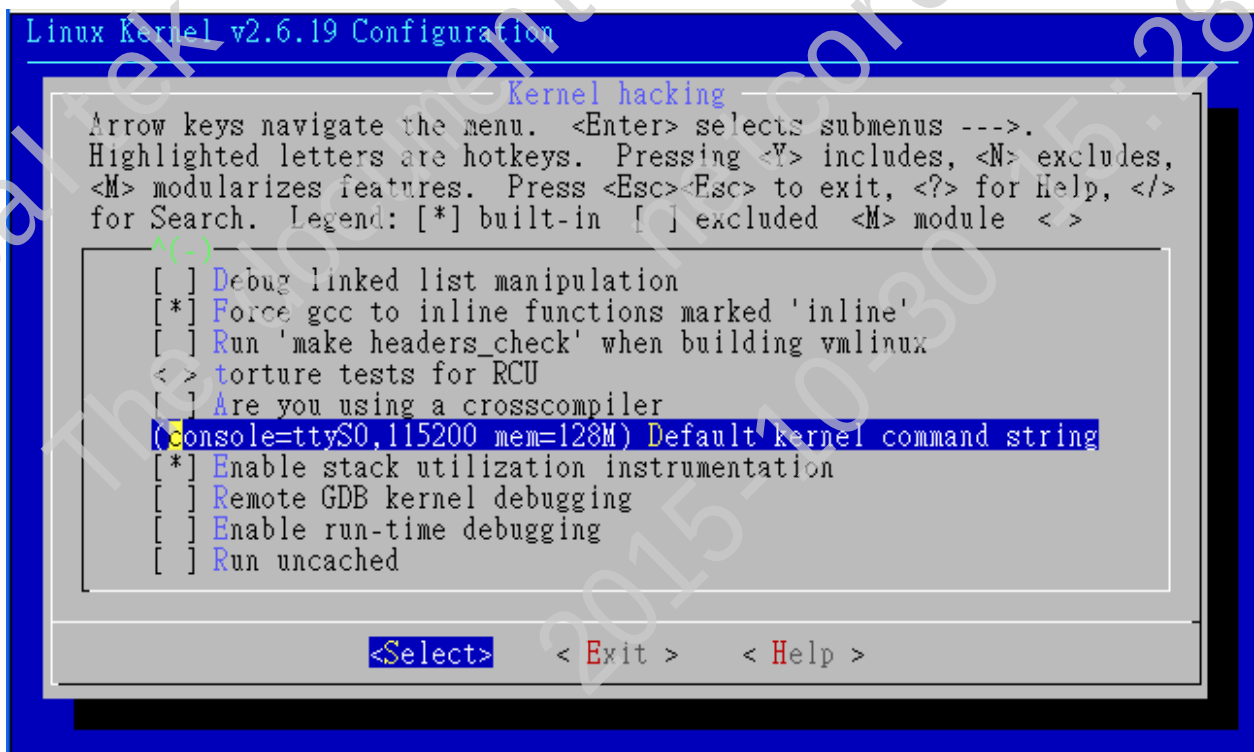
- Enter the "Linux Kernel v2.6.19 Configuration" as following:



- "RTK Universal BSP selection" setting as following: (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.

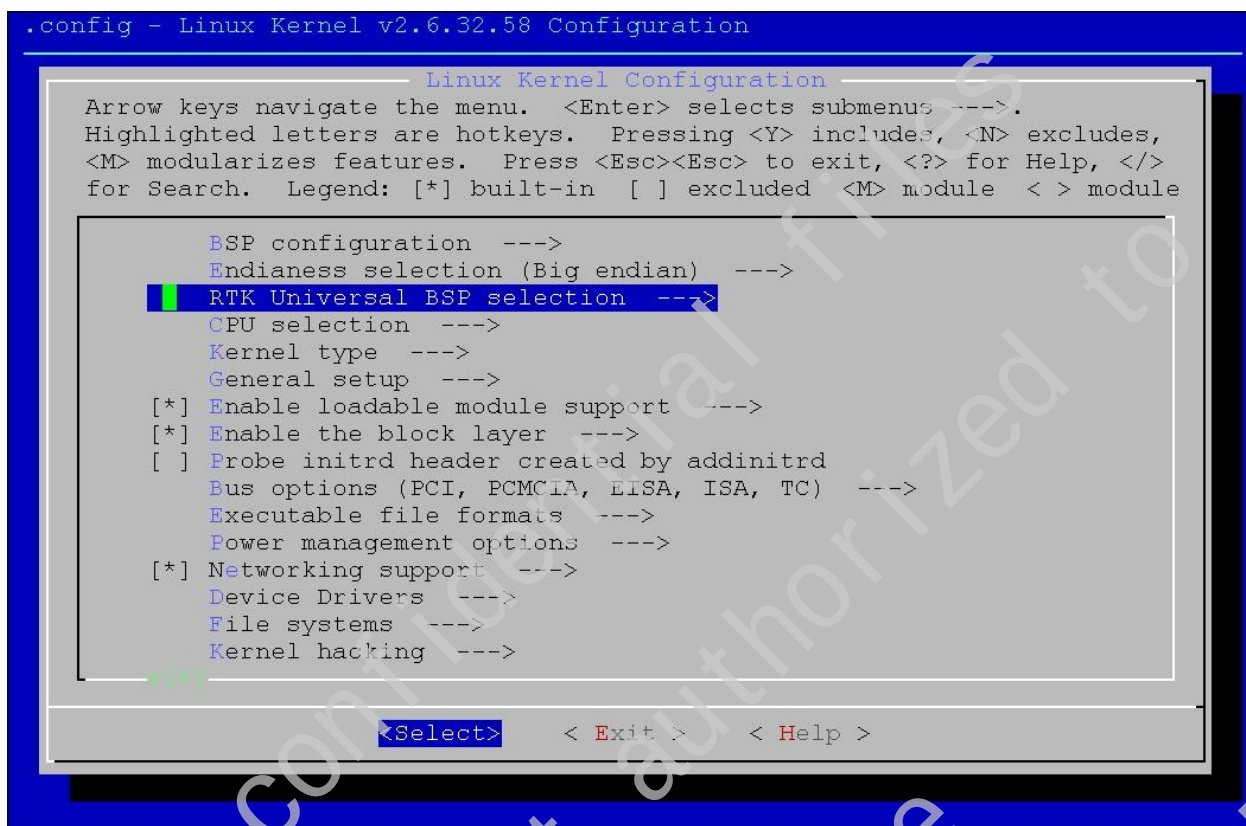


- “Kernel hacking” setting as following: (128MB DDR RAM)

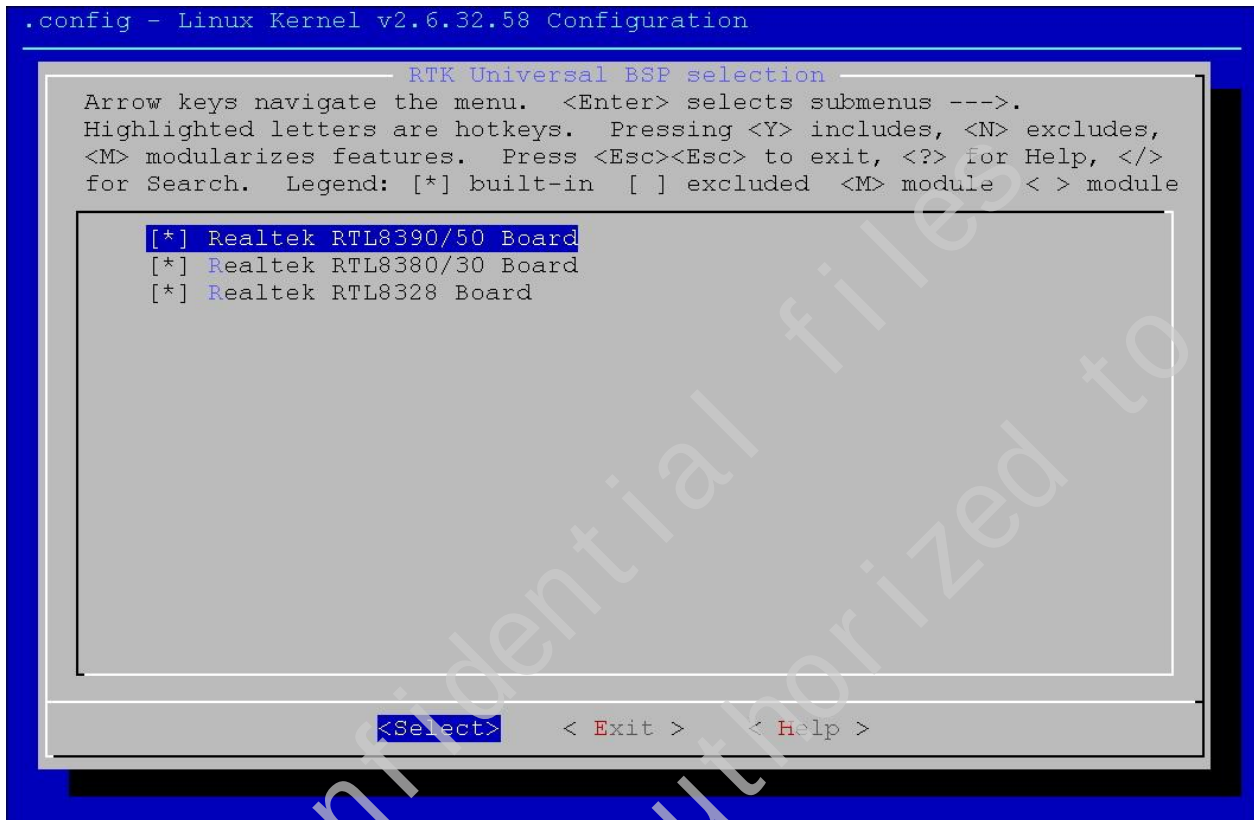


1.4.1.2 Linux v2.6.32.58 Menuconfig

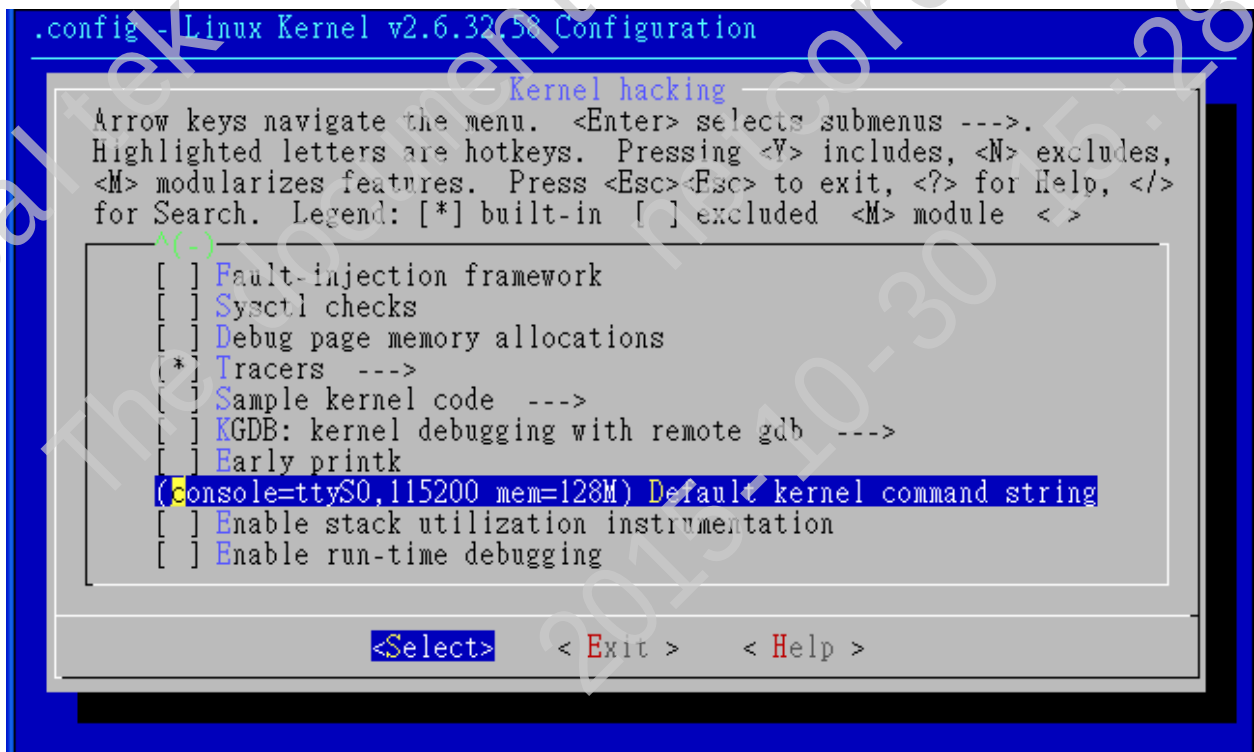
- Enter the "Linux Kernel v2.6.32.58 Configuration" as following:



- "RTK Universal BSP selection" setting as following: (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.

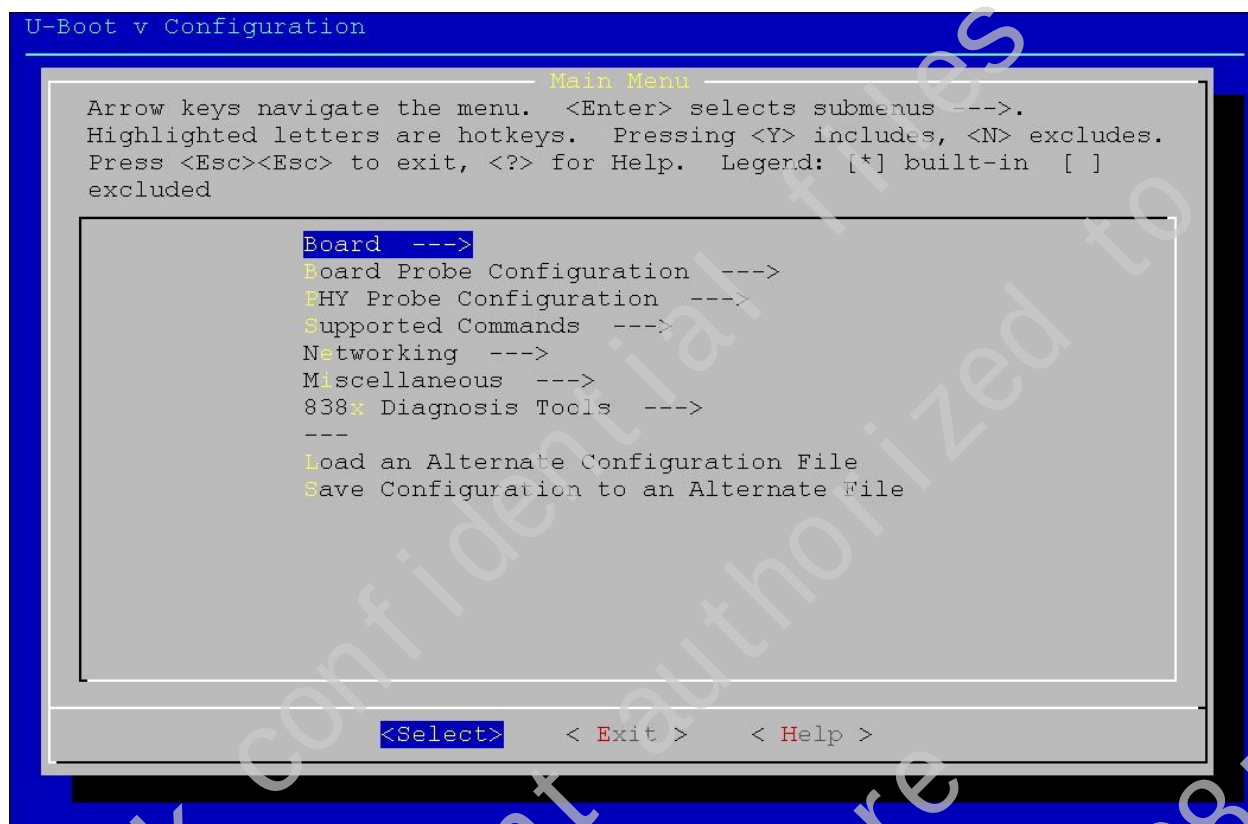


- "Kernel hacking" setting as following: (128MB DDR RAM)

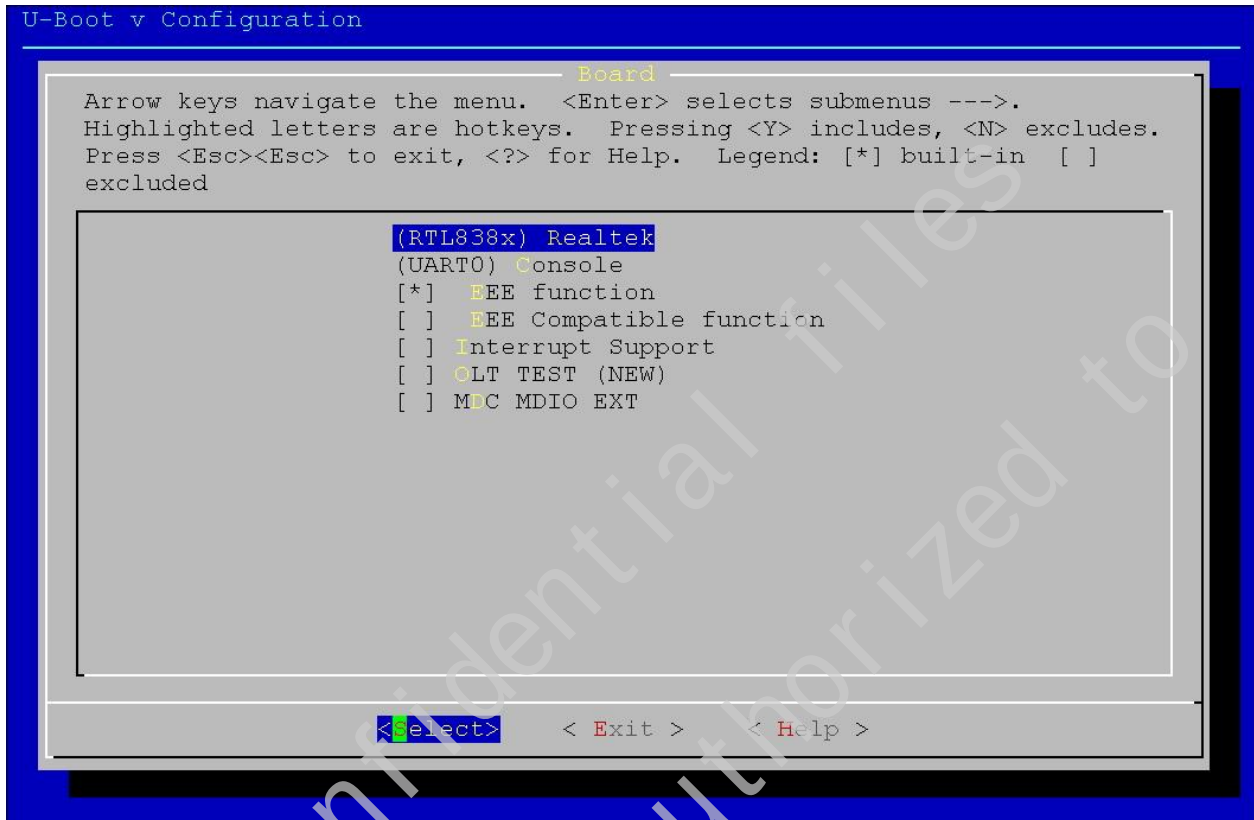


1.4.2 U-Boot 2011.12 Menuconfig

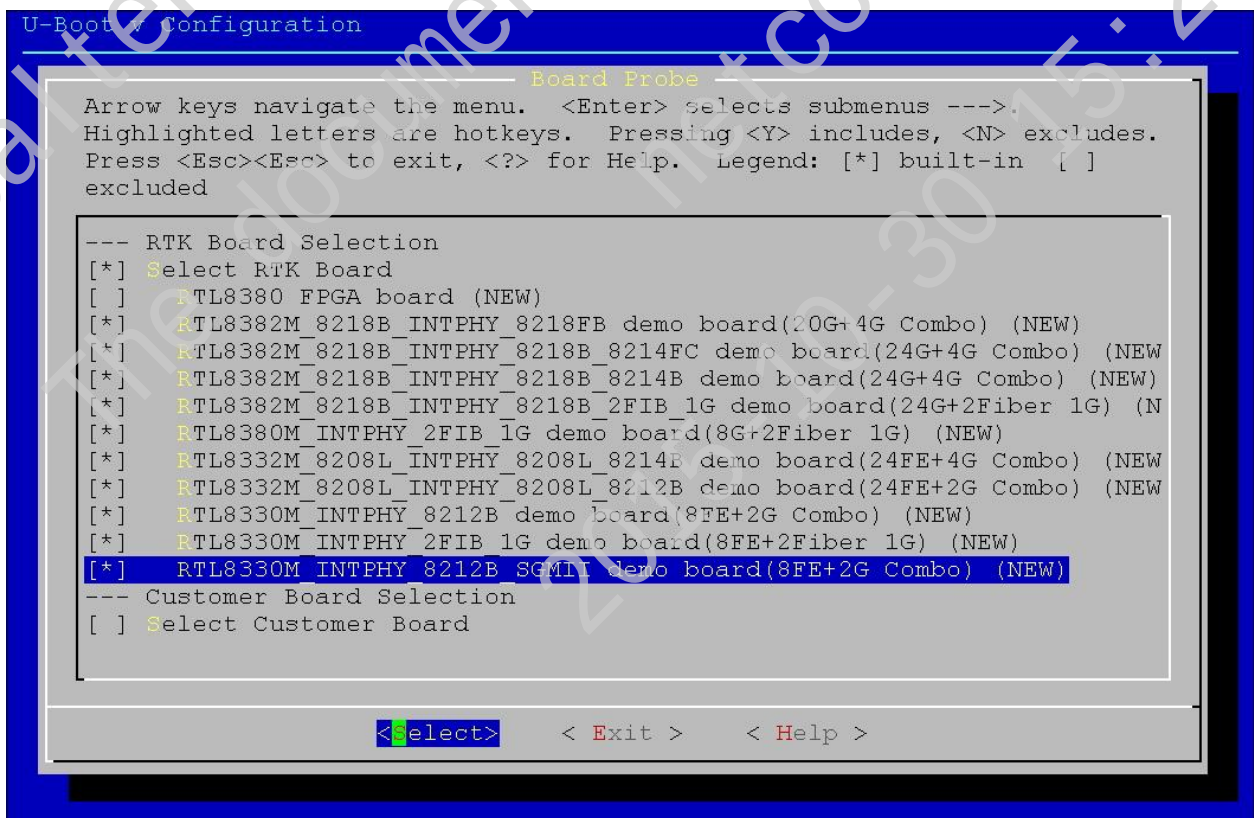
- After exit and save "Linux Kernel Configuration" setting, you can enter "U-Boot v Configuration" Main Menu as following:



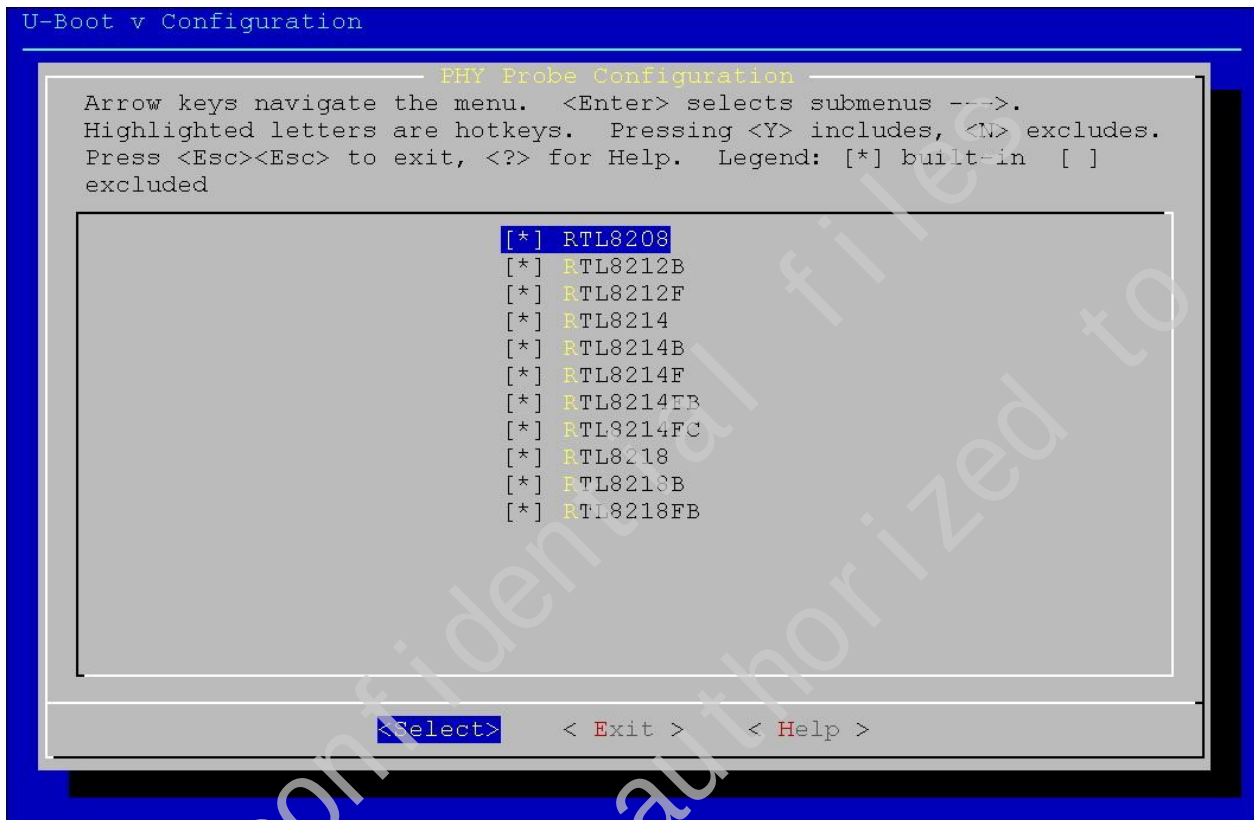
- "Board Selection" setting as following:



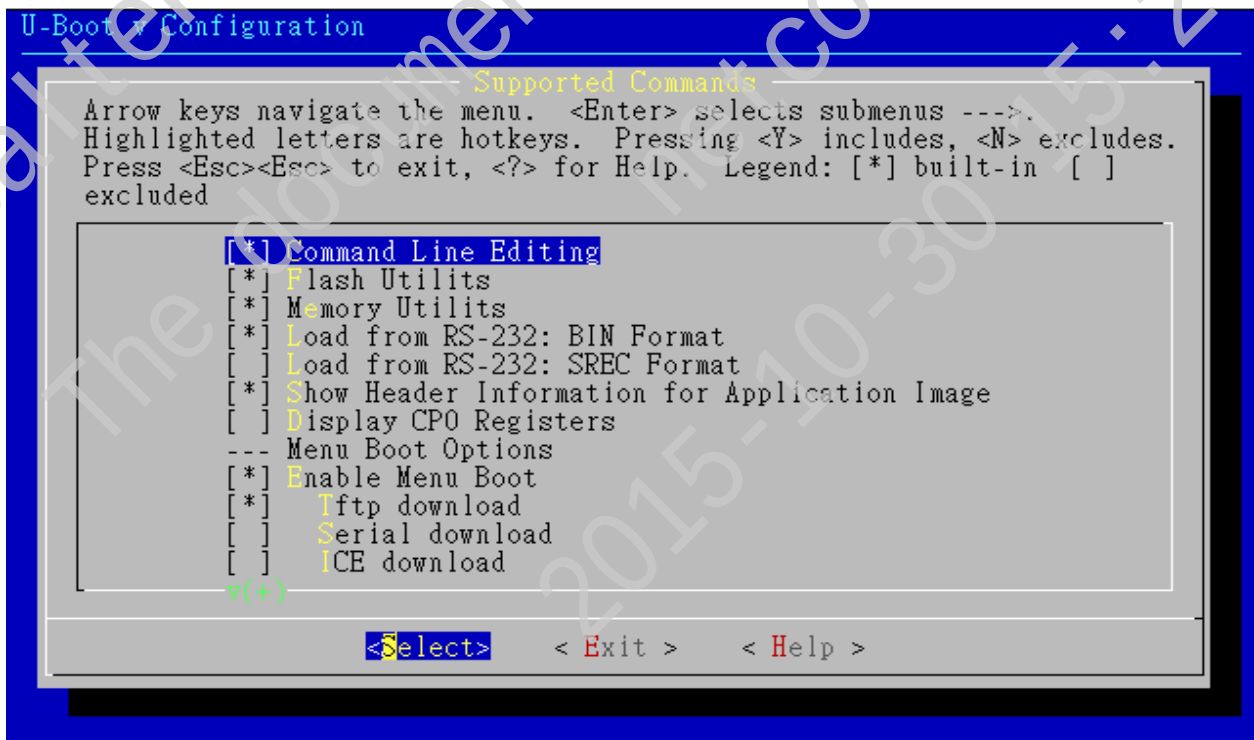
- “Board Probe Configuration” setting as following:



- “PHY Probe Configuration” setting as following:



- “Supported Commands” setting as following:



- “Networking” setting as following:

U-Boot v Configuration

Networking

Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes. Press <Esc><Esc> to exit, <?> for Help. Legend: [*] built-in [] excluded

```

MAC Address: "DE:AD:BE:EF:01:02"
[*] Network Command Support
[*] Ping
   IP Address: "192.168.1.1"
   TFTP Server IP Address: "192.168.1.111"
  
```

< Select > < Exit > < Help >

➤ "Miscellaneous" setting as following:

U-Boot v Configuration

Miscellaneous

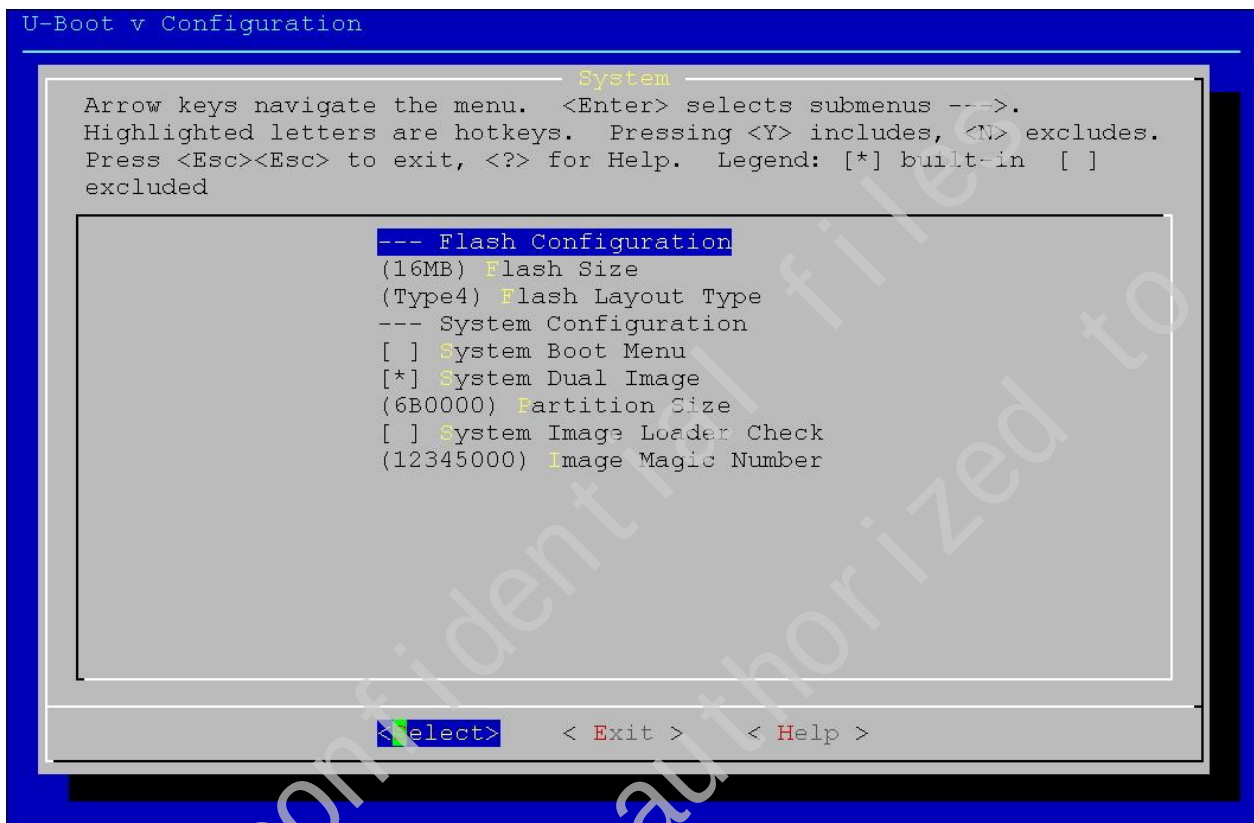
Arrow keys navigate the menu. <Enter> selects submenus --->. Highlighted letters are hotkeys. Pressing <Y> includes, <N> excludes. Press <Esc><Esc> to exit, <?> for Help. Legend: [*] built-in [] excluded

```

(83F00000) U-Boot Text Base
(-1) Boot Delay, -1 to Disable
   Boot Command: "rtk"
   Board Version: "" (NEW)
   Board Model of Platform: "RTL8382M_8218B_INTPHY_8218B_8214FC_DEMO"
   Booting Arguments: "" (NEW)
   Post Password: "" (NEW)
(115200) Baudrate (NEW)
  
```

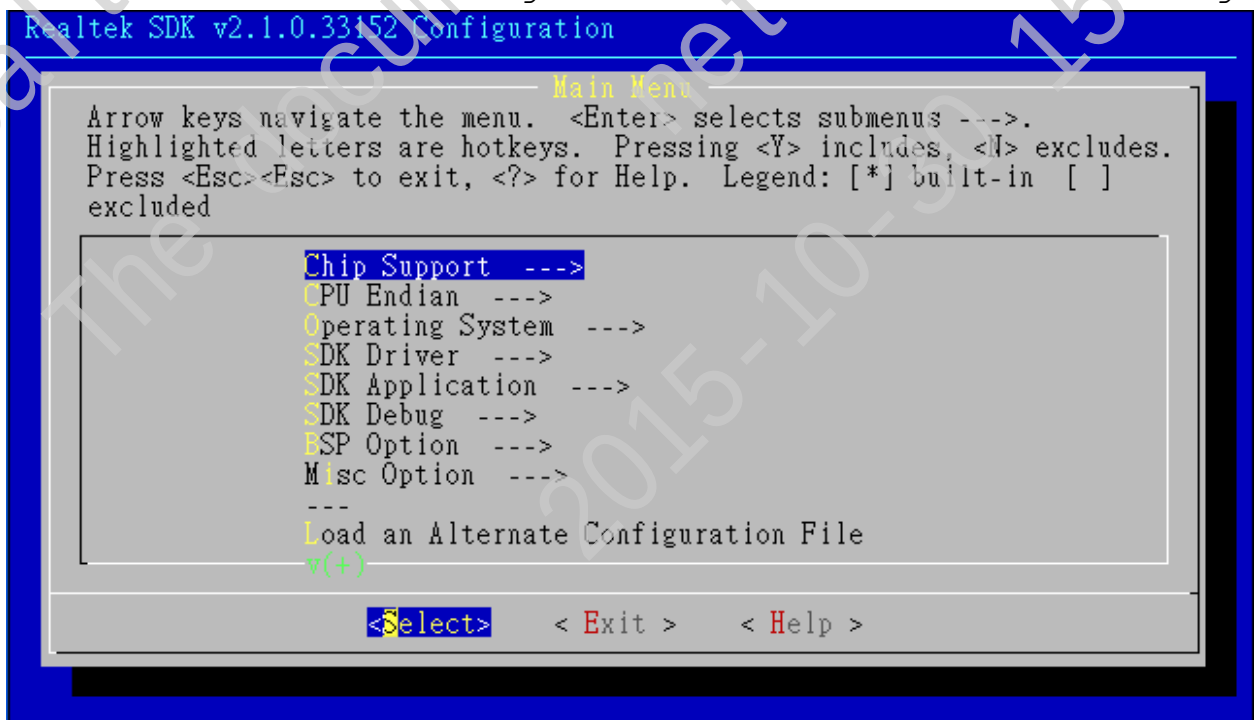
< Select > < Exit > < Help >

- “System” setting as following:

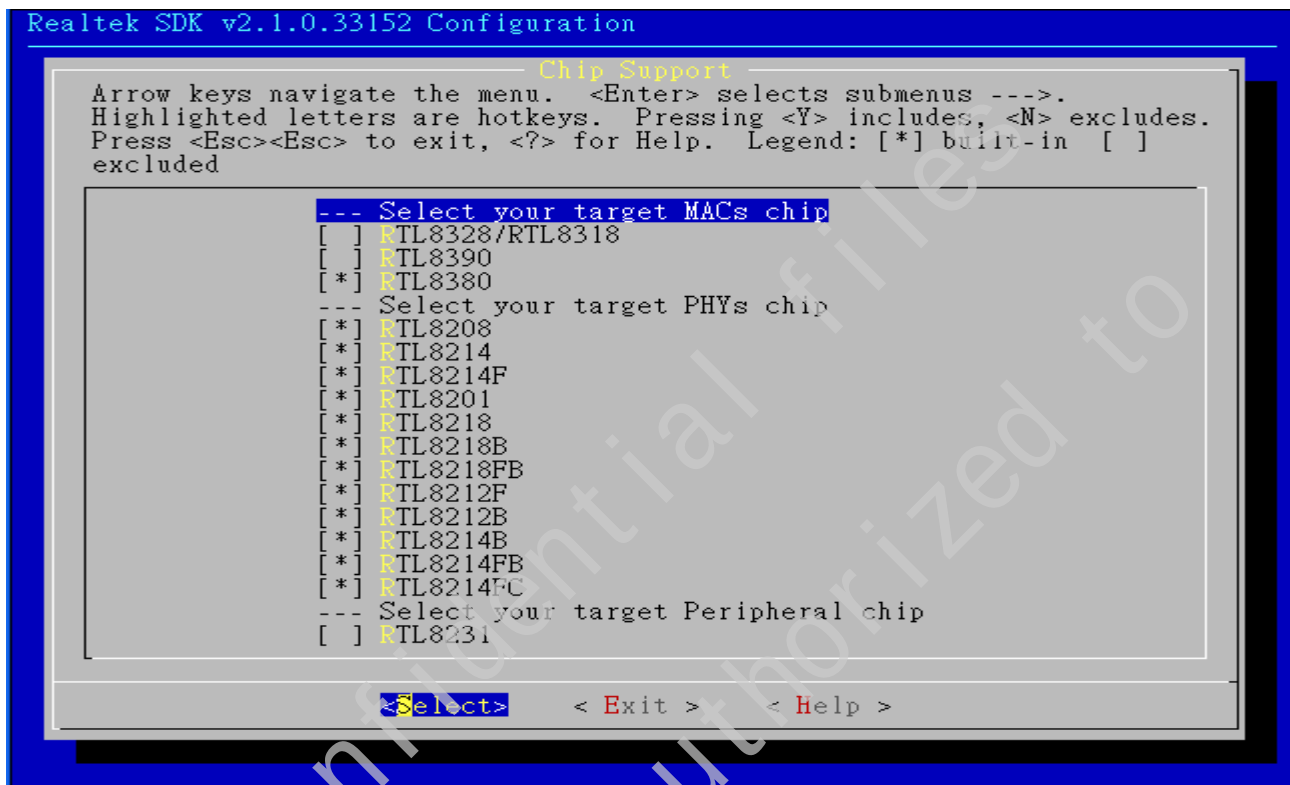


1.4.3 SDK Menuconfig

- After exit and save “U-Boot Configuration” setting, you can enter “SDK Configuration” Main Menu as following:



- “Chip Support” setting as following:

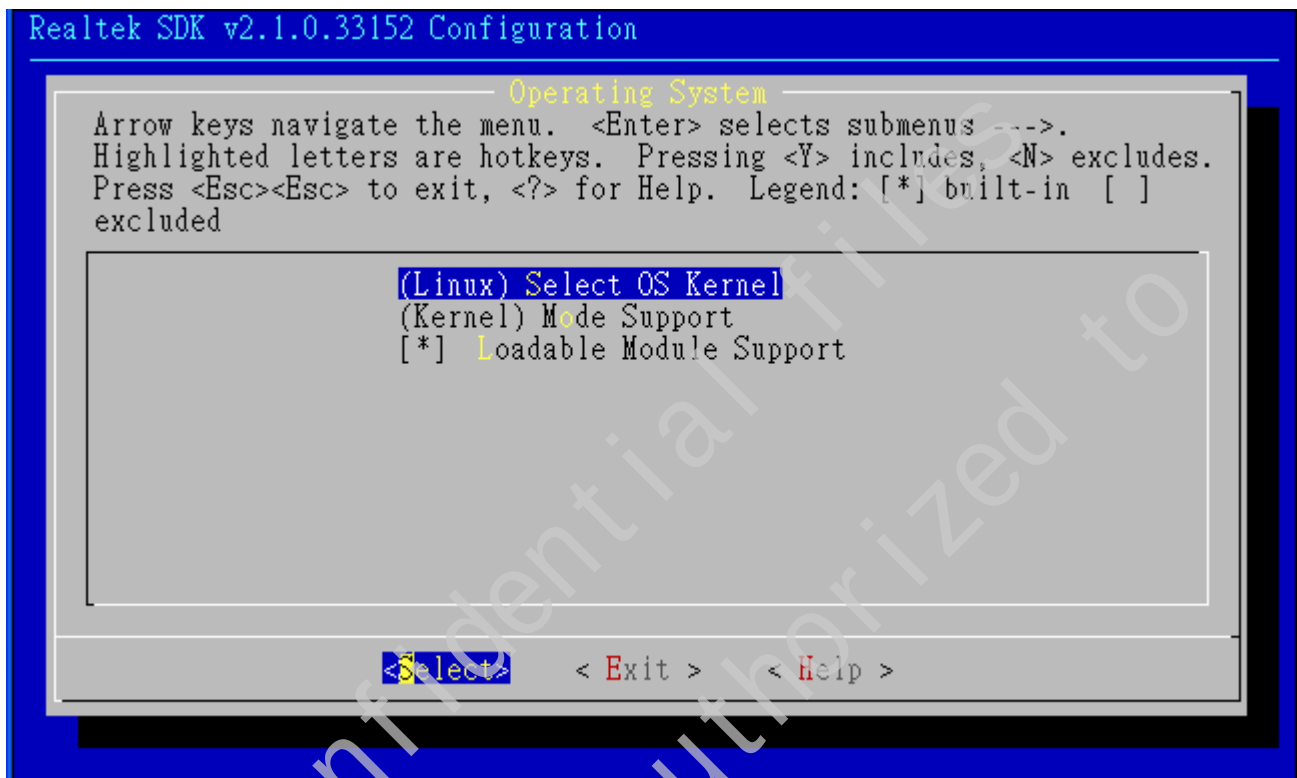


- “CPU Endian” setting as following:

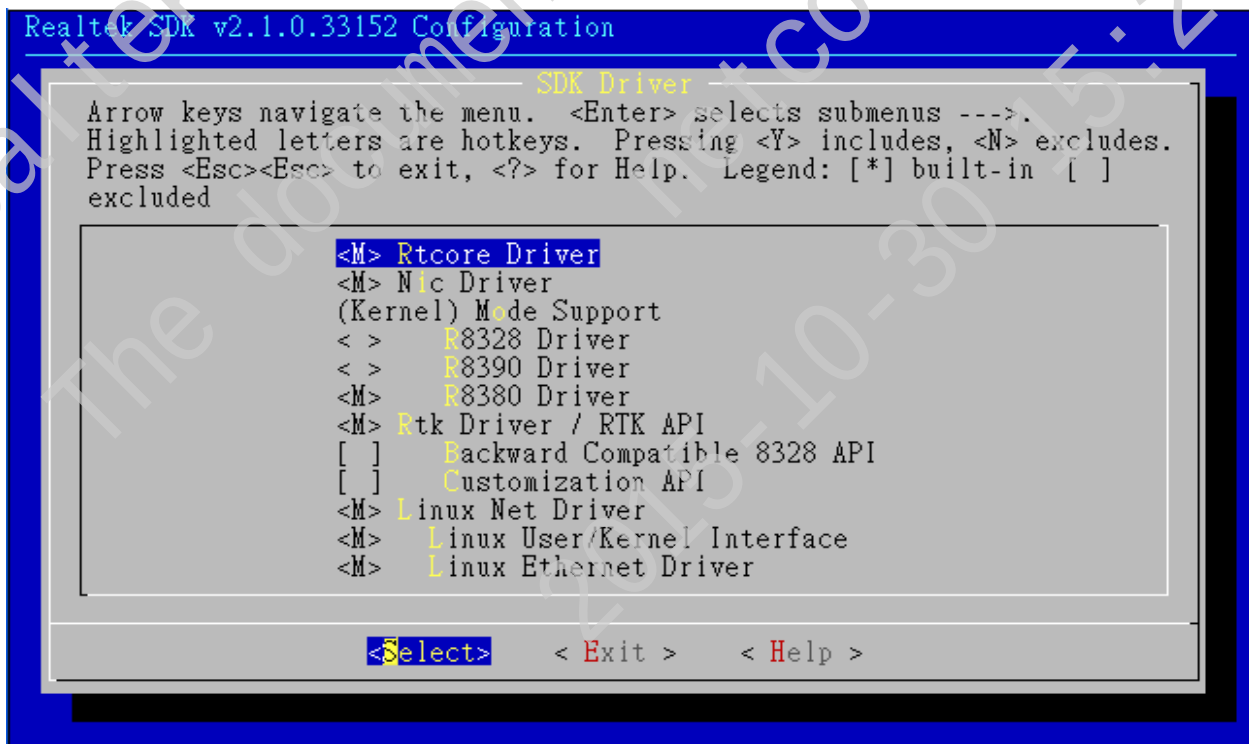


- “Operation System” setting as following for SDK driver run

at Linux kernel mode:

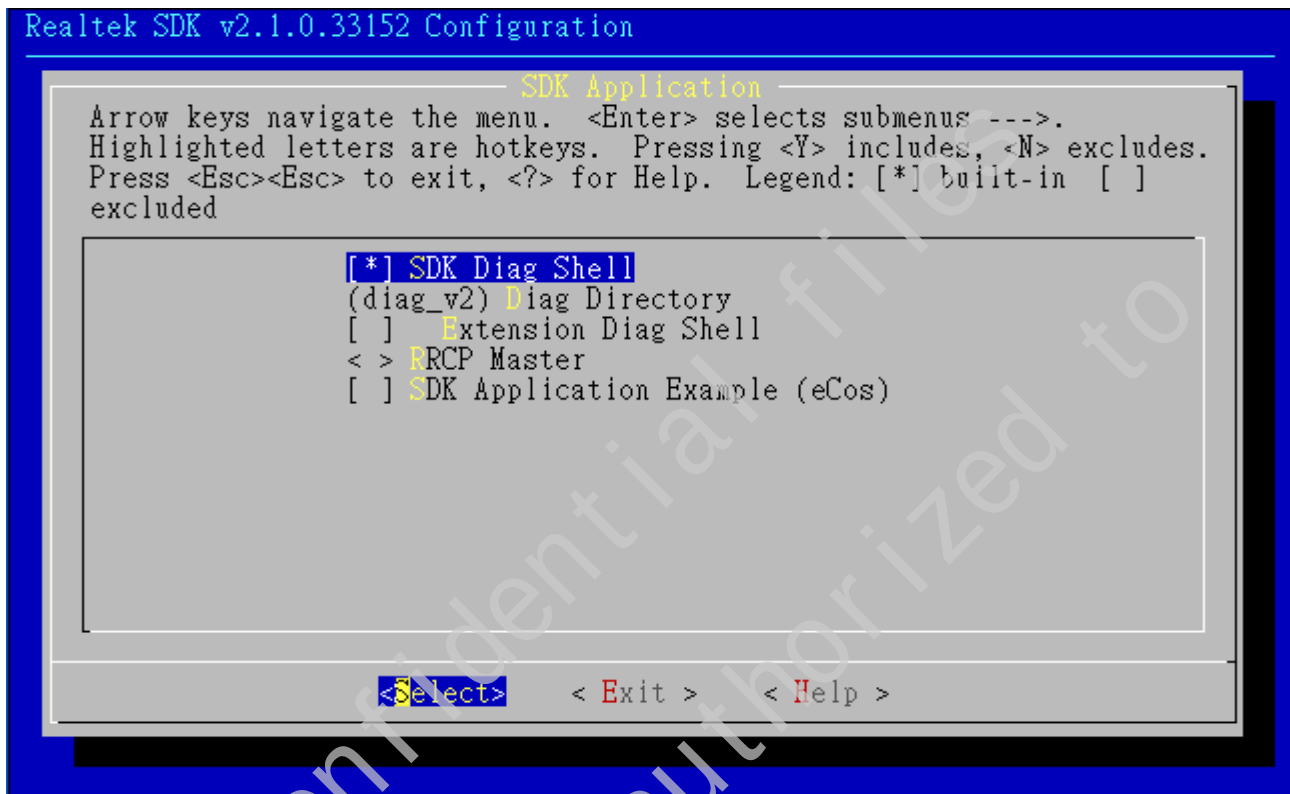


- "SDK Driver" setting as following for NIC driver run at Linux kernel mode:

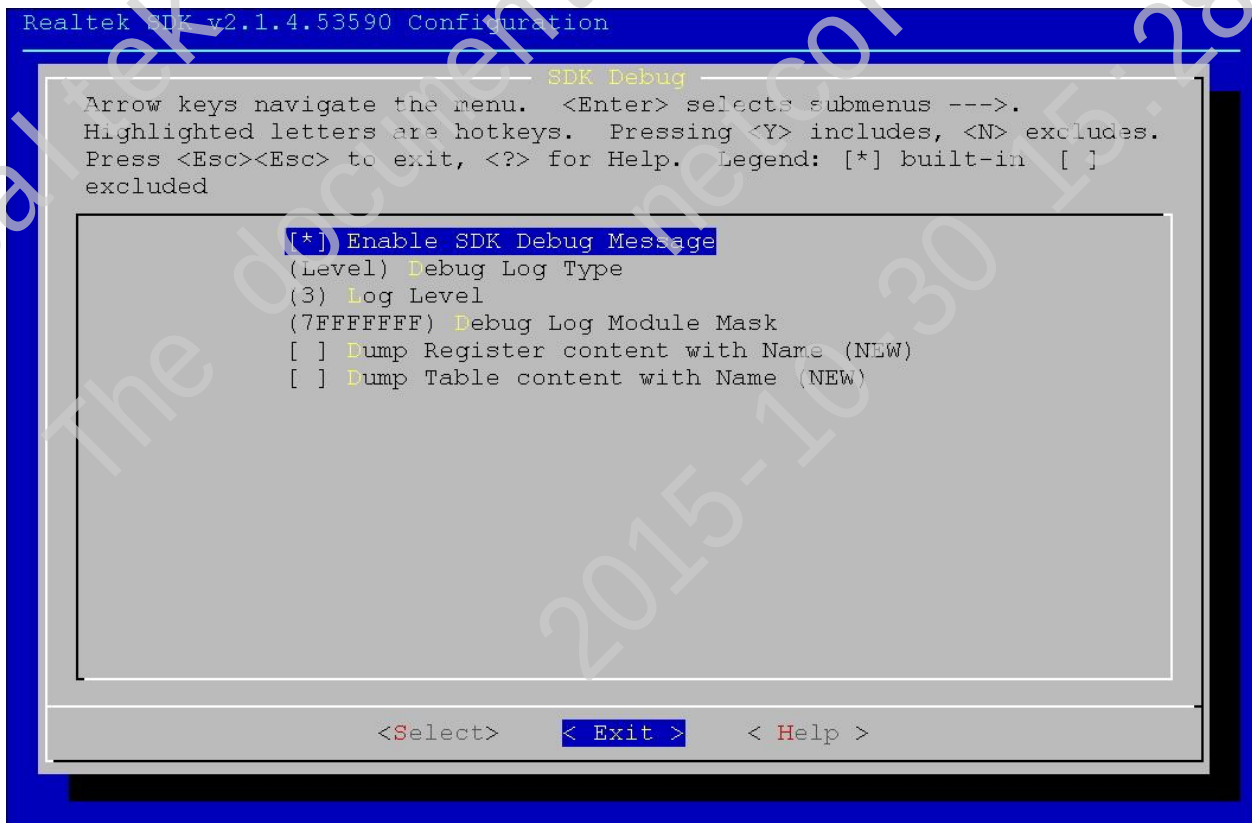


- "SDK Application" setting as following for turn on

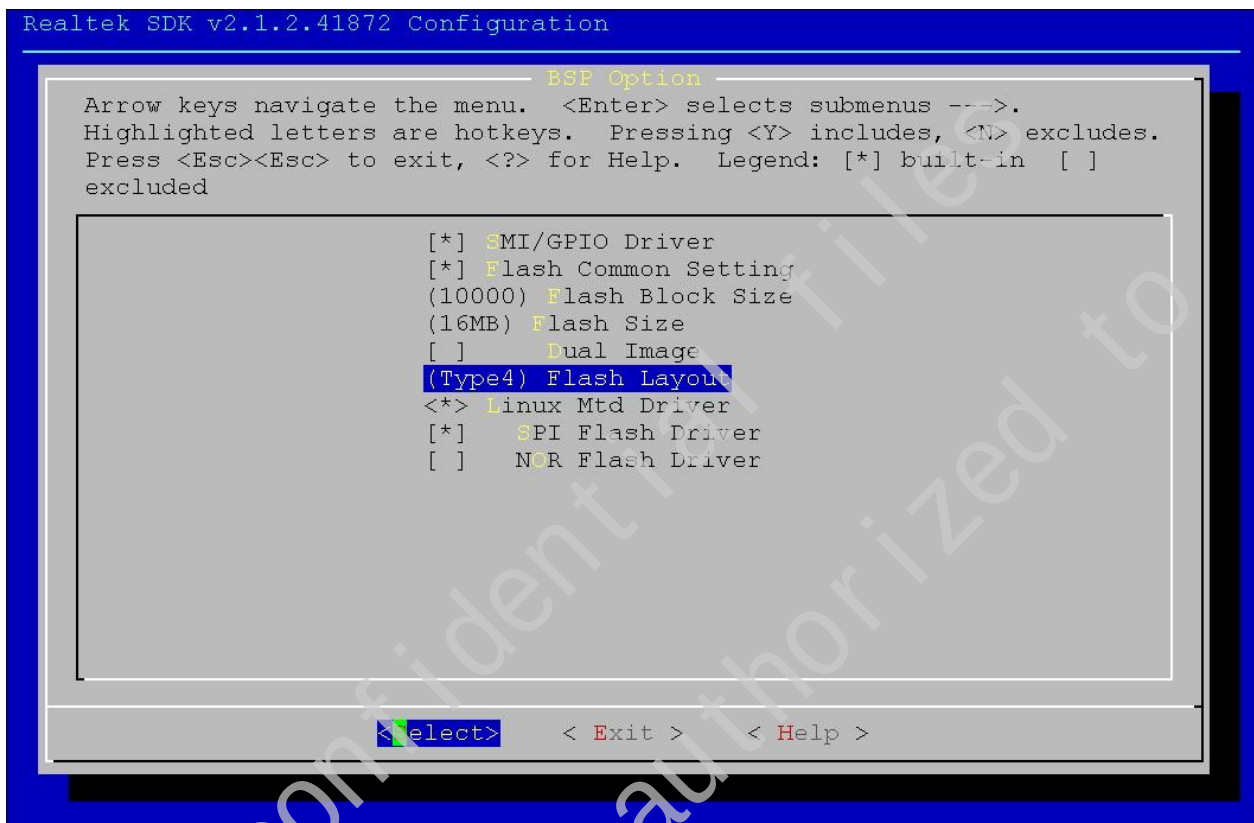
diagnostic shell: (MUST select diag_v2)



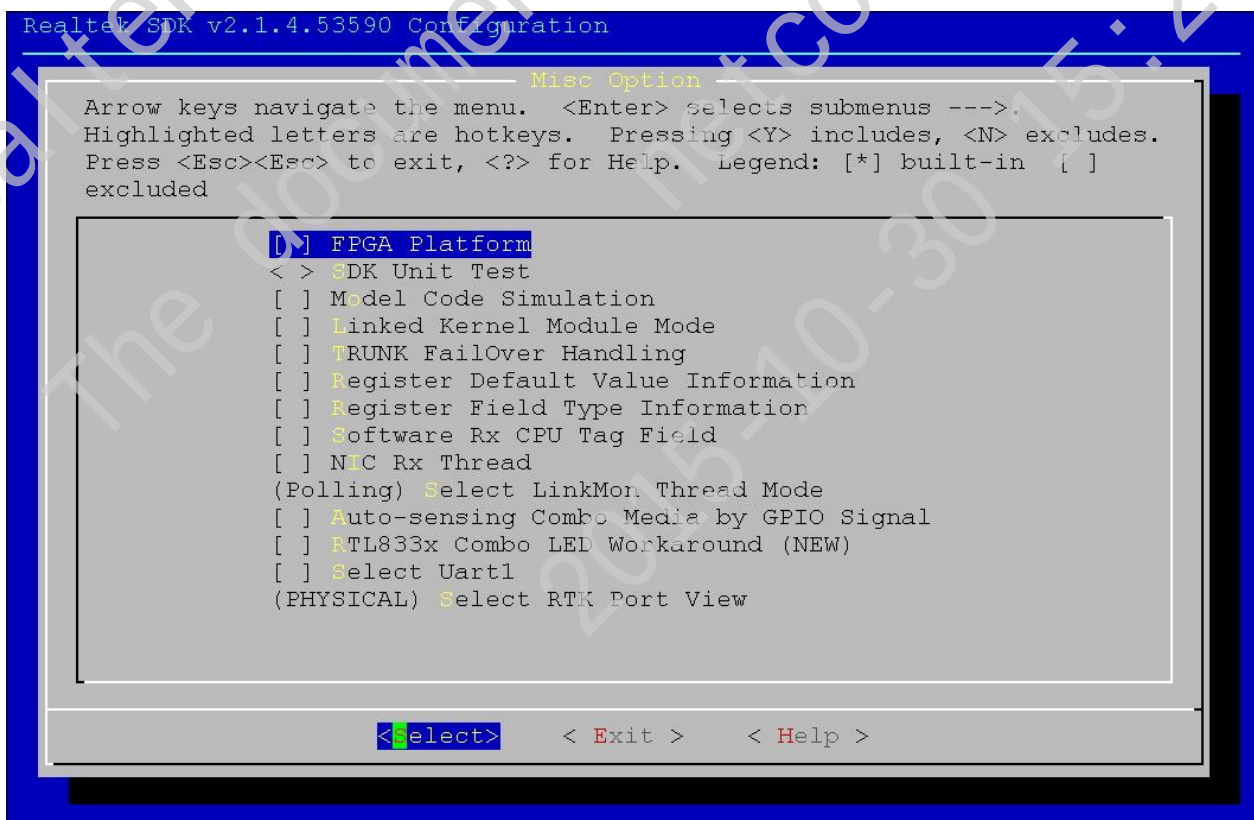
➤ “SDK Debug” setting as following:



- “BSP Option” setting as following:



- “Misc Option” setting as following:



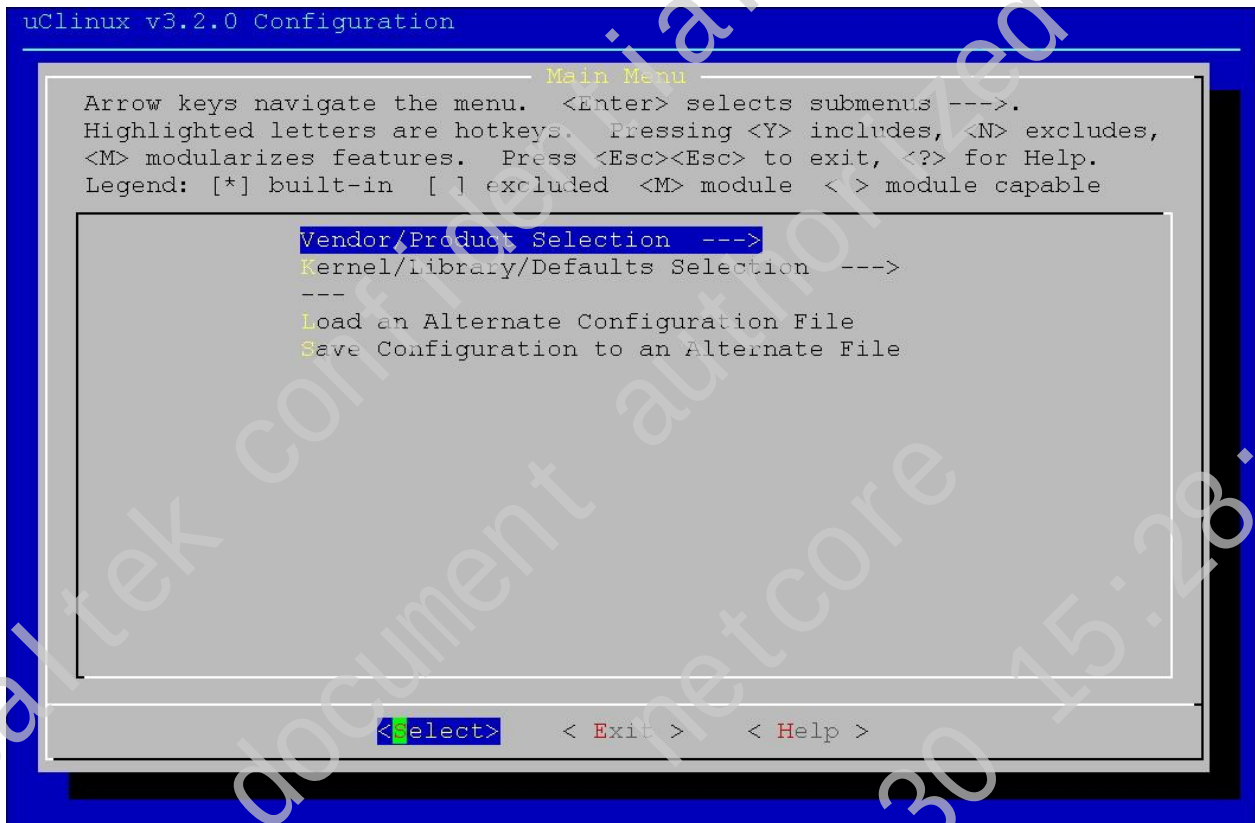
- After exit and save "SDK Configuration" setting, please type "make" for build out u-boot and runtime images

Realtek confidential files
The document authorized to
netcore
2015-10-30 15:28:26

1.5. RTL8393M/RTL8353M Demo Board

Following example is SDK driver and NIC driver run at Linux kernel mode

- When type 'make menuconfig' command, you can see the 'Main Menu' as following:



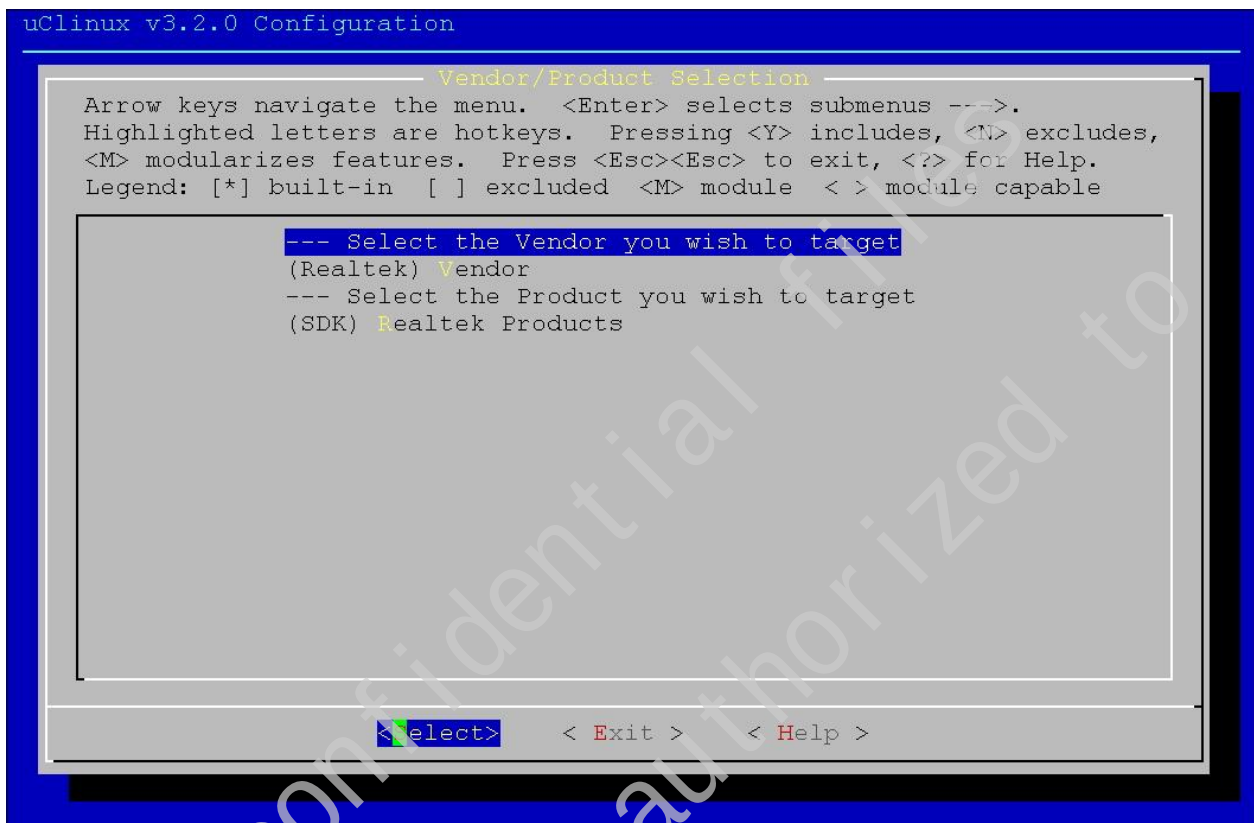
```
uClinux v3.2.0 Configuration

Main Menu
Arrow keys navigate the menu.  <Enter> selects submenus --->.
Highlighted letters are hotkeys.  Pressing <Y> includes, <N> excludes,
<M> modularizes features.  Press <Esc><Esc> to exit, <?> for Help.
Legend: [*] built-in  [ ] excluded  <M> module  < > module capable

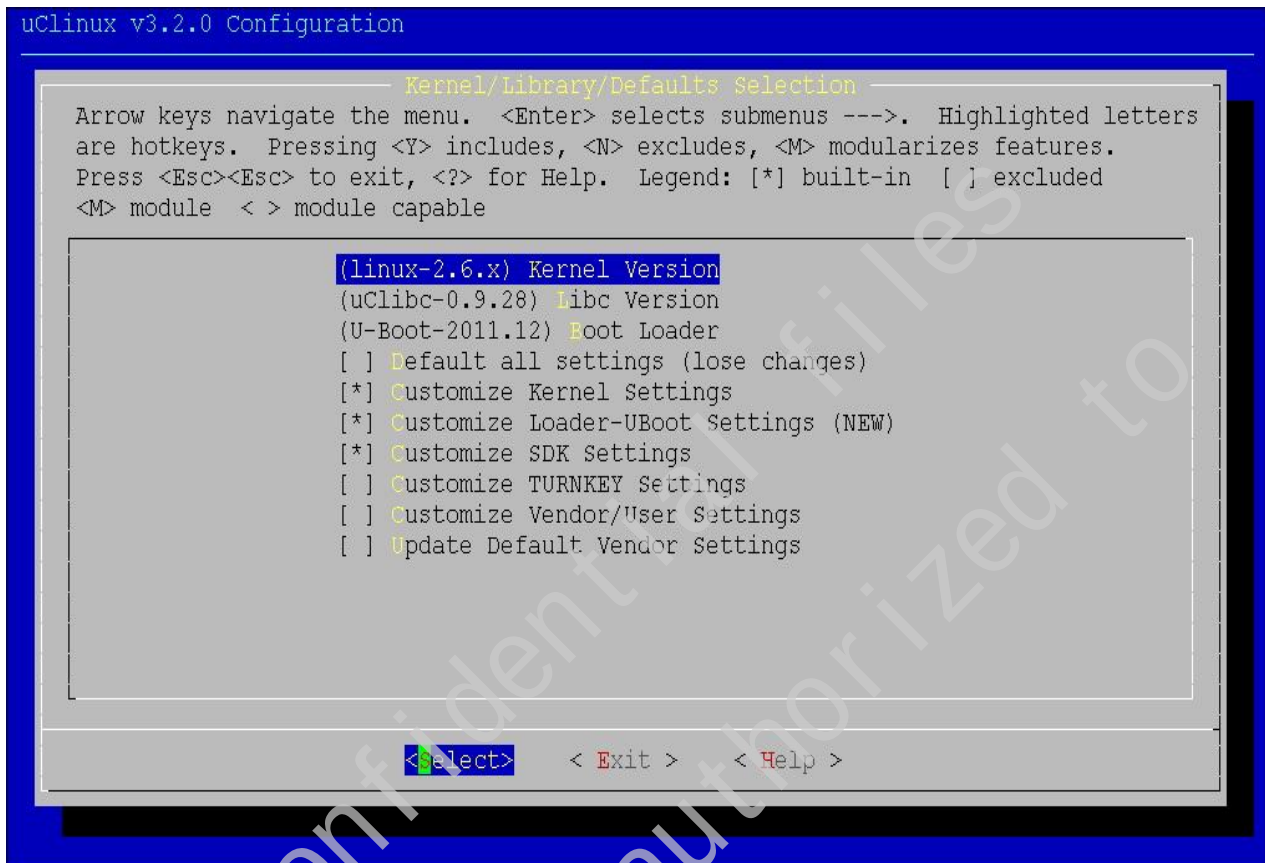
Vendor/Product Selection --->
Kernel/Library/Defaults Selection --->
---
Load an Alternate Configuration File
Save Configuration to an Alternate File

<Select>  < Exit >  < Help >
```

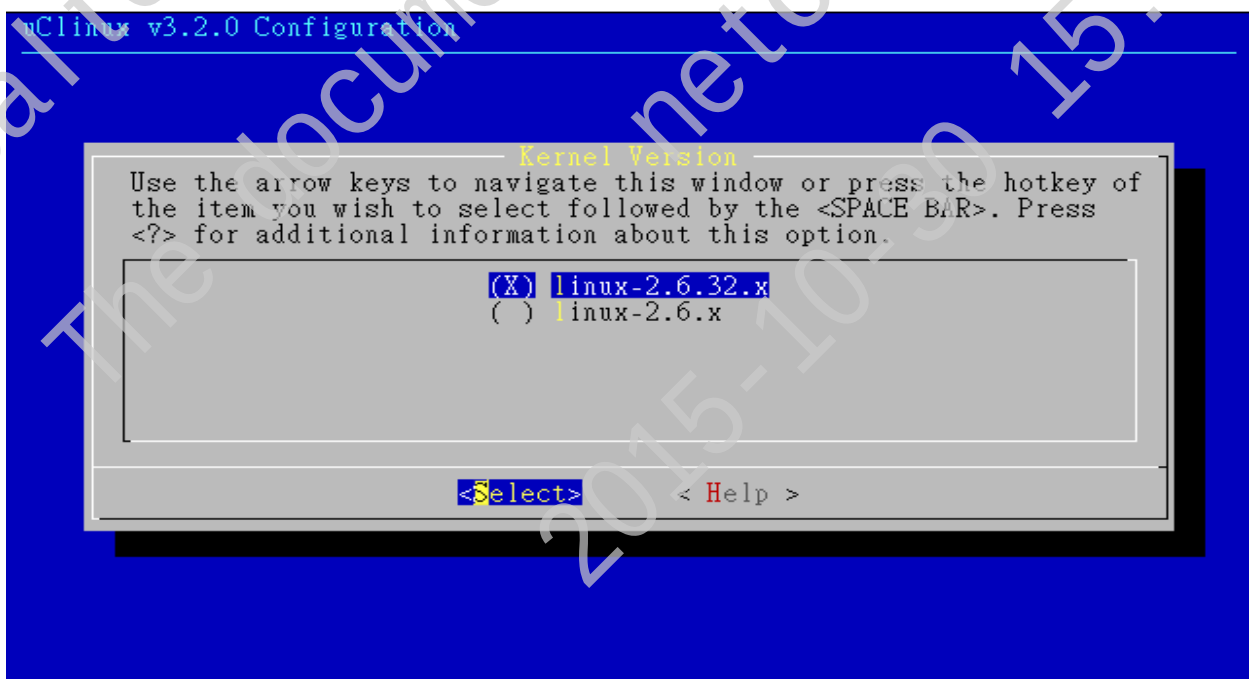
- “Vendor/Product Selection” setting as following:



- Select some items in “Kernel/Library/Defaults Selection” and “Exit” for configure Kernel, Loader-UBoot and SDK setting:

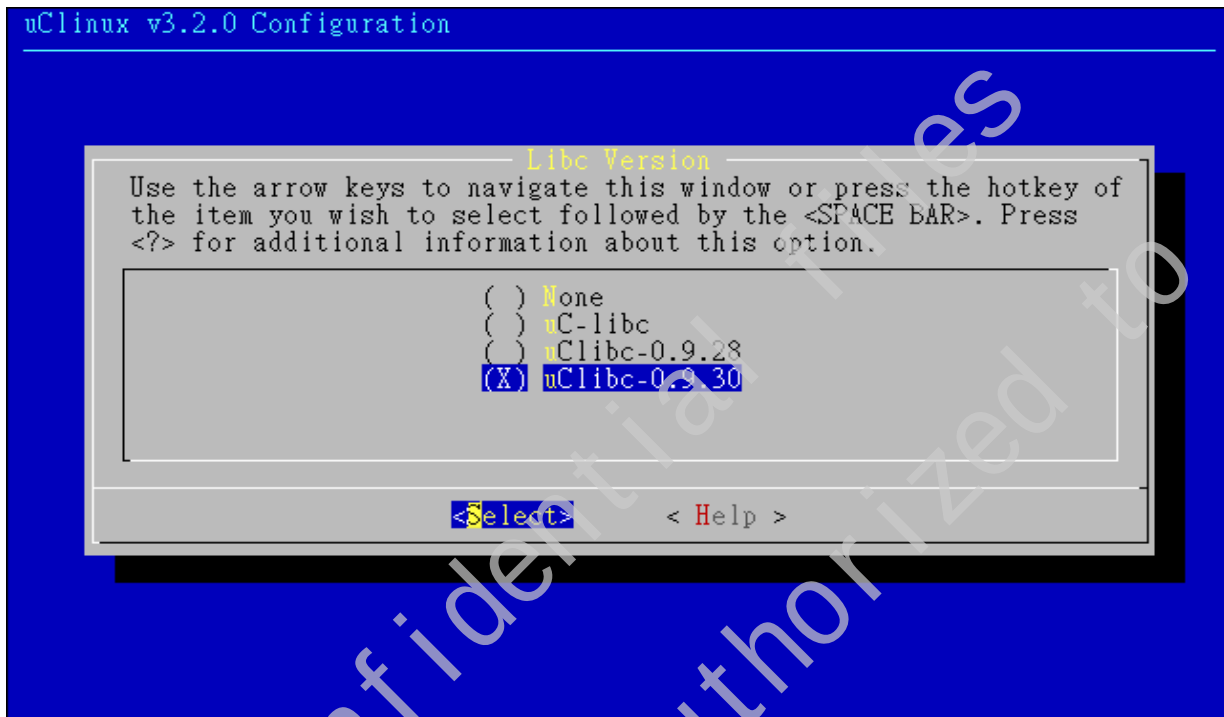


- In "Kernel/Library/Defaults Selection" menu, there are two Linux kernel version could be selected. The linux-2.6.x is 2.6.19 and the linux-2.6.32.x is 2.6.32.58:

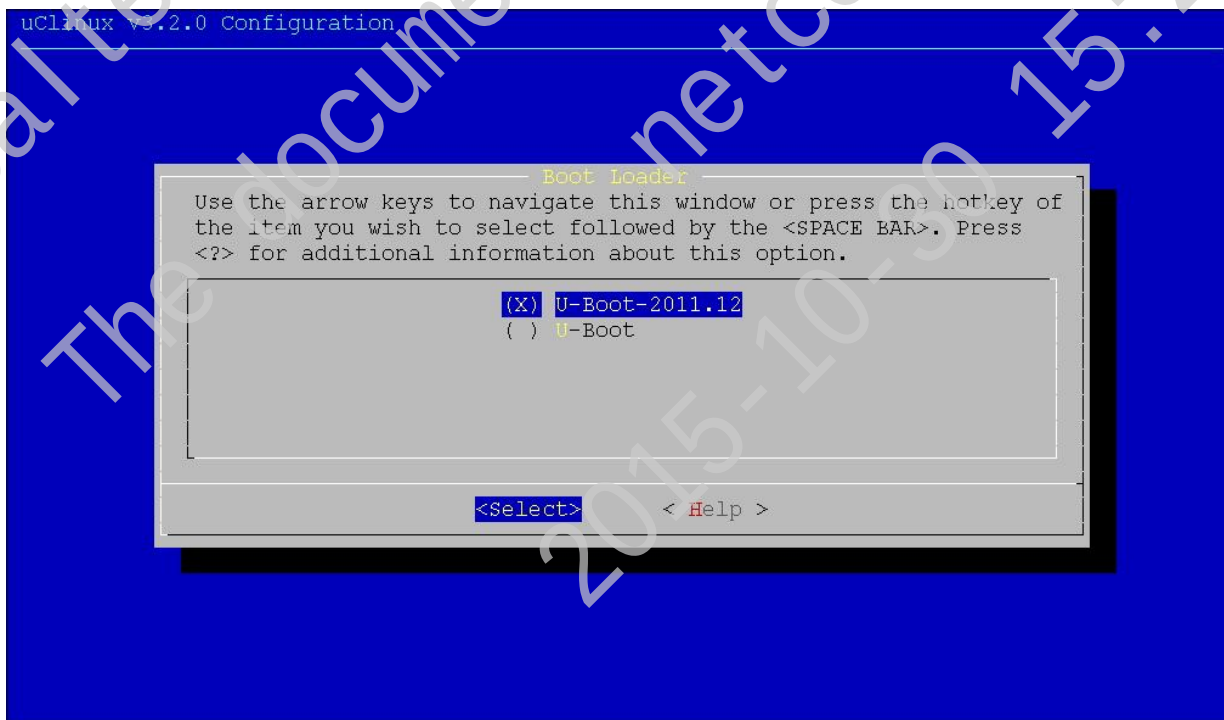


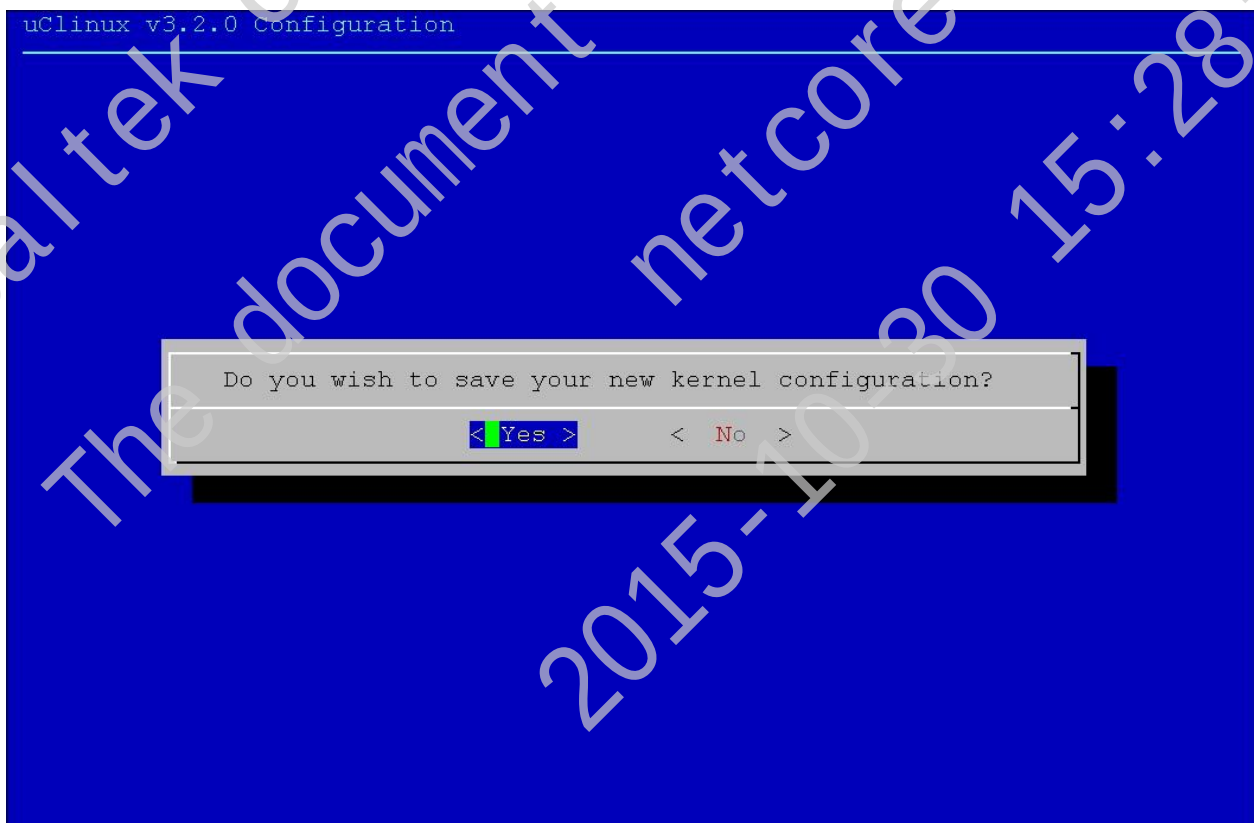
- In "Kernel/Library/Defaults Selection" menu, there are two

uClibc version could be selected. The linux-2.6.x should use 0.9.28 and the linux-2.6.32.x should use 0.9.30:



- In "Kernel/Library/Defaults Selection" menu, there are two boot code version could be selected. The "U-Boot-2011.12" is new one. For the RTL8393M/RTL8353M, please select the "U-Boot-2011.12".

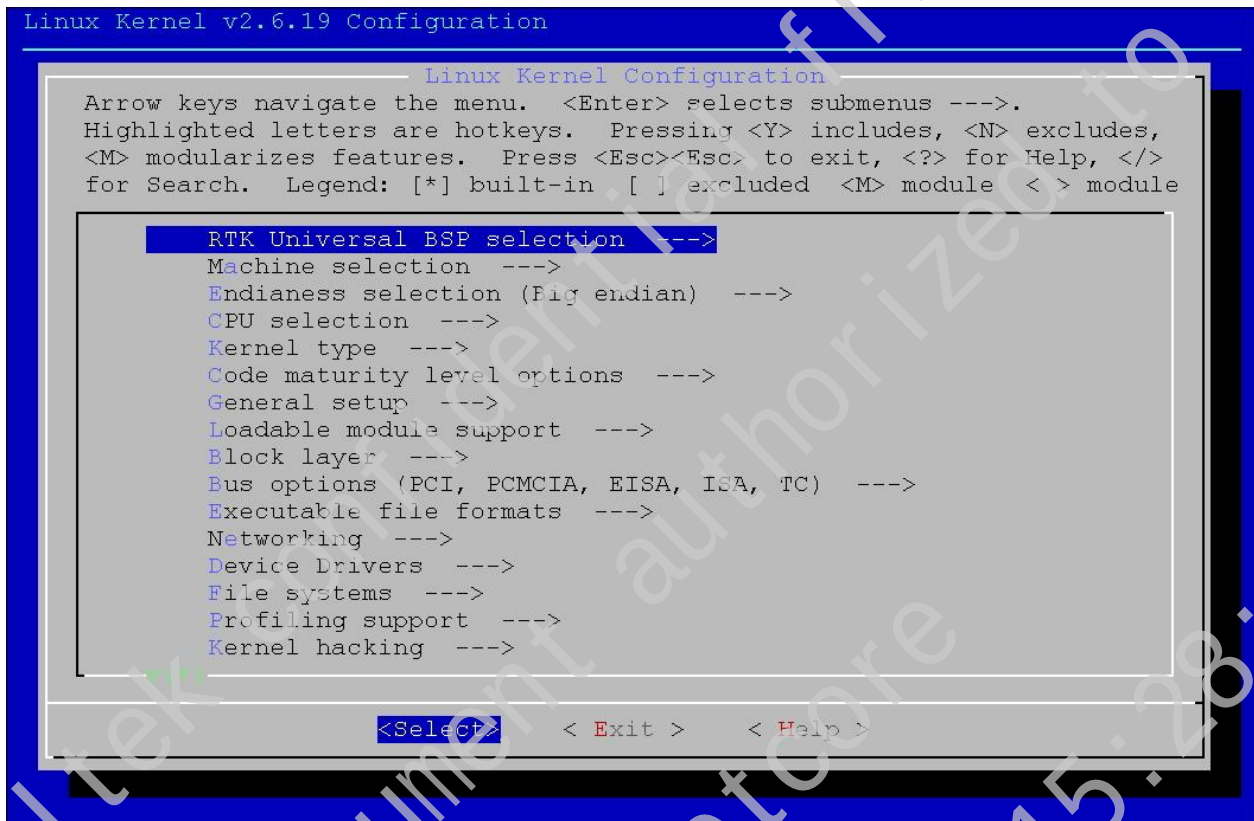




1.5.1 Kernel Menuconfig

1.5.1.1 Linux v2.6.19 Menuconfig

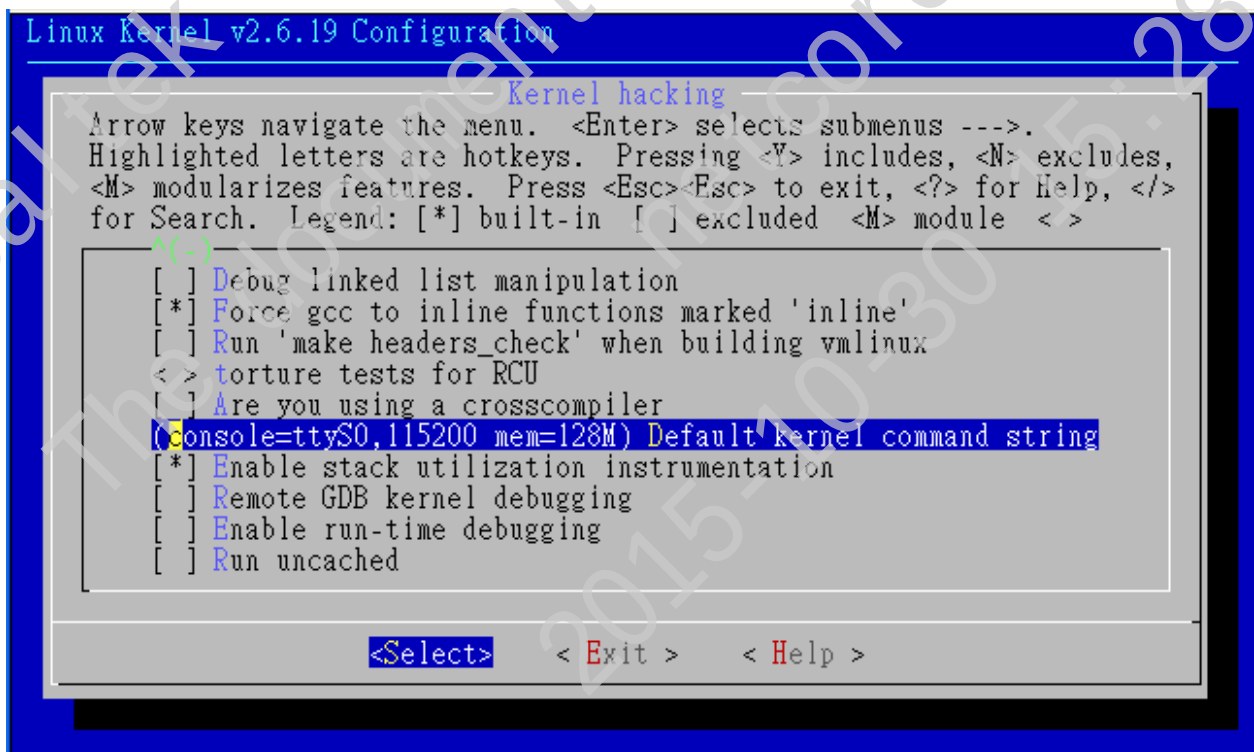
- Enter the "Linux Kernel v2.6.19 Configuration" as following:



- "RTK Universal BSP selection" setting as following: (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.

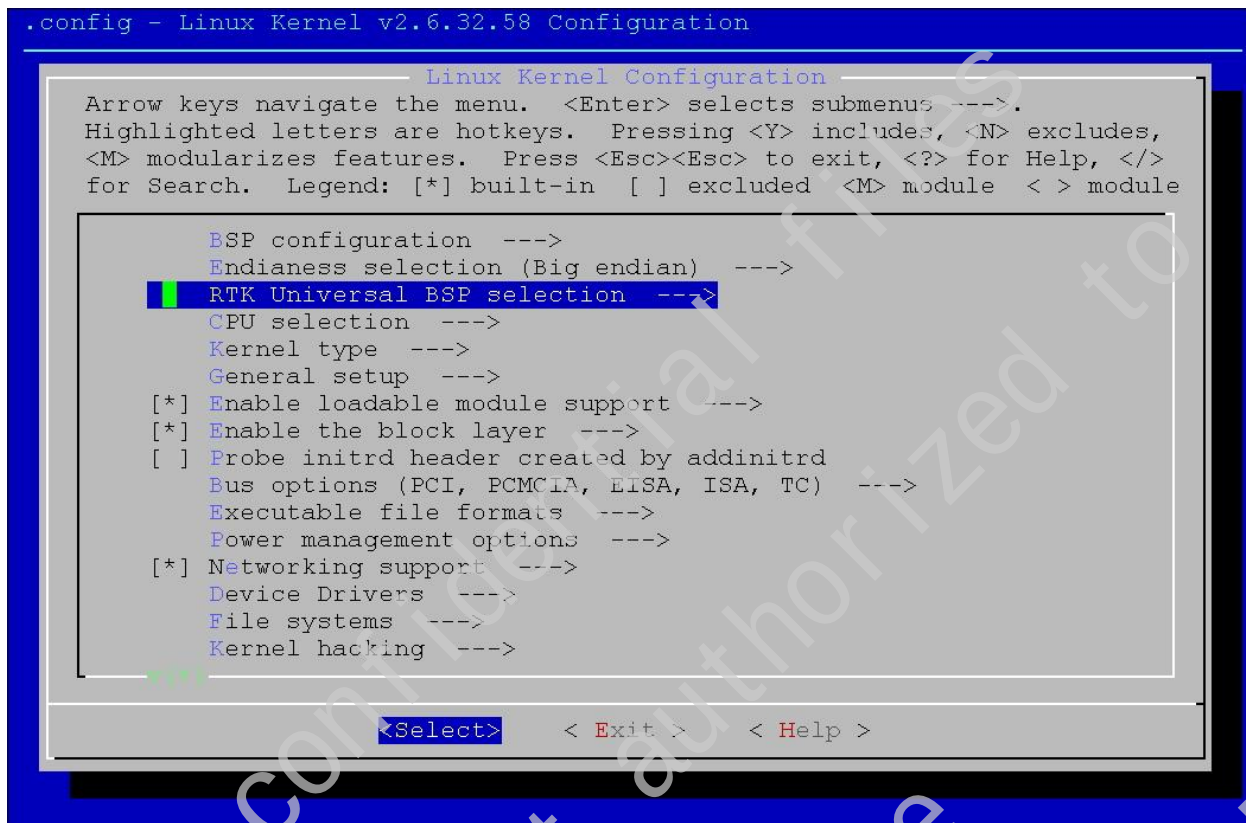


- “Kernel hacking” setting as following: (128MB DDR RAM)

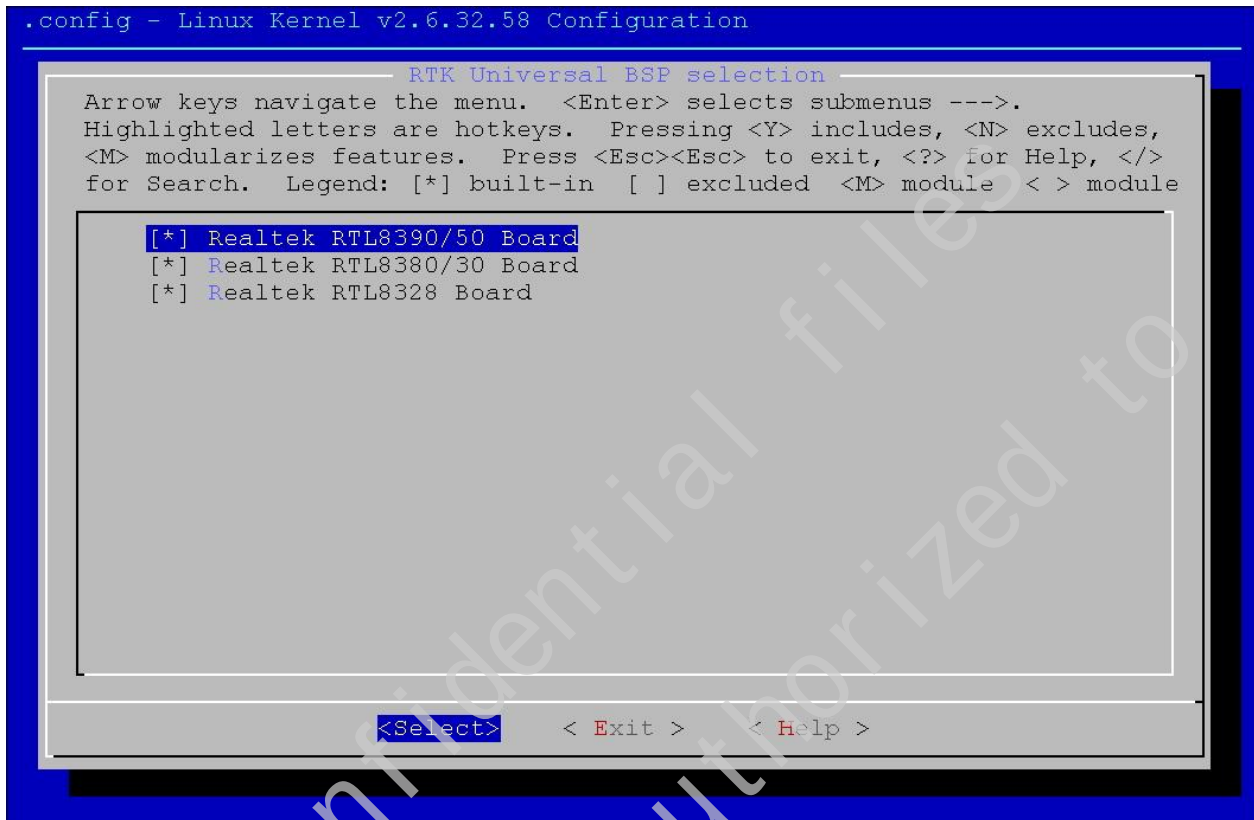


1.5.1.2 Linux v2.6.32.58 Menuconfig

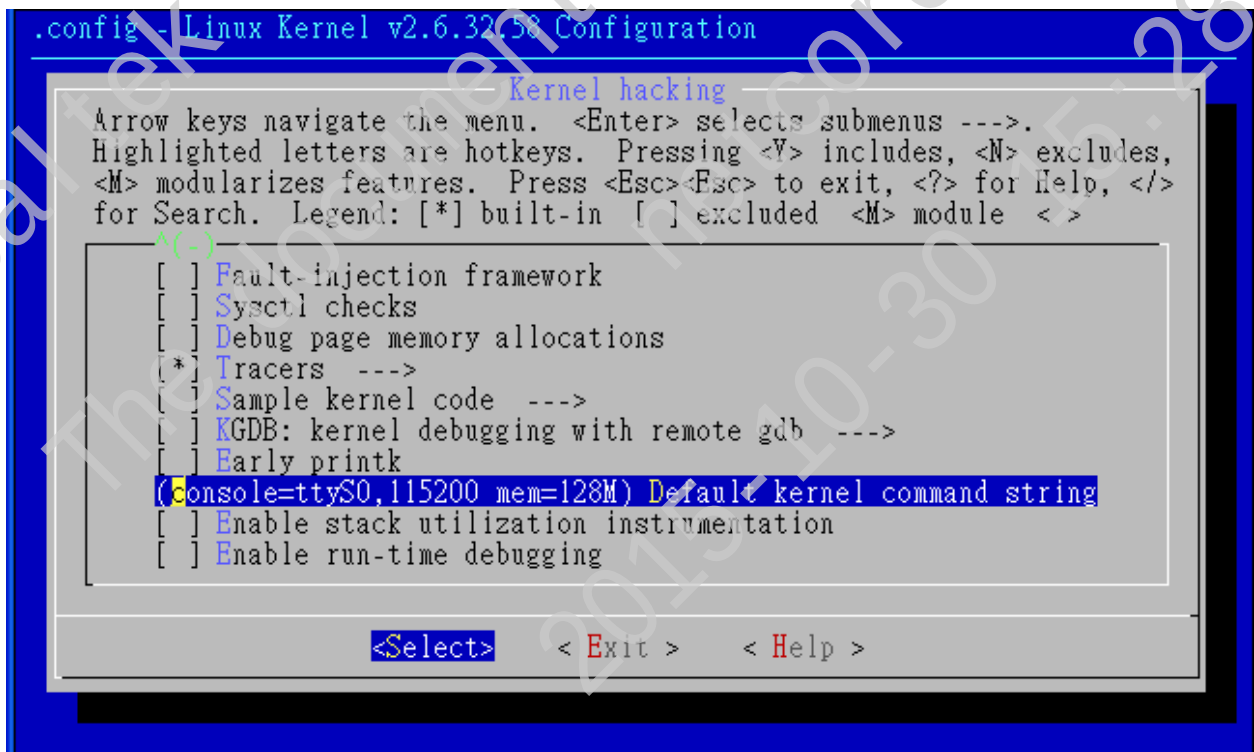
- Enter the “Linux Kernel v2.6.32.58 Configuration” as following:



- “RTK Universal BSP selection” setting as following: (16MB SPI Flash). In this menu is used to decide which chip will be supported by the Linux kernel image, and chip items could be multiple-selected.

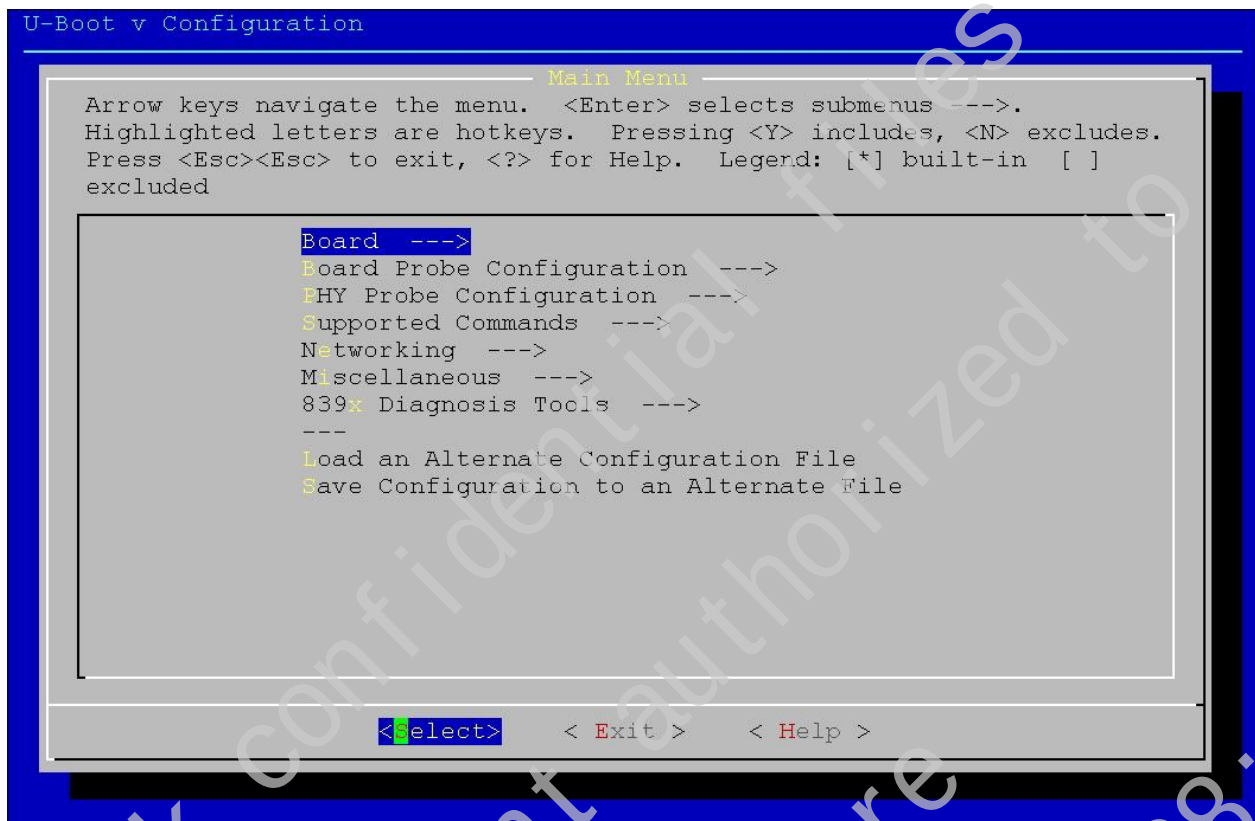


- "Kernel hacking" setting as following: (128MB DDR RAM)

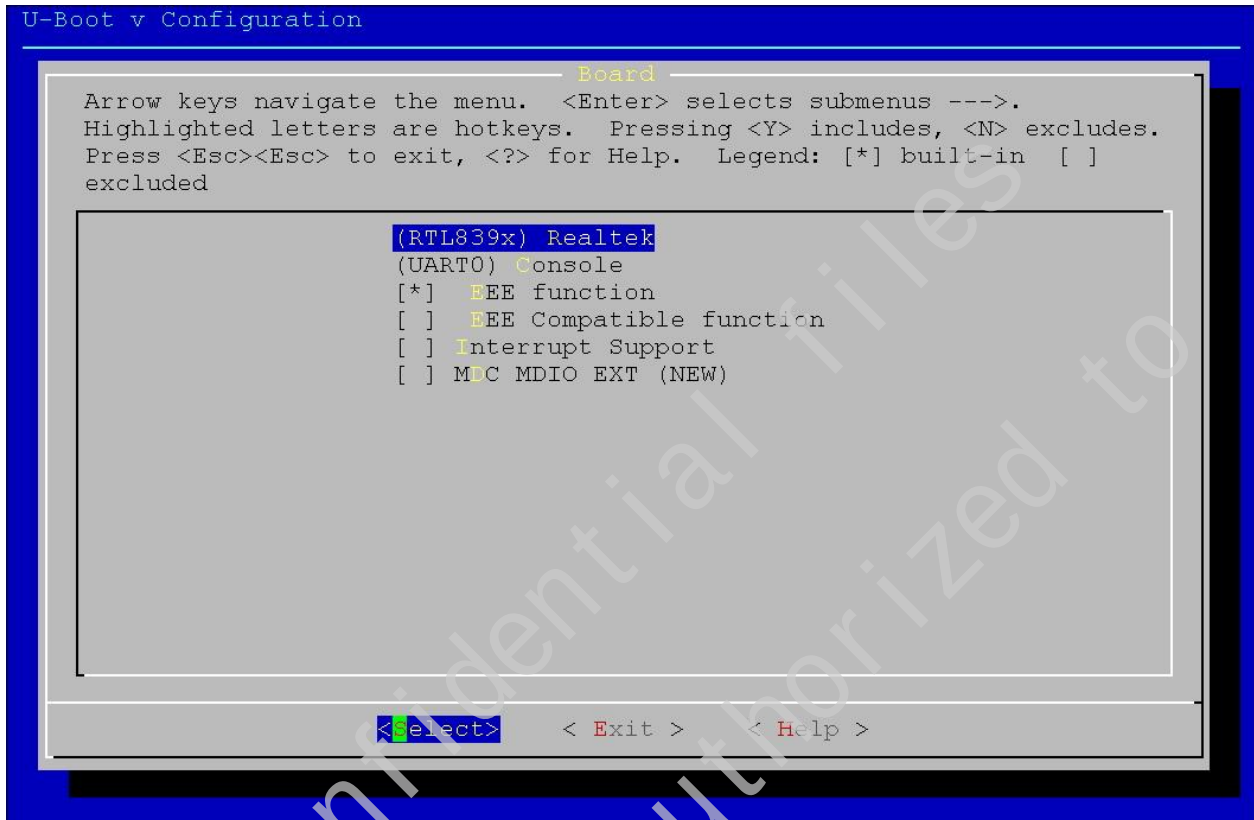


1.5.2 U-Boot 2011.12 Menuconfig

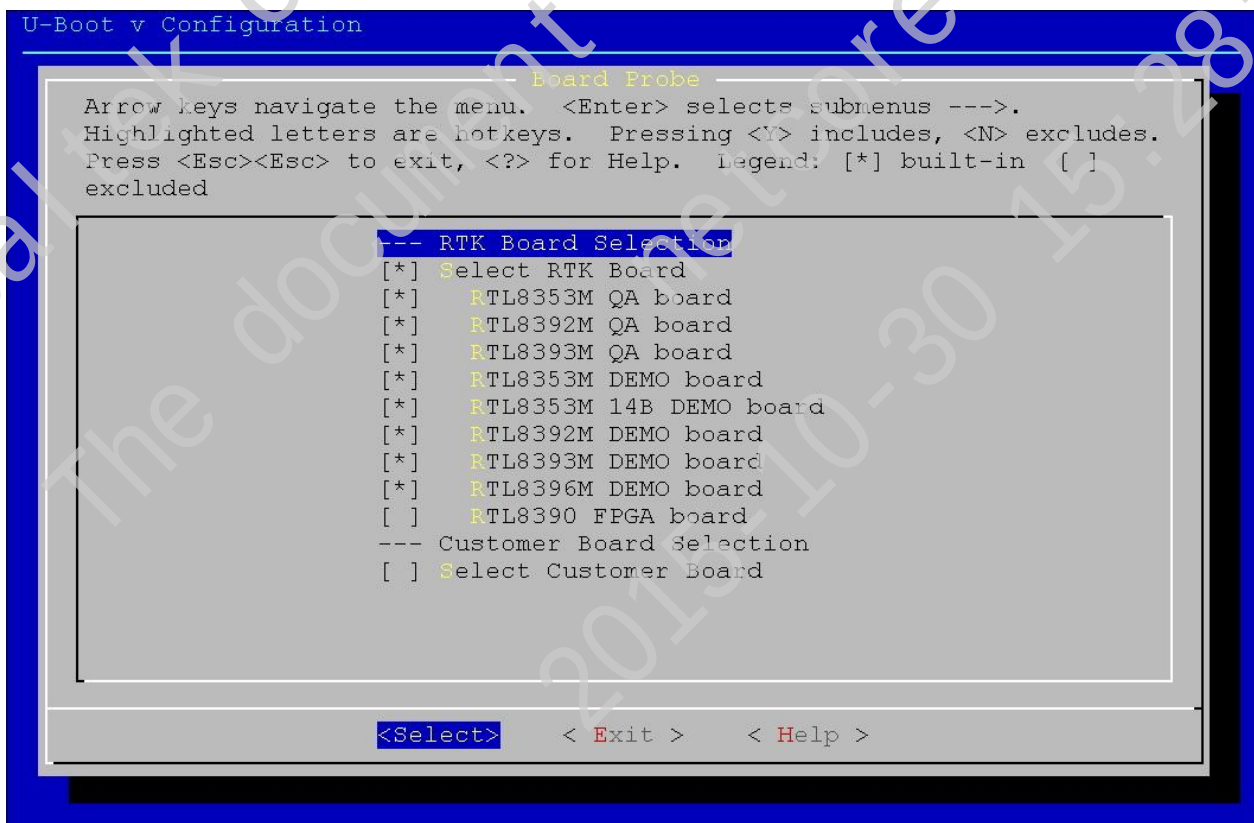
- After exit and save "Linux Kernel Configuration" setting, you can enter "U-Boot v Configuration" Main Menu as following:



- "Board Selection" setting as following:



➤ "Board Probe Configuration" setting as following:



➤ "Supported Commands" setting as following:

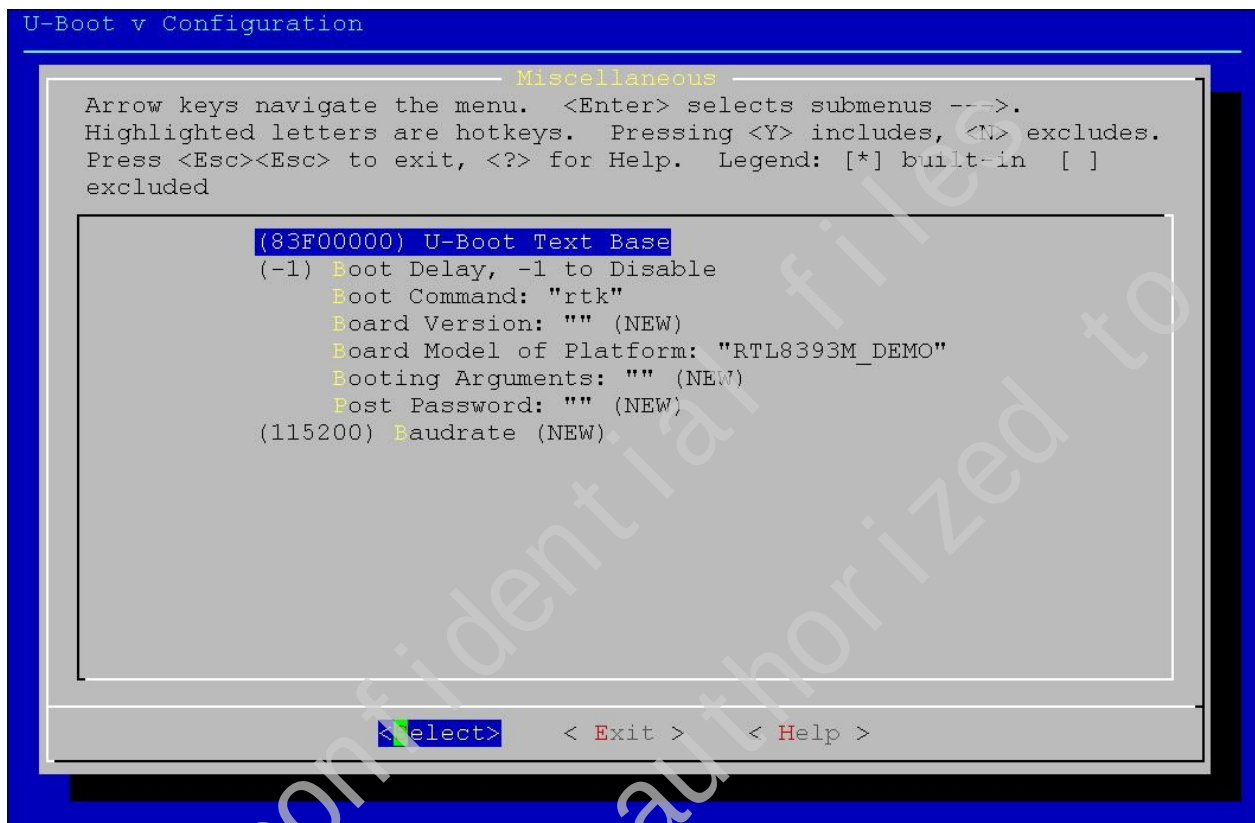
U-Boot v Configuration



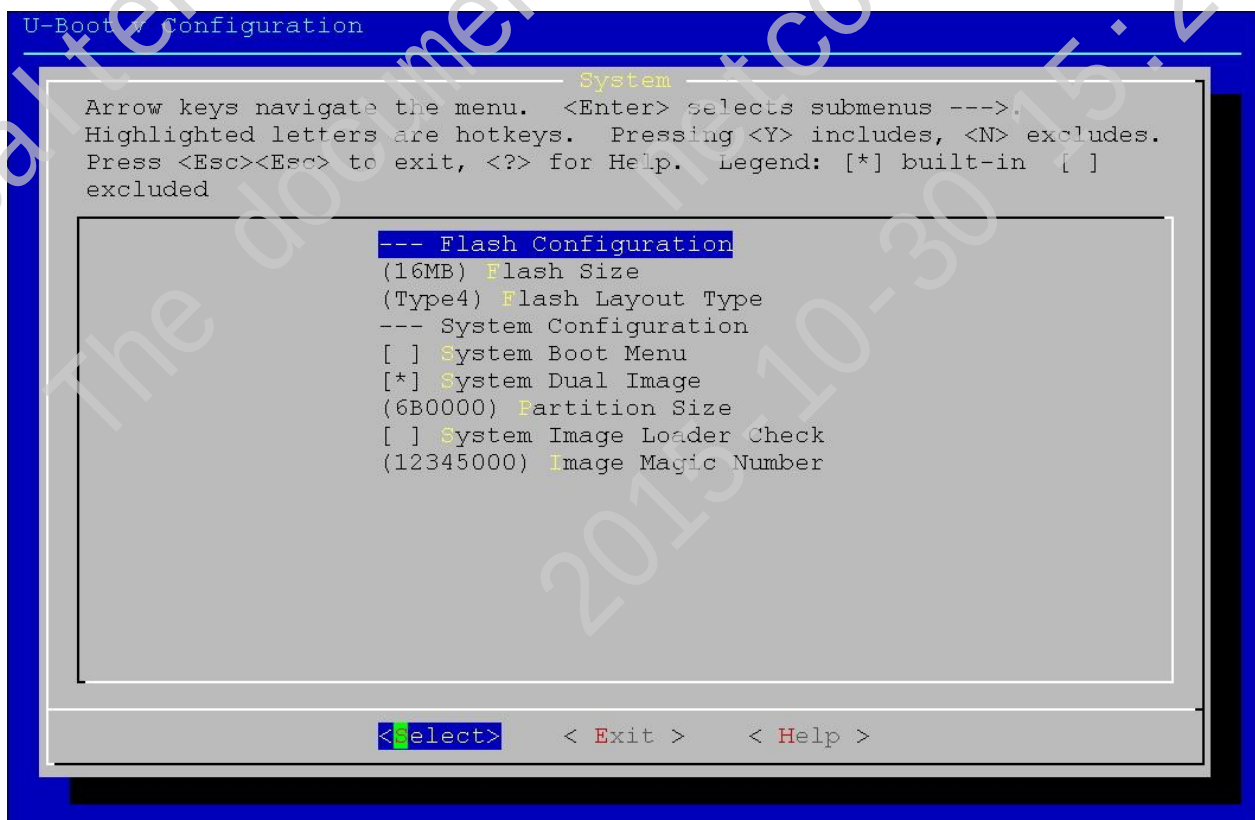
➤ "Networking" setting as following:



- “Miscellaneous” setting as following:

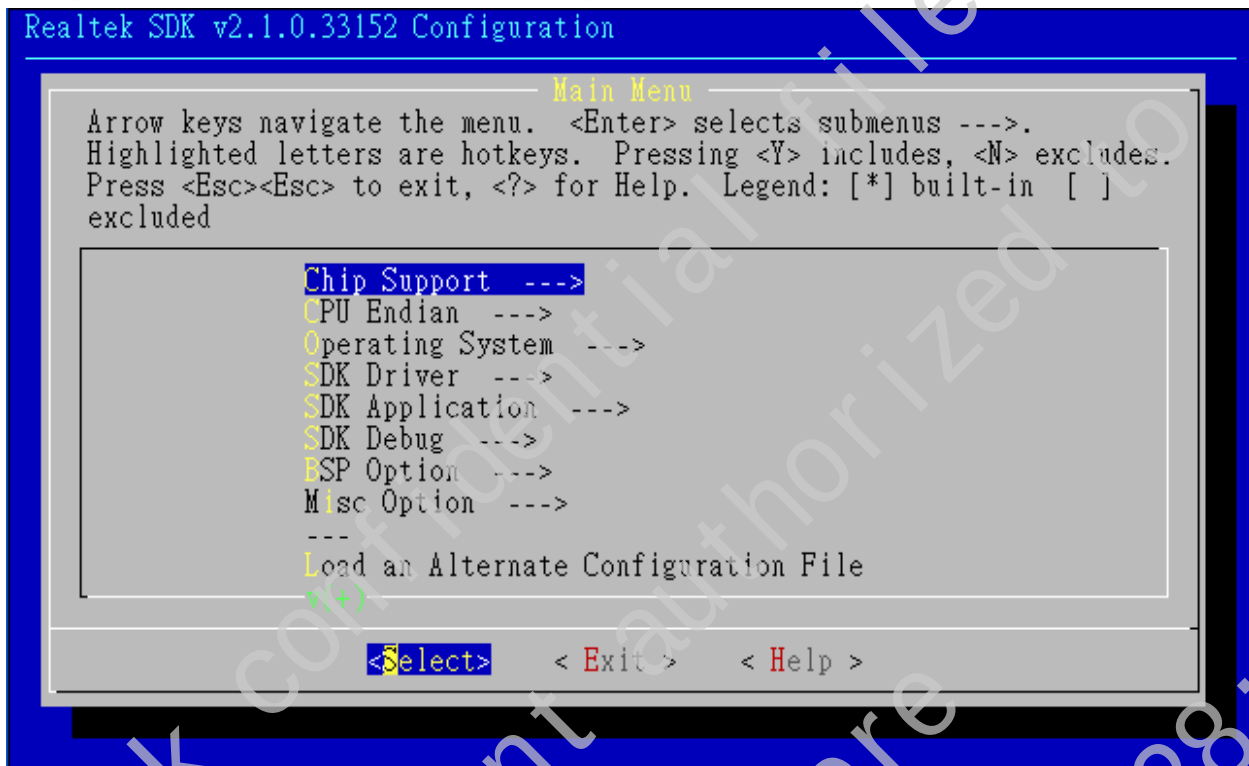


- “System” setting as following:



1.5.3 SDK Menuconfig

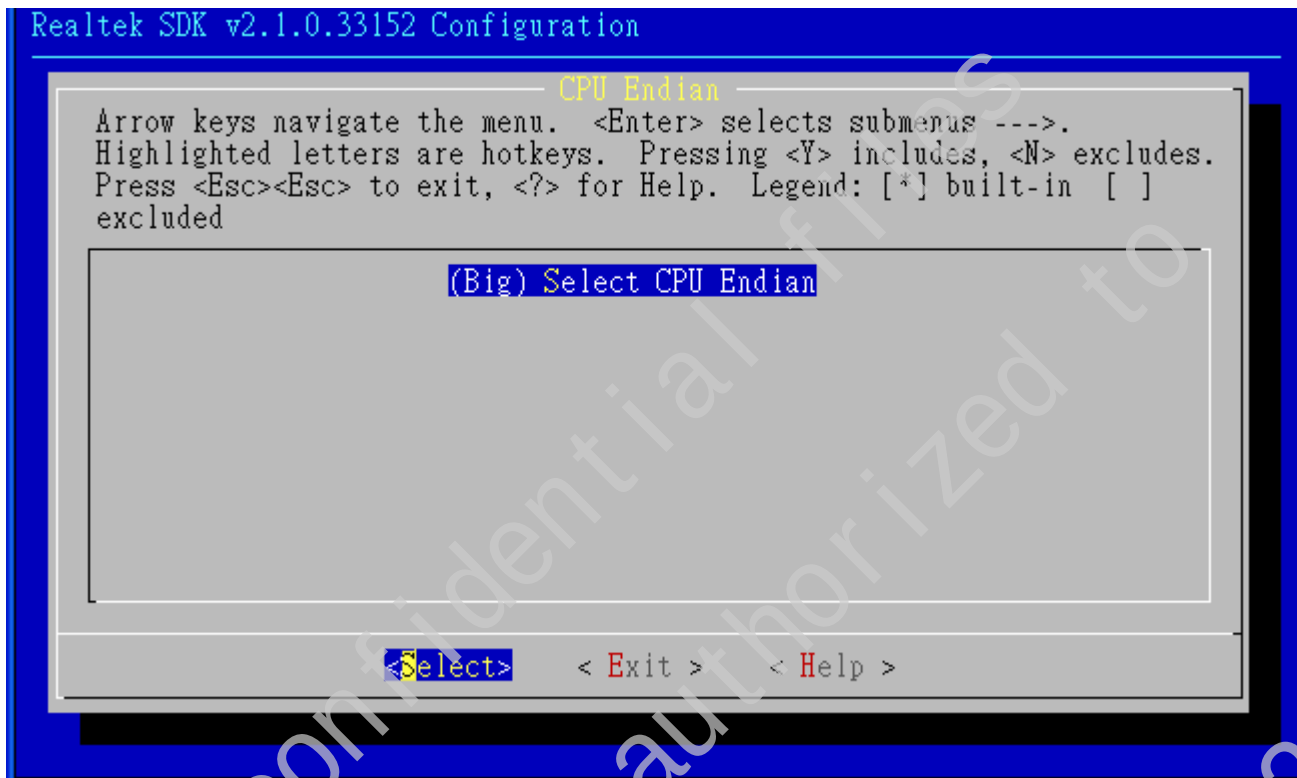
- After exit and save "U-Boot 2011.12 Configuration" setting, you can enter "SDK Configuration" Main Menu as following:



- "Chip Support" setting as following:



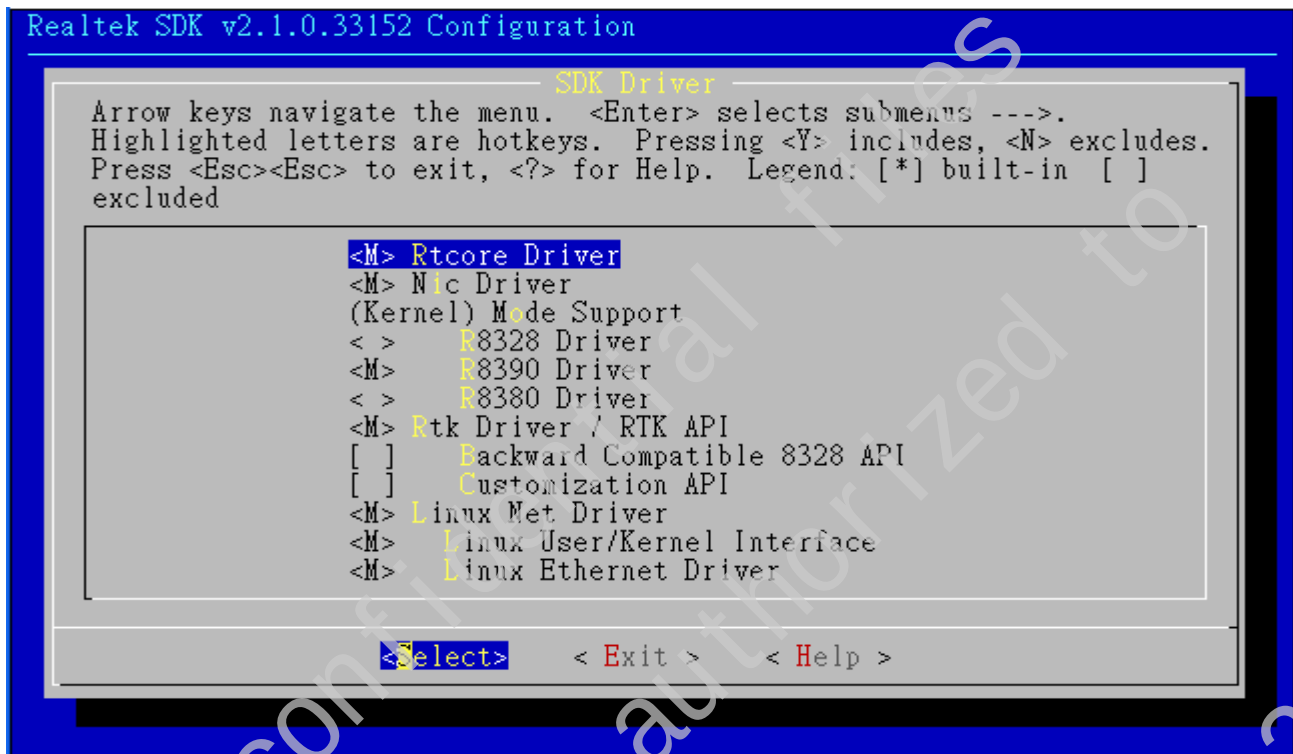
- “CPU Endian” setting as following:



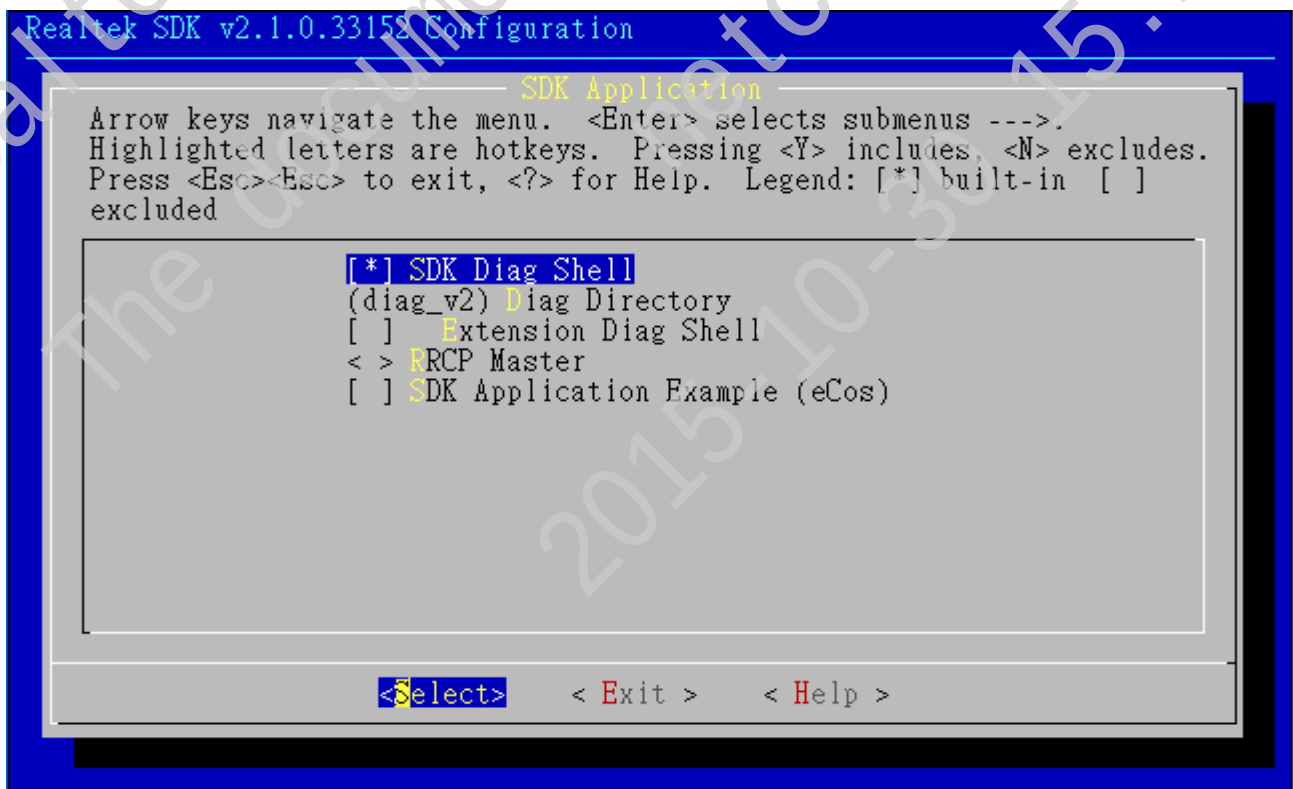
- “Operation System” setting as following for SDK driver run at Linux kernel mode:



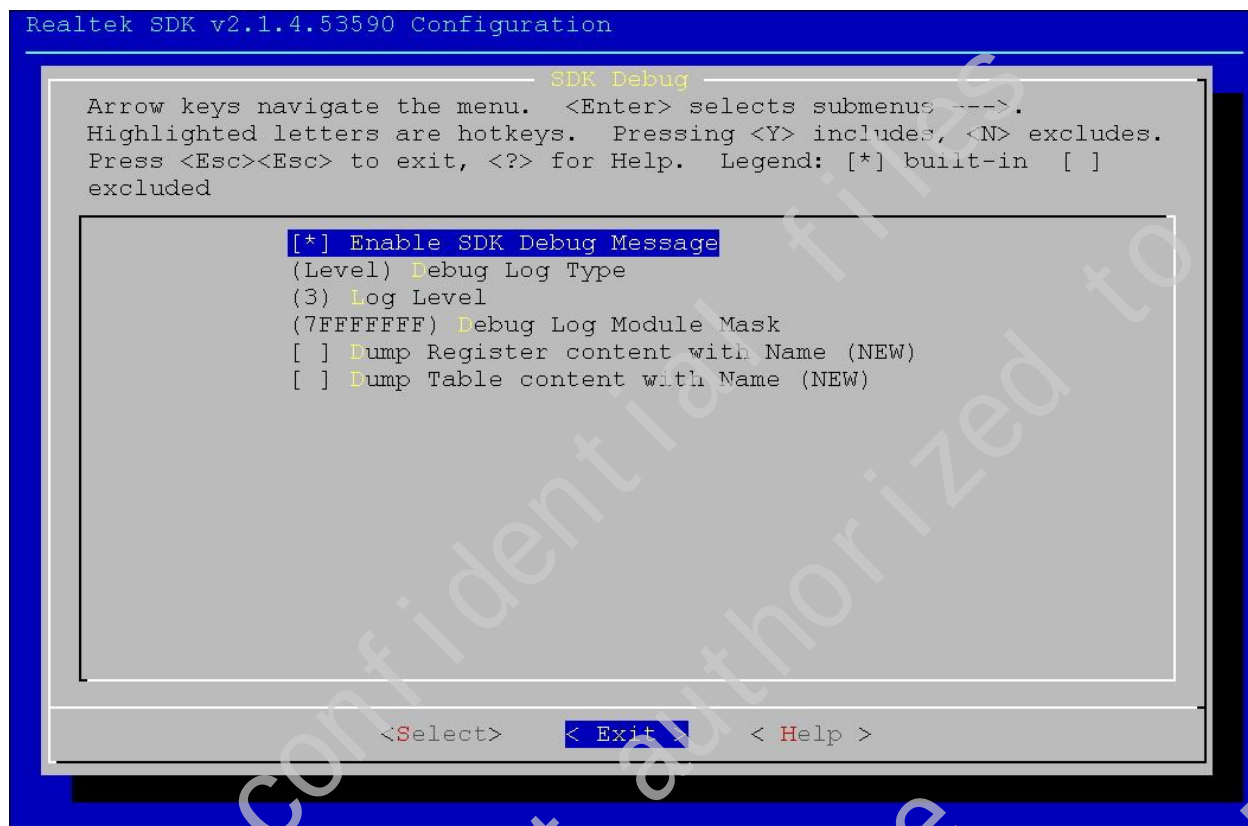
- “SDK Driver” setting as following for NIC driver run at Linux kernel mode:



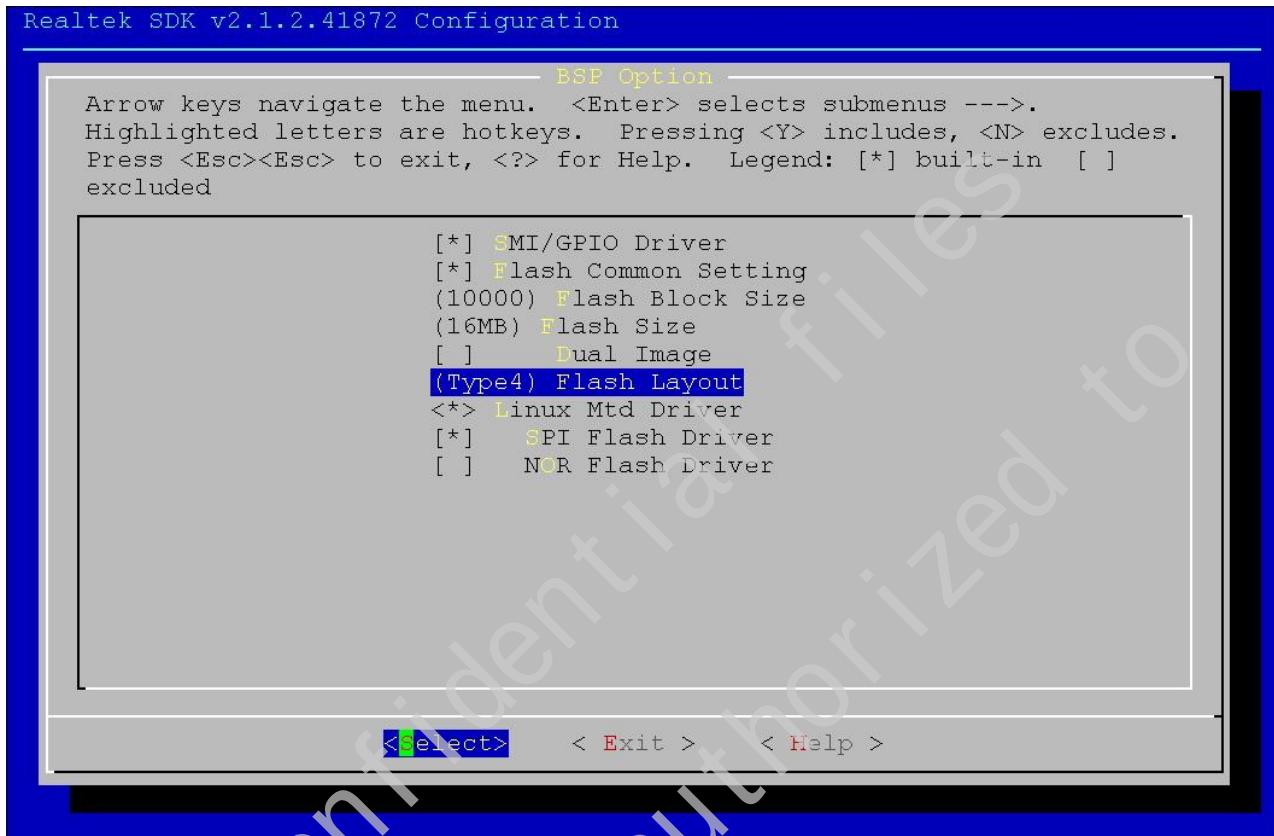
- “SDK Application” setting as following for turn on diagnostic shell: (MUST select diag_v2)



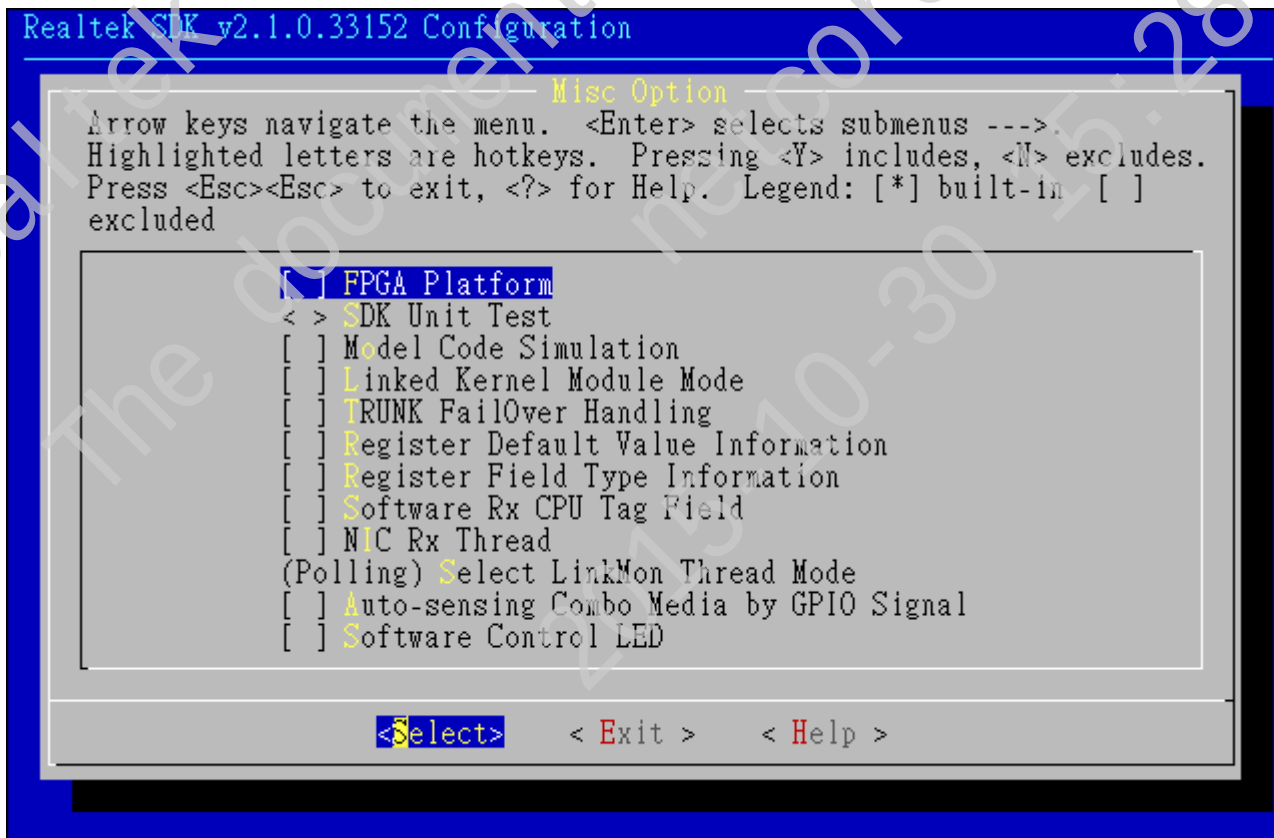
- “SDK Debug” setting as following:



- “BSP Option” setting as following:



- “Misc Option” setting as following:



- After exit and save "SDK Configuration" setting, please type "make" for build out u-boot and runtime images

Realtek Semiconductor Corp.

Headquarters

No. 2, Innovation Road II

Hsinchu Science Park, Hsinchu 300, Taiwan

Tel.: +886-3-578-0211. Fax: +886-3-577-6047

www.realtek.com